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|--------------------|------------|
| Report Reference:  | R3451      |
| Report Issue Date: | 25/06/2025 |

## **1.0 Introduction**

The Transport for London Technical Services Team are obtaining data to understand the typical interior noise levels from services on the London Underground (LU) network. These measurements are carried on specially modified trains equipped with Auto Visual Inspection (AVI) equipment and/or Asset Conditioning Monitoring (ACM) equipment.

The purpose of the measurements is to characterise and monitor the noise exposure of passengers given the current track conditions whilst travelling on the LU network. This report does not assess, evaluate or make any recommendations regarding the noise levels or whether any remedial actions need to take place.

A glossary of acoustic terms and their meanings has been included in Appendix A.

## **2.0 Test Methodology**

The acoustic tests were conducted in accordance with *BS EN ISO 2021:3381 - Railway applications — Acoustics — Noise measurement inside railbound vehicles*. It is noted that the scope within this standard "does not include all the instructions to carry out monitoring testing or evaluation of noise exposure of passengers or drivers over a whole journey".

### **2.1 Measurement Procedure**

On 16/04/2025 acoustic measurement equipment was installed in the middle passenger saloon car of a LU 1992 stock train on the Central line. The middle car was chosen as the leading carriage of the train changes depending on the route taken on the test day, therefore with the measurement equipment in the centre of the train, the position remains constant regardless of direction of travel.

Sound pressure data were acquired using a four channel integrating sound level meter with combined 1/2" microphone and preamplifier sets. The following parameters were obtained over 1-second integration periods:

- A-weighted equivalent continuous sound pressure level ( $L_{Aeq,T}$ ),
- C-weighted peak sound pressure level ( $L_{CPeak}$ ); and
- one-third octave band unweighted equivalent continuous sound pressure level ( $L_{Zeq,T}$ ) for centre band frequencies 20 Hz to 20 kHz.

Measurement positions were chosen to represent both seated and standing positions within the saloon. Microphones were positioned using clamps and/or tripods at a relative height of 1.2 m and 1.6 m for seating and standing positions, respectively. All positions were more than 0.3 m distance from any reflecting surface.

Figure 1 below shows a generic passenger car with the positions of a typical set-up. Depending on the equipment limitations at the time of the test, not all locations were utilised. Table 1 shows the equipment used at the relevant position.

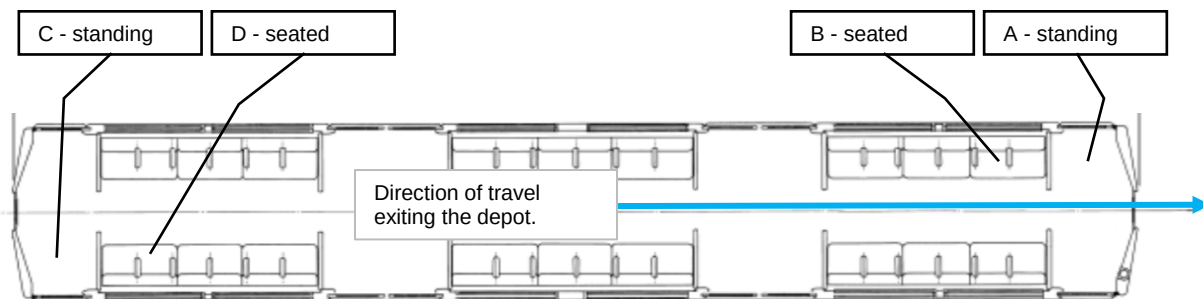


Figure 1 - Typical measurement locations in passenger car



## **2.2 Equipment**

Field calibration of the operation channels of sound level meter were performed using an acoustic calibrator which gave a sinusoidal calibration tone of 94 dB at a frequency of 1000 Hz. The field calibration was carried out before and after the survey; the measurements remained stable throughout. The SLM and acoustic calibrator conform to 'class 1' specifications for corresponding equipment defined in British Standards BS EN 61672-1:2013, BS EN 61260-3:2016 and BS EN 60942:2018. The SLM, microphones and pre-amps have been calibrated to traceable standards within the preceding two years, and the acoustic calibrators within the preceding 12 months.

*Table 1 - Equipment*

| Manufacturer | Item        | Type  | Serial Number | Location     |
|--------------|-------------|-------|---------------|--------------|
| Svantek      | Sound Meter | 958A  | 69835         | n/a          |
| Svantek      | Microphone  | SV12L | 77976         | A - Standing |
| Svantek      | Microphone  | SV12L | 75577         | B - Seated   |

## **2.3 Test Conditions**

The test operating conditions for the vehicle are representative of normal service and may have included passenger announcements. Microphones were placed away from any speakers to minimise the influence. Windows and ventilation grilles remained closed during the exercise. Test engineers may have also accessed the carriage during the measurement and as a result there may be some influence from doors slamming or localised speech.

Generally, the services would follow the Driving Technique Guidance where possible, however, invariably there may be deviations or significant changes to the speed due to existing operational issues on the day of the test run, these should therefore be used with caution.

Presented within Appendix B, is the test run order, the corresponding track Location Coding System (LCS), the direction of travel, the interstation section station code and the interstation section description.

## **2.4 Tachometer**

A tachometer was fitted to an axle-box on train. A signal from the tachometer was obtained and synchronised with the sound level meter in the middle carriage. Speed and distance were derived allowing for an approximate location along each interstation section.

## **2.5 Limitations**

Depending on the direction of the leading car, the noise levels in the middle carriage could be +/- 70 m of the stated measurement position along with the potentially distance covered over the 1 second integration period (i.e. speed of 20 m/s would mean the train could be +/- 10 m over the integration period).

## **3.0 Analysis**

A spatial average of the  $L_{Aeq}$  and the highest  $L_{CPeak}$  measured was derived from the positions listed in Table 1. For each interstation section, which includes the departure from the station and arrival into the next station, the following parameters were determined: the A-weighted equivalent continuous sound pressure level ( $L_{Aeq}$ ), the C-weighted peak sound pressure level ( $L_{CPeak}$ ), the sound exposure level (SEL); and the Daily Personal Exposure level ( $L_{EP,d}$ ) as defined by the Control of Noise at Work Regulations 2005. Additionally, one-third octave band unweighted equivalent continuous sound pressure level ( $L_{Zeq}$ ) for centre band frequencies 20 Hz to 20 kHz were processed but not provided within the report.

## **4.0 Results**

Presented below, is a graphical and tabulated summary of the  $L_{Aeq,T}$ ,  $L_{CPeak,T}$ ,  $L_{EP,d}$  and SEL for each interstation section. The station codes are presented and the interstation section description can be found in Appendix B.

Appendix C presents a distance vs. noise plot for each interstation section.

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

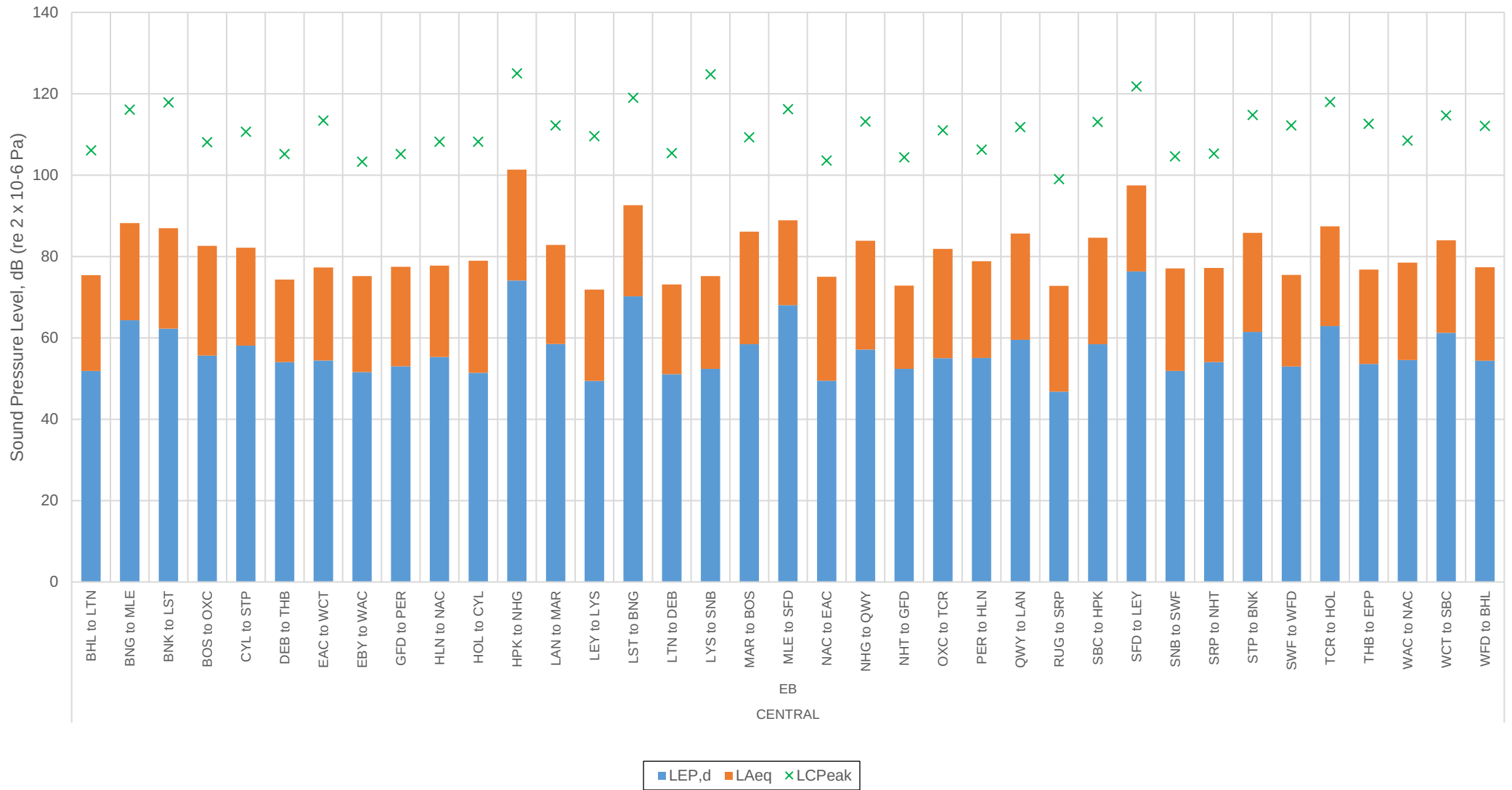


Figure 2 - Interstation noise levels on the Central eastbound

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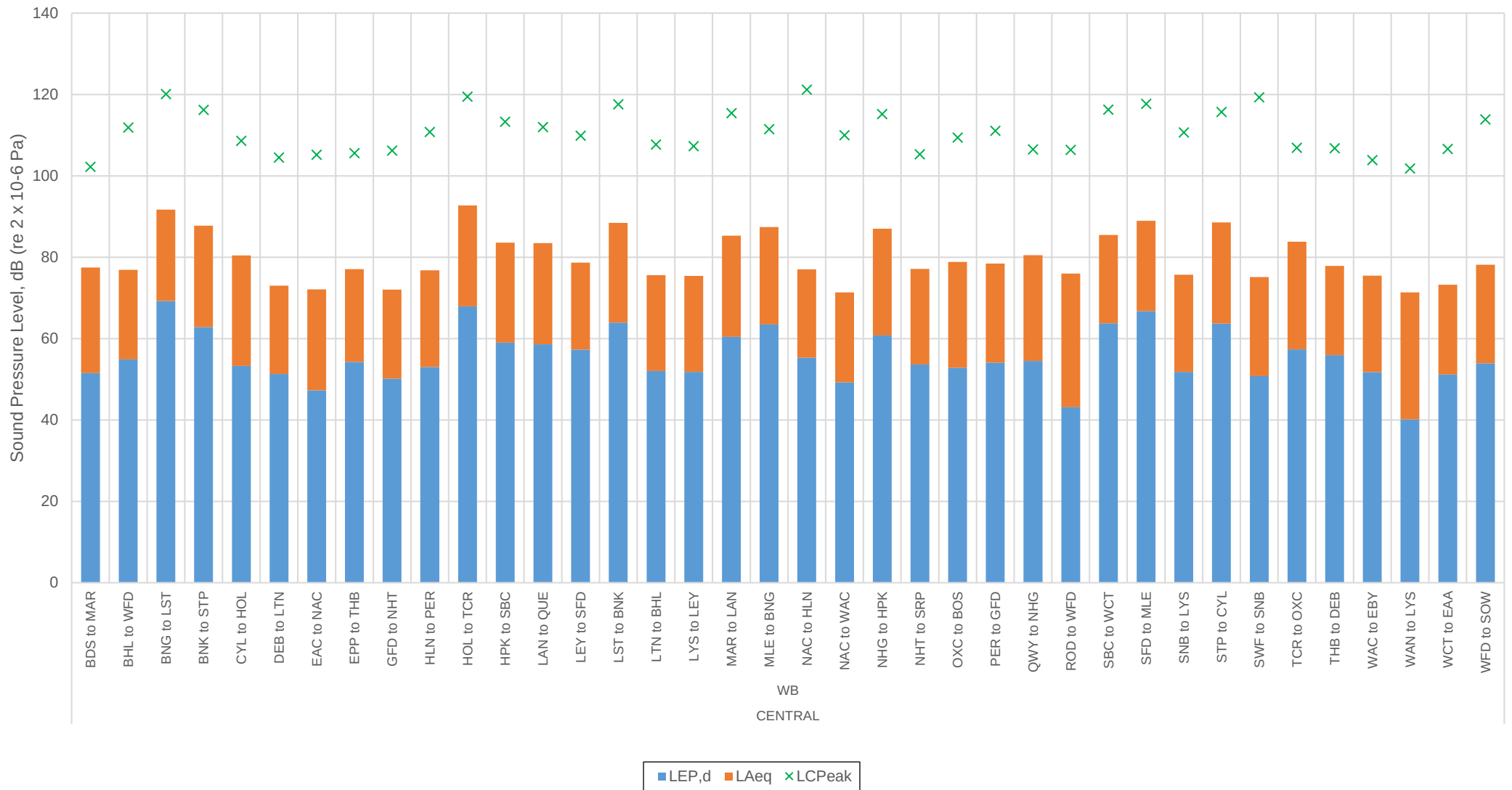


Figure 3 - Interstation noise levels on the Central westbound

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

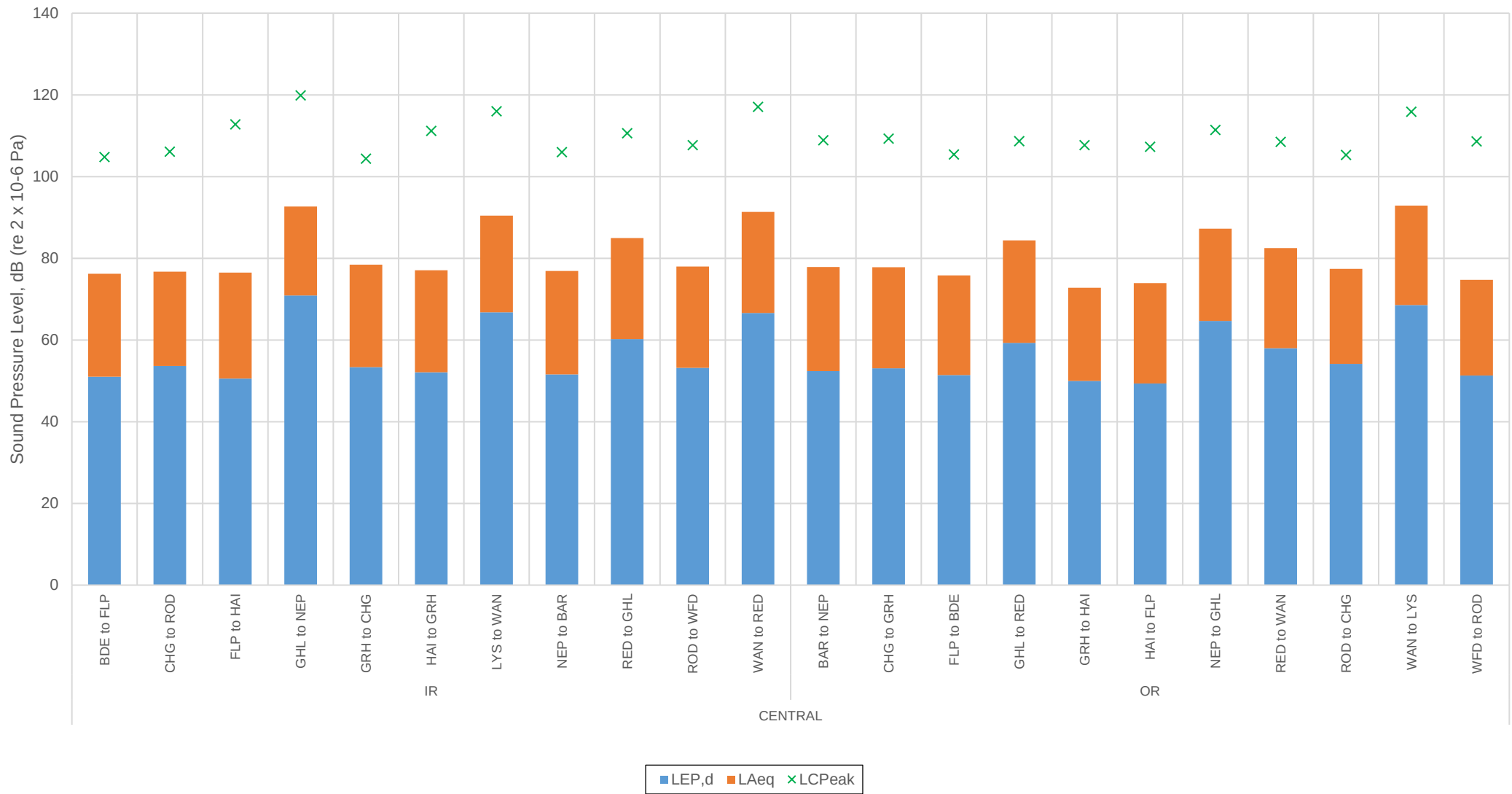


Figure 3 - Interstation noise levels on the Central inner and outer road

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**Interior Noise Measurement of a Test Vehicle**



Table 2 - Summary Table

| Interstation Section | Duration<br>(secs) (T) | LAeq,T | LEP,d | SEL | LCPeak,T |
|----------------------|------------------------|--------|-------|-----|----------|
| <b>CENTRAL</b>       |                        |        |       |     |          |
| <b>EB</b>            |                        |        |       |     |          |
| BHL to LTN           | 128                    | 75     | 52    | 96  | 106      |
| BNG to MLE           | 117                    | 88     | 64    | 109 | 116      |
| BNK to LST           | 98                     | 87     | 62    | 107 | 118      |
| BOS to OXC           | 58                     | 83     | 56    | 100 | 108      |
| CYL to STP           | 114                    | 82     | 58    | 103 | 111      |
| DEB to THB           | 273                    | 74     | 54    | 99  | 105      |
| EAC to WCT           | 148                    | 77     | 54    | 99  | 113      |
| EBY to WAC           | 127                    | 75     | 52    | 96  | 103      |
| GFD to PER           | 103                    | 77     | 53    | 98  | 105      |
| HLN to NAC           | 163                    | 78     | 55    | 100 | 108      |
| HOL to CYL           | 51                     | 79     | 51    | 96  | 108      |
| HPK to NHG           | 54                     | 101    | 74    | 119 | 125      |
| LAN to MAR           | 106                    | 83     | 59    | 103 | 112      |
| LEY to LYS           | 164                    | 72     | 49    | 94  | 110      |
| LST to BNG           | 167                    | 93     | 70    | 115 | 119      |
| LTN to DEB           | 181                    | 73     | 51    | 96  | 105      |
| LYS to SNB           | 151                    | 75     | 52    | 97  | 125      |
| MAR to BOS           | 50                     | 86     | 58    | 103 | 109      |
| MLE to SFD           | 237                    | 89     | 68    | 113 | 116      |
| NAC to EAC           | 81                     | 75     | 49    | 94  | 104      |
| NHG to QWY           | 61                     | 84     | 57    | 102 | 113      |
| NHT to GFD           | 257                    | 73     | 52    | 97  | 104      |
| OXC to TCR           | 60                     | 82     | 55    | 100 | 111      |
| PER to HLN           | 121                    | 79     | 55    | 100 | 106      |
| QWY to LAN           | 70                     | 86     | 60    | 104 | 112      |
| RUG to SRP           | 73                     | 73     | 47    | 91  | 99       |
| SBC to HPK           | 70                     | 85     | 58    | 103 | 113      |
| SFD to LEY           | 221                    | 97     | 76    | 121 | 122      |
| SNB to SWF           | 86                     | 77     | 52    | 96  | 105      |
| SRP to NHT           | 140                    | 77     | 54    | 99  | 105      |
| STP to BNK           | 106                    | 86     | 61    | 106 | 115      |
| SWF to WFD           | 163                    | 75     | 53    | 98  | 112      |
| TCR to HOL           | 102                    | 87     | 63    | 108 | 118      |
| THB to EPP           | 137                    | 77     | 54    | 98  | 113      |
| WAC to NAC           | 116                    | 78     | 55    | 99  | 109      |
| WCT to SBC           | 153                    | 84     | 61    | 106 | 115      |
| WFD to BHL           | 145                    | 77     | 54    | 99  | 112      |
| <b>IR</b>            |                        |        |       |     |          |
| BDE to FLP           | 87                     | 76     | 51    | 96  | 105      |
| CHG to ROD           | 142                    | 77     | 54    | 98  | 106      |
| FLP to HAI           | 73                     | 77     | 51    | 95  | 113      |
| GHL to NEP           | 193                    | 93     | 71    | 116 | 120      |
| GRH to CHG           | 89                     | 78     | 53    | 98  | 104      |
| HAI to GRH           | 92                     | 77     | 52    | 97  | 111      |

**Acoustic Test Report:**  
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|            |     |    |    |     |     |
|------------|-----|----|----|-----|-----|
| LYS to WAN | 124 | 90 | 67 | 111 | 116 |
| NEP to BAR | 85  | 77 | 52 | 96  | 106 |
| RED to GHL | 97  | 85 | 60 | 105 | 111 |
| ROD to WFD | 95  | 78 | 53 | 98  | 108 |
| WAN to RED | 97  | 91 | 67 | 111 | 117 |
| <b>OR</b>  |     |    |    |     |     |
| BAR to NEP | 82  | 78 | 52 | 97  | 109 |
| CHG to GRH | 97  | 78 | 53 | 98  | 109 |
| FLP to BDE | 105 | 76 | 51 | 96  | 105 |
| GHL to RED | 90  | 84 | 59 | 104 | 109 |
| GRH to HAI | 150 | 73 | 50 | 95  | 108 |
| HAI to FLP | 100 | 74 | 49 | 94  | 107 |
| NEP to GHL | 159 | 87 | 65 | 109 | 111 |
| RED to WAN | 102 | 82 | 58 | 103 | 109 |
| ROD to CHG | 136 | 77 | 54 | 99  | 105 |
| WAN to LYS | 106 | 93 | 69 | 113 | 116 |
| WFD to ROD | 130 | 75 | 51 | 96  | 109 |
| <b>WB</b>  |     |    |    |     |     |
| BDS to MAR | 74  | 77 | 52 | 96  | 102 |
| BHL to WFD | 183 | 77 | 55 | 100 | 112 |
| BNG to LST | 163 | 92 | 69 | 114 | 120 |
| BNK to STP | 93  | 88 | 63 | 107 | 116 |
| CYL to HOL | 56  | 80 | 53 | 98  | 109 |
| DEB to LTN | 193 | 73 | 51 | 96  | 105 |
| EAC to NAC | 95  | 72 | 47 | 92  | 105 |
| EPP to THB | 151 | 77 | 54 | 99  | 106 |
| GFD to NHT | 188 | 72 | 50 | 95  | 106 |
| HLN to PER | 121 | 77 | 53 | 98  | 111 |
| HOL to TCR | 96  | 93 | 68 | 113 | 120 |
| HPK to SBC | 100 | 84 | 59 | 104 | 113 |
| LAN to QUE | 94  | 83 | 59 | 103 | 112 |
| LEY to SFD | 209 | 79 | 57 | 102 | 110 |
| LST to BNK | 101 | 88 | 64 | 109 | 118 |
| LTN to BHL | 127 | 76 | 52 | 97  | 108 |
| LYS to LEY | 126 | 75 | 52 | 96  | 107 |
| MAR to LAN | 93  | 85 | 60 | 105 | 115 |
| MLE to BNG | 117 | 87 | 63 | 108 | 112 |
| NAC to HLN | 193 | 77 | 55 | 100 | 121 |
| NAC to WAC | 177 | 71 | 49 | 94  | 110 |
| NHG to HPK | 69  | 87 | 61 | 105 | 115 |
| NHT to SRP | 130 | 77 | 54 | 98  | 105 |
| OXC to BOS | 73  | 79 | 53 | 97  | 109 |
| PER to GFD | 105 | 78 | 54 | 99  | 111 |
| QWY to NHG | 72  | 81 | 54 | 99  | 107 |
| ROD to WFD | 15  | 76 | 43 | 88  | 106 |
| SBC to WCT | 192 | 85 | 64 | 108 | 116 |
| SFD to MLE | 172 | 89 | 67 | 111 | 118 |
| SNB to LYS | 118 | 76 | 52 | 96  | 111 |
| SRP to RUG | 68  | 75 | 49 | 93  | 120 |
| STP to CYL | 94  | 89 | 64 | 108 | 116 |
| SWF to SNB | 106 | 75 | 51 | 95  | 119 |

**Acoustic Test Report:**  
**Interior Noise Measurement of a Test Vehicle**



|            |     |    |    |     |     |
|------------|-----|----|----|-----|-----|
| TCR to OXC | 65  | 84 | 57 | 102 | 107 |
| THB to DEB | 185 | 78 | 56 | 101 | 107 |
| WAC to EBY | 122 | 75 | 52 | 96  | 104 |
| WAN to LYS | 22  | 71 | 40 | 85  | 102 |
| WCT to EAA | 179 | 73 | 51 | 96  | 107 |
| WFD to SOW | 109 | 78 | 54 | 99  | 114 |





## **Appendix A - Terminology**

### ***A-weighting***

a frequency weighting devised to attempt to take into account the fact that human response to sound is not equally sensitive to all frequencies

### ***C-weighting***

a frequency weighting that correlates better with the human response to high noise levels and therefore noise induced hearing loss.

### ***Continuous equivalent A-weighted sound pressure level, $L_{Aeq}$***

the steady noise level which, over the period of time under consideration, contains the same amount of A-weighted sound energy as the time-varying noise, over the same period of time.

### ***Daily personal noise exposure ( $L_{EP'd}$ )***

the time-averaged, A-weighted noise level for a nominal 8-hour working day used for assessing the noise exposure of a worker during a working day.

### ***Maximum A-weighted sound pressure level, $L_{AFMax}$***

the maximum RMS A-weighted sound pressure level occurring within a specified time period with a fast time weighting.

### ***Peak sound pressure level ( $L_{CPeak}$ )***

the maximum instantaneous C-weighted sound pressure level within a specified time period.

### ***Sound exposure level, SEL ( $L_{AE}$ )***

the A-weighted sound pressure level which, if occurring over a period of one second, would contain the same amount of A-weighted sound energy as the event.



## **Appendix B - Test Run Order**

*Table B1 - Test Run Order*

| <b>Run Order</b> | <b>LCS</b>  | <b>Road</b> | <b>STN TO STN</b> | <b>Interstation Section</b>                     |
|------------------|-------------|-------------|-------------------|-------------------------------------------------|
| 2                | C166-C-EBFA | EB          | RUG to SRP        | Ruislip Gardens (EB) TO South Ruislip (EB)      |
| 3                | C165-C-EBFA | EB          | RUG to SRP        | South Ruislip (EB)                              |
| 4                | C164-C-EBFA | EB          | SRP to NHT        | South Ruislip (EB) TO Northolt (EB)             |
| 5                | C163-C-EBFA | EB          | SRP to NHT        | Northolt (EB)                                   |
| 6                | C162-C-EBFA | EB          | NHT to GFD        | Northolt (EB) TO Greenford (EB)                 |
| 7                | C157-C-EBFA | EB          | NHT to GFD        | Greenford (EB)                                  |
| 8                | C156-C-EBFA | EB          | GFD to PER        | Greenford (EB) TO Perivale (EB)                 |
| 9                | C155-C-EBFA | EB          | GFD to PER        | Perivale (EB)                                   |
| 10               | C154-C-EBFA | EB          | PER to HLN        | Perivale (EB) TO Hanger Lane (EB)               |
| 11               | C153-C-EBFA | EB          | PER to HLN        | Hanger Lane (EB)                                |
| 12               | C152-C-EBFA | EB          | HLN to NAC        | Hanger Lane (EB) TO North Acton (EB)            |
| 13               | C147-C-EBFA | EB          | HLN to NAC        | North Acton (EB)                                |
| 14               | C146-C-EBFA | EB          | NAC to EAC        | North Acton (EB) TO East Acton (EB)             |
| 15               | C145-C-EBFA | EB          | NAC to EAC        | East Acton (EB)                                 |
| 16               | C144-C-EBFA | EB          | EAC to WCT        | East Acton (EB) TO White City (EB)              |
| 17               | C143-C-EBFA | EB          | EAC to WCT        | White City (EB)                                 |
| 18               | C142-C-EBFA | EB          | WCT to SBC        | White City (EB) TO Shepherd's Bush (EB)         |
| 19               | C139-C-EBFA | EB          | WCT to SBC        | Shepherd's Bush (EB)                            |
| 20               | C138-C-EBFA | EB          | SBC to HPK        | Shepherd's Bush (EB) TO Holland Park (EB)       |
| 21               | C137-C-EBFA | EB          | SBC to HPK        | Holland Park (EB)                               |
| 22               | C136-C-EBFA | EB          | HPK to NHG        | Holland Park (EB) TO Notting Hill Gate (EB)     |
| 23               | C135-C-EBFA | EB          | HPK to NHG        | Notting Hill Gate (EB)                          |
| 24               | C134-C-EBFA | EB          | NHG to QWY        | Notting Hill Gate (EB) TO Queensway (EB)        |
| 25               | C133-C-EBFA | EB          | NHG to QWY        | Queensway (EB)                                  |
| 26               | C132-C-EBFA | EB          | QWY to LAN        | Queensway (EB) TO Lancaster Gate (EB)           |
| 27               | C129-C-EBFA | EB          | QWY to LAN        | Lancaster Gate (EB)                             |
| 28               | C128-C-EBFA | EB          | LAN to MAR        | Lancaster Gate (EB) TO Marble Arch (EB)         |
| 29               | C127-C-EBFA | EB          | LAN to MAR        | Marble Arch (EB)                                |
| 30               | C126-C-EBFA | EB          | MAR to BOS        | Marble Arch (EB) TO Bond Street (EB)            |
| 31               | C125-C-EBFA | EB          | MAR to BOS        | Bond Street (EB)                                |
| 32               | C124-C-EBFA | EB          | BOS to OXC        | Bond Street (EB) TO Oxford Circus (EB)          |
| 33               | C123-C-EBFA | EB          | BOS to OXC        | Oxford Circus (EB)                              |
| 34               | C122-C-EBFA | EB          | OXC to TCR        | Oxford Circus (EB) TO Tottenham Court Road (EB) |
| 35               | N105-C-EBFA | EB          | OXC to TCR        | Tottenham Court Road (EB)                       |
| 36               | C118-C-EBFA | EB          | TCR to HOL        | Tottenham Court Road (EB) TO Holborn (EB)       |
| 37               | P055-C-EBFA | EB          | TCR to HOL        | Holborn (EB)                                    |
| 38               | C116-C-EBFA | EB          | HOL to CYL        | Holborn (EB) TO Chancery Lane (EB)              |
| 39               | C115-C-EBFA | EB          | HOL to CYL        | Chancery Lane (EB)                              |
| 40               | C114-C-EBFA | EB          | CYL to STP        | Chancery Lane (EB) TO St. Paul's (EB)           |
| 41               | C113-C-EBFA | EB          | CYL to STP        | St. Paul's (EB)                                 |
| 42               | C112-C-EBFA | EB          | STP to BNK        | St. Paul's (EB) TO Bank (EB)                    |
| 43               | N133-C-EBFA | EB          | STP to BNK        | Bank (EB)                                       |
| 44               | C102-C-EBFA | EB          | BNK to LST        | Bank (EB) TO Liverpool Street (EB)              |
| 45               | C101-C-EBFA | EB          | BNK to LST        | Liverpool Street (EB)                           |
| 46               | C096-C-EBFA | EB          | LST to BNG        | Liverpool Street (EB) TO Bethnal Green (EB)     |
| 47               | C095-C-EBFA | EB          | LST to BNG        | Bethnal Green (EB)                              |

**Acoustic Test Report:**  
**Interior Noise Measurement of a Test Vehicle**



|    |             |    |            |                                           |
|----|-------------|----|------------|-------------------------------------------|
| 48 | C094-C-EBFA | EB | BNG to MLE | Bethnal Green (EB) TO Mile End (EB)       |
| 49 | D053-C-EBFA | EB | BNG to MLE | Mile End (EB)                             |
| 50 | C092-C-EBFA | EB | MLE to SFD | Mile End (EB) TO Stratford (EB)           |
| 51 | C091-C-EBFA | EB | MLE to SFD | Stratford (EB)                            |
| 52 | C084-C-EBFA | EB | SFD to LEY | Stratford (EB) TO Leyton (EB)             |
| 53 | C083-C-EBFA | EB | SFD to LEY | Leyton (EB)                               |
| 54 | C082-C-EBFA | EB | LEY to LYS | Leyton (EB) TO Leytonstone (EB)           |
| 55 | C081-C-EBFA | EB | LEY to LYS | Leytonstone (EB)                          |
| 56 | C076-C-IRLO | IR | LYS to WAN | Leytonstone (EB) TO Wanstead (IR)         |
| 57 | C075-C-IRLO | IR | LYS to WAN | Wanstead (IR)                             |
| 58 | C074-C-IRLO | IR | WAN to RED | Wanstead (IR) TO Redbridge (IR)           |
| 59 | C073-C-IRLO | IR | WAN to RED | Redbridge (IR)                            |
| 60 | C072-C-IRLO | IR | RED to GHL | Redbridge (IR) TO Gants Hill (IR)         |
| 61 | C069-C-IRLO | IR | RED to GHL | Gants Hill (IR)                           |
| 62 | C068-C-IRLO | IR | GHL to NEP | Gants Hill (IR) TO Newbury Park (IR)      |
| 63 | C067-C-IRLO | IR | GHL to NEP | Newbury Park (IR)                         |
| 64 | C066-C-IRLO | IR | NEP to BAR | Newbury Park (IR) TO Barkingside (IR)     |
| 65 | C065-C-IRLO | IR | NEP to BAR | Barkingside (IR)                          |
| 66 | C064-C-IRLO | IR | BDE to FLP | Barkingside (IR) TO Fairlop (IR)          |
| 67 | C063-C-IRLO | IR | BDE to FLP | Fairlop (IR)                              |
| 68 | C062-C-IRLO | IR | FLP to HAI | Fairlop (IR) TO Hainault (IR)             |
| 69 | C059-C-IRLO | IR | FLP to HAI | Hainault (IR)                             |
| 70 | C058-C-IRLO | IR | HAI to GRH | Hainault (IR) TO Grange Hill (IR)         |
| 71 | C057-C-IRLO | IR | HAI to GRH | Grange Hill (IR)                          |
| 72 | C056-C-IRLO | IR | GRH to CHG | Grange Hill (IR) TO Chigwell (IR)         |
| 73 | C055-C-IRLO | IR | GRH to CHG | Chigwell (IR)                             |
| 74 | C054-C-IRLO | IR | CHG to ROD | Chigwell (IR) TO Roding Valley (IR)       |
| 75 | C053-C-IRLO | IR | CHG to ROD | Roding Valley (IR)                        |
| 76 | C052-C-IRLO | IR | ROD to WFD | Roding Valley (IR) TO Woodford (WB)       |
| 77 | C035-C-WBFA | WB | ROD to WFD | Woodford (WB)                             |
| 78 | C036-C-WBFA | WB | WFD to SOW | Woodford (WB) TO South Woodford (WB)      |
| 79 | C037-C-WBFA | WB | WFD to SOW | South Woodford (WB)                       |
| 80 | C038-C-WBFA | WB | SWF to SNB | South Woodford (WB) TO Snaresbrook (WB)   |
| 81 | C039-C-WBFA | WB | SWF to SNB | Snaresbrook (WB)                          |
| 82 | C042-C-WBFA | WB | SNB to LYS | Snaresbrook (WB) TO Leytonstone (WB)      |
| 83 | C042-C-EBFA | EB | LYS to SNB | Leytonstone (WB) TO Snaresbrook (EB)      |
| 84 | C039-C-EBFA | EB | LYS to SNB | Snaresbrook (EB)                          |
| 85 | C038-C-EBFA | EB | SNB to SWF | Snaresbrook (EB) TO South Woodford (EB)   |
| 86 | C037-C-EBFA | EB | SNB to SWF | South Woodford (EB)                       |
| 87 | C036-C-EBFA | EB | SWF to WFD | South Woodford (EB) TO Woodford (EB)      |
| 88 | C035-C-EBFA | EB | SWF to WFD | Woodford (EB)                             |
| 89 | C034-C-EBFA | EB | WFD to BHL | Woodford (EB) TO Buckhurst Hill (EB)      |
| 90 | C033-C-EBFA | EB | WFD to BHL | Buckhurst Hill (EB)                       |
| 91 | C032-C-EBFA | EB | BHL to LTN | Buckhurst Hill (EB) TO Loughton (EB)      |
| 92 | C031-C-EBFA | EB | BHL to LTN | Loughton (EB)                             |
| 93 | C024-C-EBFA | EB | LTN to DEB | Loughton (EB) TO Debden (EB)              |
| 94 | C023-C-EBFA | EB | LTN to DEB | Debden (EB)                               |
| 95 | C022-C-EBFA | EB | DEB to THB | Debden (EB) TO Theydon Bois (EB)          |
| 96 | C019-C-EBFA | EB | DEB to THB | Theydon Bois (EB)                         |
| 97 | C018-C-EBFA | EB | THB to EPP | Theydon Bois (EB) TO Epping (Terminating) |
| 98 | C018-C-WBFA | WB | EPP to THB | Epping (Terminating) TO Theydon Bois (WB) |
| 99 | C019-C-WBFA | WB | EPP to THB | Theydon Bois (WB)                         |

**Acoustic Test Report:**  
**Interior Noise Measurement of a Test Vehicle**



|     |             |    |            |                                                 |
|-----|-------------|----|------------|-------------------------------------------------|
| 100 | C022-C-WBFA | WB | THB to DEB | Theydon Bois (WB) TO Debden (WB)                |
| 101 | C023-C-WBFA | WB | THB to DEB | Debden (WB)                                     |
| 102 | C024-C-WBFA | WB | DEB to LTN | Debden (WB) TO Loughton (WB)                    |
| 103 | C031-C-WBFA | WB | DEB to LTN | Loughton (WB)                                   |
| 104 | C032-C-WBFA | WB | LTN to BHL | Loughton (WB) TO Buckhurst Hill (WB)            |
| 105 | C033-C-WBFA | WB | LTN to BHL | Buckhurst Hill (WB)                             |
| 106 | C034-C-WBFA | WB | BHL to WFD | Buckhurst Hill (WB) TO Woodford (WB)            |
| 107 | C052-C-ORLO | OR | WFD to ROD | Woodford (EB) TO Roding Valley (OR)             |
| 108 | C053-C-ORLO | OR | WFD to ROD | Roding Valley (OR)                              |
| 109 | C054-C-ORLO | OR | ROD to CHG | Roding Valley (OR) TO Chigwell (OR)             |
| 110 | C055-C-ORLO | OR | ROD to CHG | Chigwell (OR)                                   |
| 111 | C056-C-ORLO | OR | CHG to GRH | Chigwell (OR) TO Grange Hill (OR)               |
| 112 | C057-C-ORLO | OR | CHG to GRH | Grange Hill (OR)                                |
| 113 | C058-C-ORLO | OR | GRH to HAI | Grange Hill (OR) TO Hainault (OR)               |
| 114 | C059-C-ORLO | OR | GRH to HAI | Hainault (OR)                                   |
| 115 | C062-C-ORLO | OR | HAI to FLP | Hainault (OR) TO Fairlop (OR)                   |
| 116 | C063-C-ORLO | OR | HAI to FLP | Fairlop (OR)                                    |
| 117 | C064-C-ORLO | OR | FLP to BDE | Fairlop (OR) TO Barkingside (OR)                |
| 118 | C065-C-ORLO | OR | FLP to BDE | Barkingside (OR)                                |
| 119 | C066-C-ORLO | OR | BAR to NEP | Barkingside (OR) TO Newbury Park (OR)           |
| 120 | C067-C-ORLO | OR | BAR to NEP | Newbury Park (OR)                               |
| 121 | C068-C-ORLO | OR | NEP to GHL | Newbury Park (OR) TO Gants Hill (OR)            |
| 122 | C069-C-ORLO | OR | NEP to GHL | Gants Hill (OR)                                 |
| 123 | C072-C-ORLO | OR | GHL to RED | Gants Hill (OR) TO Redbridge (OR)               |
| 124 | C073-C-ORLO | OR | GHL to RED | Redbridge (OR)                                  |
| 125 | C074-C-ORLO | OR | RED to WAN | Redbridge (OR) TO Wanstead (OR)                 |
| 126 | C075-C-ORLO | OR | RED to WAN | Wanstead (OR)                                   |
| 127 | C076-C-ORLO | OR | WAN to LYS | Wanstead (OR) TO Leytonstone (WB)               |
| 128 | C081-C-WBFA | WB | WAN to LYS | Leytonstone (WB)                                |
| 129 | C082-C-WBFA | WB | LYS to LEY | Leytonstone (WB) TO Leyton (WB)                 |
| 130 | C083-C-WBFA | WB | LYS to LEY | Leyton (WB)                                     |
| 131 | C084-C-WBFA | WB | LEY to SFD | Leyton (WB) TO Stratford (WB)                   |
| 132 | C091-C-WBFA | WB | LEY to SFD | Stratford (WB)                                  |
| 133 | C092-C-WBFA | WB | SFD to MLE | Stratford (WB) TO Mile End (WB)                 |
| 134 | D053-C-WBFA | WB | SFD to MLE | Mile End (WB)                                   |
| 135 | C094-C-WBFA | WB | MLE to BNG | Mile End (WB) TO Bethnal Green (WB)             |
| 136 | C095-C-WBFA | WB | MLE to BNG | Bethnal Green (WB)                              |
| 137 | C096-C-WBFA | WB | BNG to LST | Bethnal Green (WB) TO Liverpool Street (WB)     |
| 138 | C101-C-WBFA | WB | BNG to LST | Liverpool Street (WB)                           |
| 139 | C102-C-WBFA | WB | LST to BNK | Liverpool Street (WB) TO Bank (WB)              |
| 140 | N133-C-WBFA | WB | LST to BNK | Bank (WB)                                       |
| 141 | C112-C-WBFA | WB | BNK to STP | Bank (WB) TO St. Paul's (WB)                    |
| 142 | C113-C-WBFA | WB | BNK to STP | St. Paul's (WB)                                 |
| 143 | C114-C-WBFA | WB | STP to CYL | St. Paul's (WB) TO Chancery Lane (WB)           |
| 144 | C115-C-WBFA | WB | STP to CYL | Chancery Lane (WB)                              |
| 145 | C116-C-WBFA | WB | CYL to HOL | Chancery Lane (WB) TO Holborn (WB)              |
| 146 | P055-C-WBFA | WB | CYL to HOL | Holborn (WB)                                    |
| 147 | C118-C-WBFA | WB | HOL to TCR | Holborn (WB) TO Tottenham Court Road (WB)       |
| 148 | N105-C-WBFA | WB | HOL to TCR | Tottenham Court Road (WB)                       |
| 149 | C122-C-WBFA | WB | TCR to OXC | Tottenham Court Road (WB) TO Oxford Circus (WB) |
| 150 | C123-C-WBFA | WB | TCR to OXC | Oxford Circus (WB)                              |
| 151 | C124-C-WBFA | WB | OXC to BOS | Oxford Circus (WB) TO Bond Street (WB)          |

**Acoustic Test Report:**  
**Interior Noise Measurement of a Test Vehicle**



|     |             |    |            |                                                  |
|-----|-------------|----|------------|--------------------------------------------------|
| 152 | C125-C-WBFA | WB | OXF to BOS | Bond Street (WB)                                 |
| 153 | C126-C-WBFA | WB | BDS to MAR | Bond Street (WB) TO Marble Arch (WB)             |
| 154 | C127-C-WBFA | WB | BDS to MAR | Marble Arch (WB)                                 |
| 155 | C128-C-WBFA | WB | MAR to LAN | Marble Arch (WB) TO Lancaster Gate (WB)          |
| 156 | C129-C-WBFA | WB | MAR to LAN | Lancaster Gate (WB)                              |
| 157 | C132-C-WBFA | WB | LAN to QUE | Lancaster Gate (WB) TO Queensway (WB)            |
| 158 | C133-C-WBFA | WB | LAN to QUE | Queensway (WB)                                   |
| 159 | C134-C-WBFA | WB | QWY to NHG | Queensway (WB) TO Notting Hill Gate (WB)         |
| 160 | C135-C-WBFA | WB | QWY to NHG | Notting Hill Gate (WB)                           |
| 161 | C136-C-WBFA | WB | NHG to HPK | Notting Hill Gate (WB) TO Holland Park (WB)      |
| 162 | C137-C-WBFA | WB | NHG to HPK | Holland Park (WB)                                |
| 163 | C138-C-WBFA | WB | HPK to SBC | Holland Park (WB) TO Shepherd's Bush (WB)        |
| 164 | C139-C-WBFA | WB | HPK to SBC | Shepherd's Bush (WB)                             |
| 165 | C142-C-WBFA | WB | SBC to WCT | Shepherd's Bush (WB) TO White City (WB)          |
| 166 | C143-C-WBFA | WB | SBC to WCT | White City (WB)                                  |
| 167 | C144-C-WBFA | WB | WCT to EAA | White City (WB) TO East Acton (WB)               |
| 168 | C145-C-WBFA | WB | WCT to EAA | East Acton (WB)                                  |
| 169 | C146-C-WBFA | WB | EAC to NAC | East Acton (WB) TO North Acton (WB)              |
| 170 | C147-C-WBFA | WB | EAC to NAC | North Acton (WB)                                 |
| 171 | C172-C-WBLO | WB | NAC to WAC | North Acton (WB) TO West Acton (WB)              |
| 172 | C173-C-WBLO | WB | NAC to WAC | West Acton (WB)                                  |
| 173 | C174-C-WBLO | WB | WAC to EBY | West Acton (WB) TO Ealing Broadway (Terminating) |
| 174 | C174-C-EBLO | EB | EBY to WAC | Ealing Broadway (Terminating) TO West Acton (EB) |
| 175 | C173-C-EBLO | EB | EBY to WAC | West Acton (EB)                                  |
| 176 | C172-C-EBLO | EB | WAC to NAC | West Acton (EB) TO North Acton (EB)              |
| 177 | C152-C-WBFA | WB | NAC to HLN | North Acton (EB) TO Hanger Lane (WB)             |
| 178 | C153-C-WBFA | WB | NAC to HLN | Hanger Lane (WB)                                 |
| 179 | C154-C-WBFA | WB | HLN to PER | Hanger Lane (WB) TO Perivale (WB)                |
| 180 | C155-C-WBFA | WB | HLN to PER | Perivale (WB)                                    |
| 181 | C156-C-WBFA | WB | PER to GFD | Perivale (WB) TO Greenford (WB)                  |
| 182 | C157-C-WBFA | WB | PER to GFD | Greenford (WB)                                   |
| 183 | C162-C-WBFA | WB | GFD to NHT | Greenford (WB) TO Northolt (WB)                  |
| 184 | C163-C-WBFA | WB | GFD to NHT | Northolt (WB)                                    |
| 185 | C164-C-WBFA | WB | NHT to SRP | Northolt (WB) TO South Ruislip (WB)              |
| 186 | C165-C-WBFA | WB | NHT to SRP | South Ruislip (WB)                               |
| 187 | C166-C-WBFA | WB | SRP to RUG | South Ruislip (WB) TO Ruislip Gardens (WB)       |
| 188 | C167-C-WBFA | WB | SRP to RUG | Ruislip Gardens (WB)                             |



## **Appendix C - Distance vs. Noise Graphs**

The graphs below are presented in the test run order and show the following at each interstation section according to it's LCS:

- the maximum noise level in terms of  $L_{AFMax}$ ;
- the peak noise level in terms of  $L_{CPeak}$ . The data points presented only show the top 1<sup>st</sup> percentile or levels above 120 dB, whichever is lower; and
- the speed in metres per second.

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

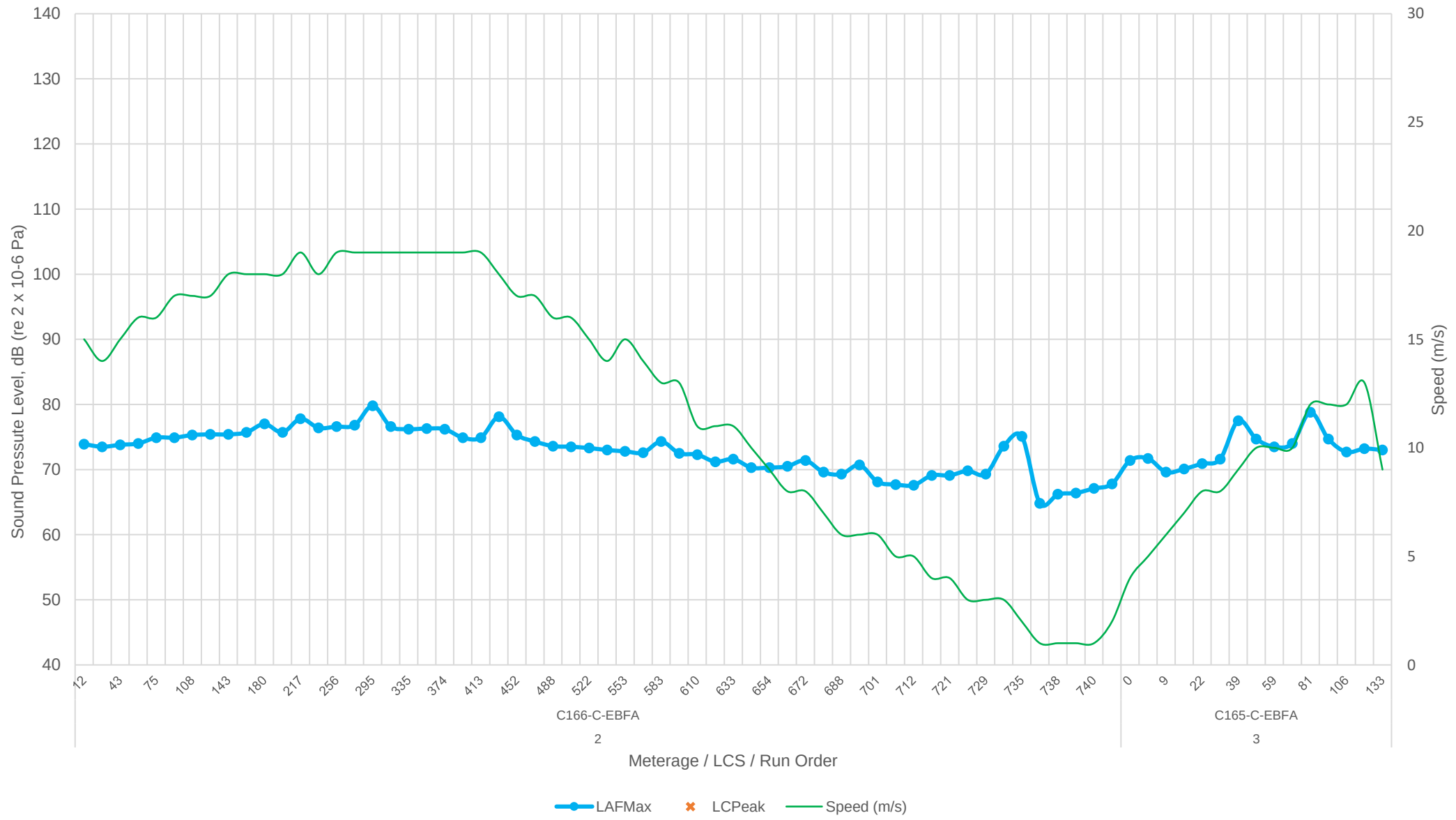


Figure C1 - Ruislip Gardens (EB) to South Ruislip (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

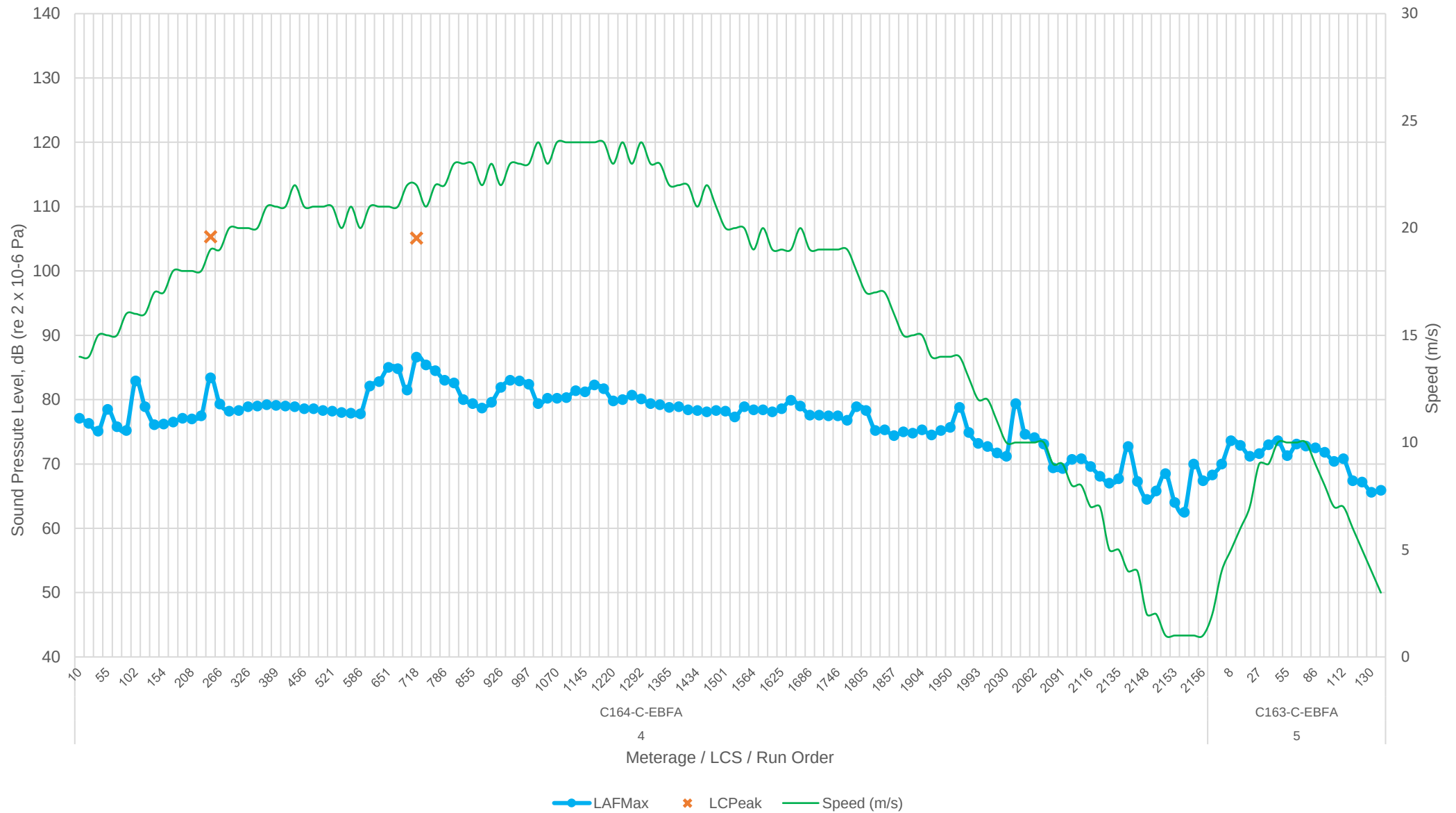


Figure C2 - South Ruislip (EB) to Northolt (EB)



# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

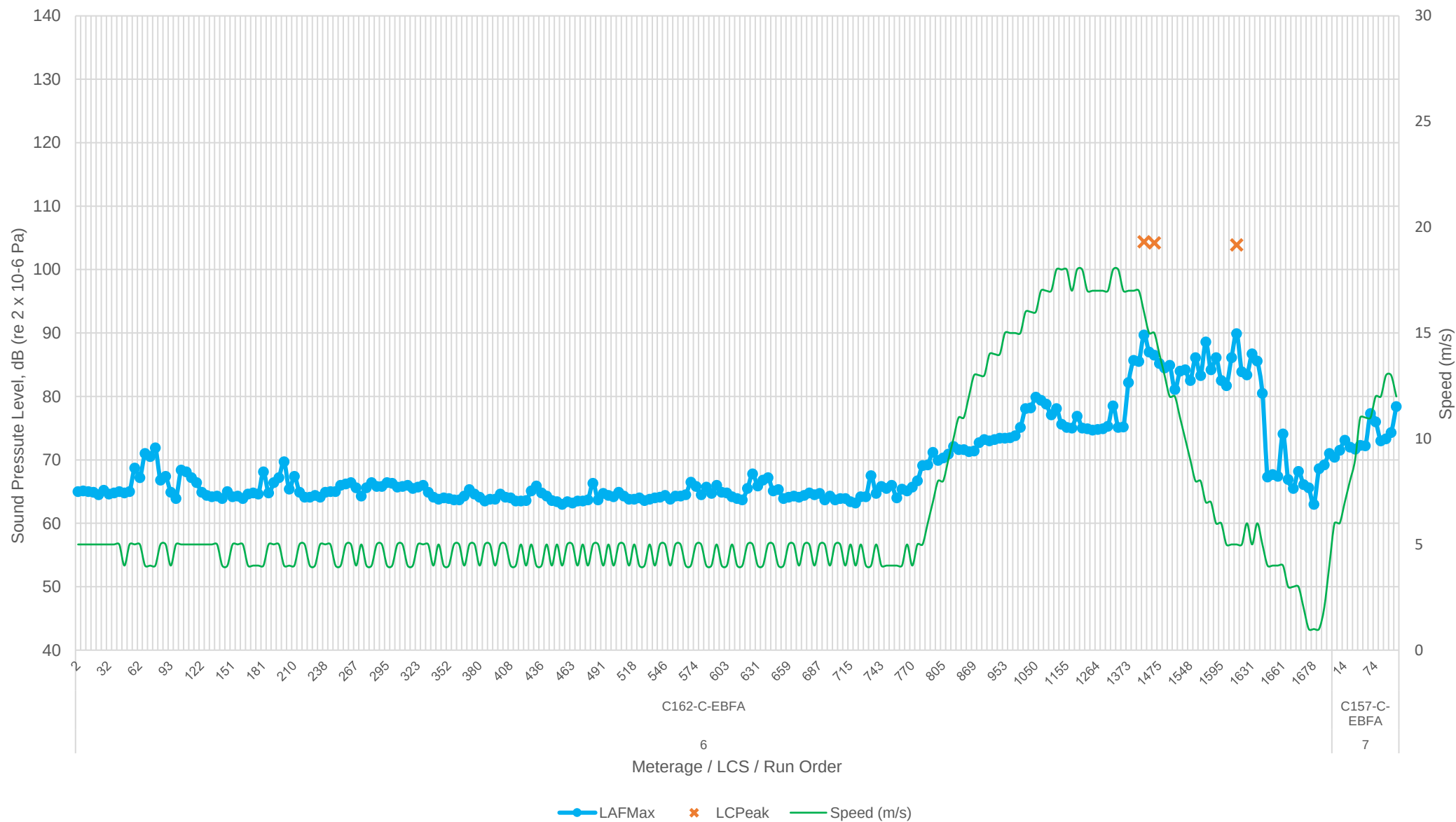


Figure C3 - Northolt (EB) to Greenford (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

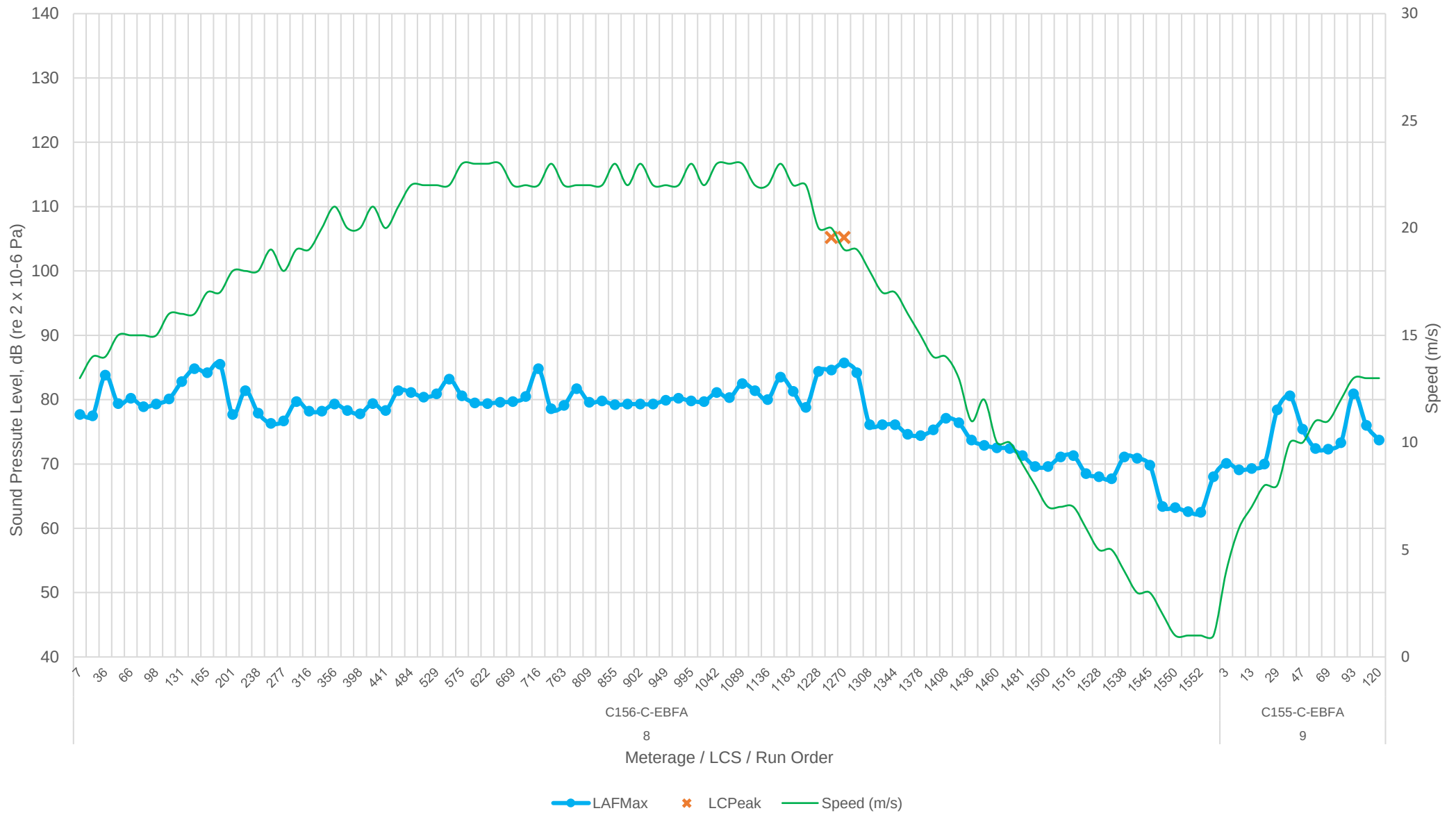


Figure C4 - Greenford (EB) to Perivale (EB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

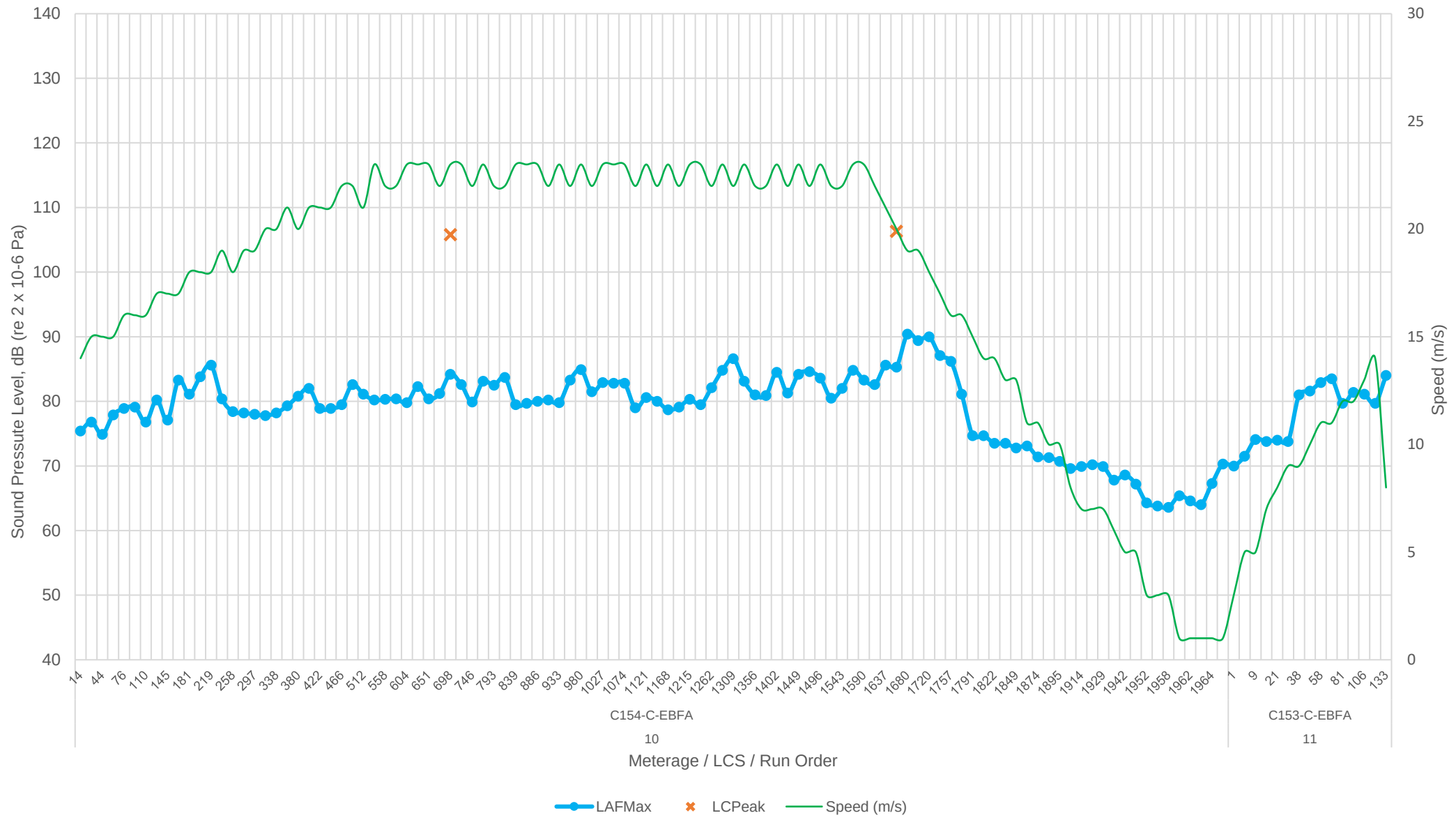


Figure C5 - Perivale (EB) to Hanger Lane (EB)

**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

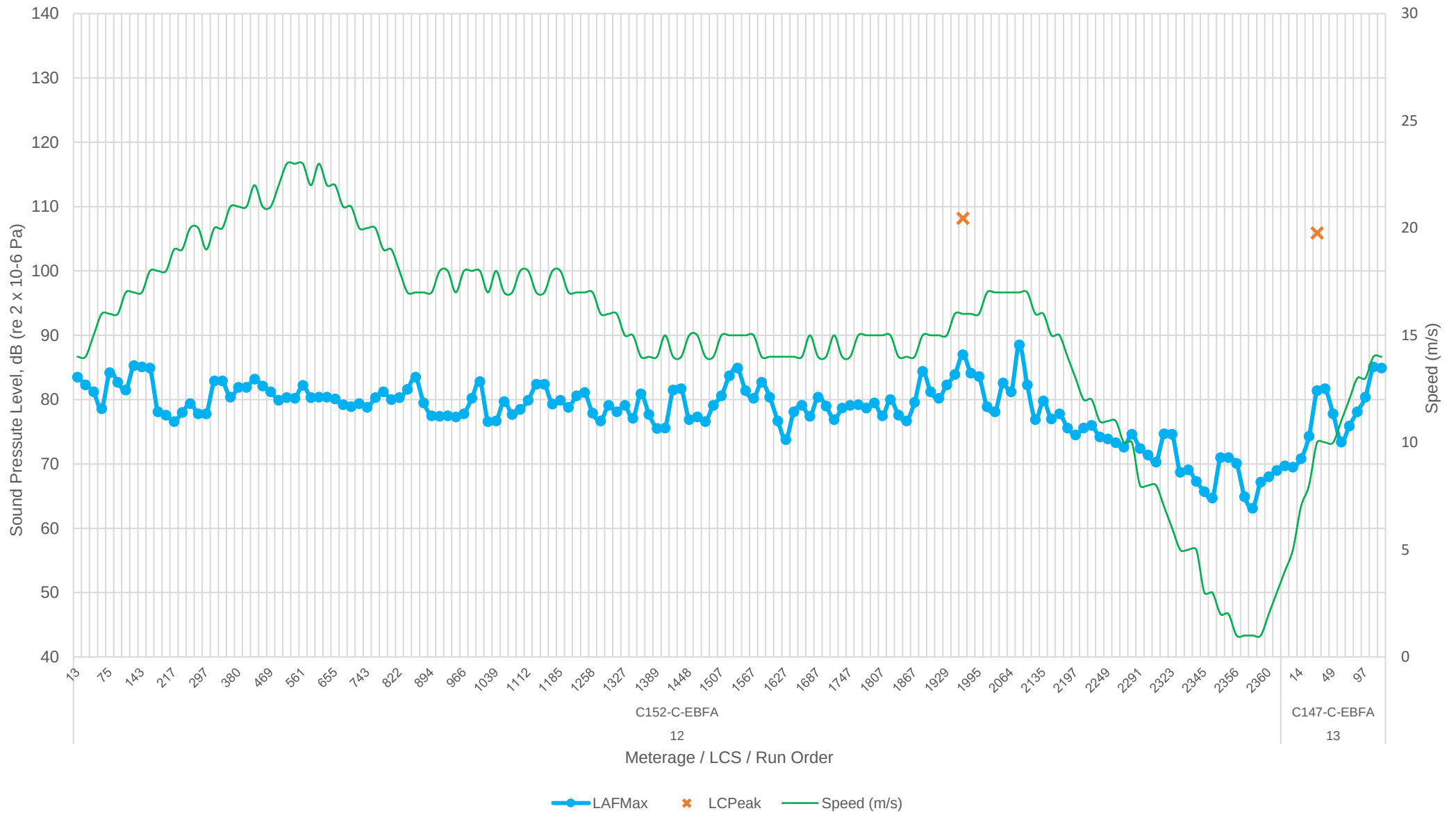


Figure C6 - Hanger Lane (EB) to North Acton (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

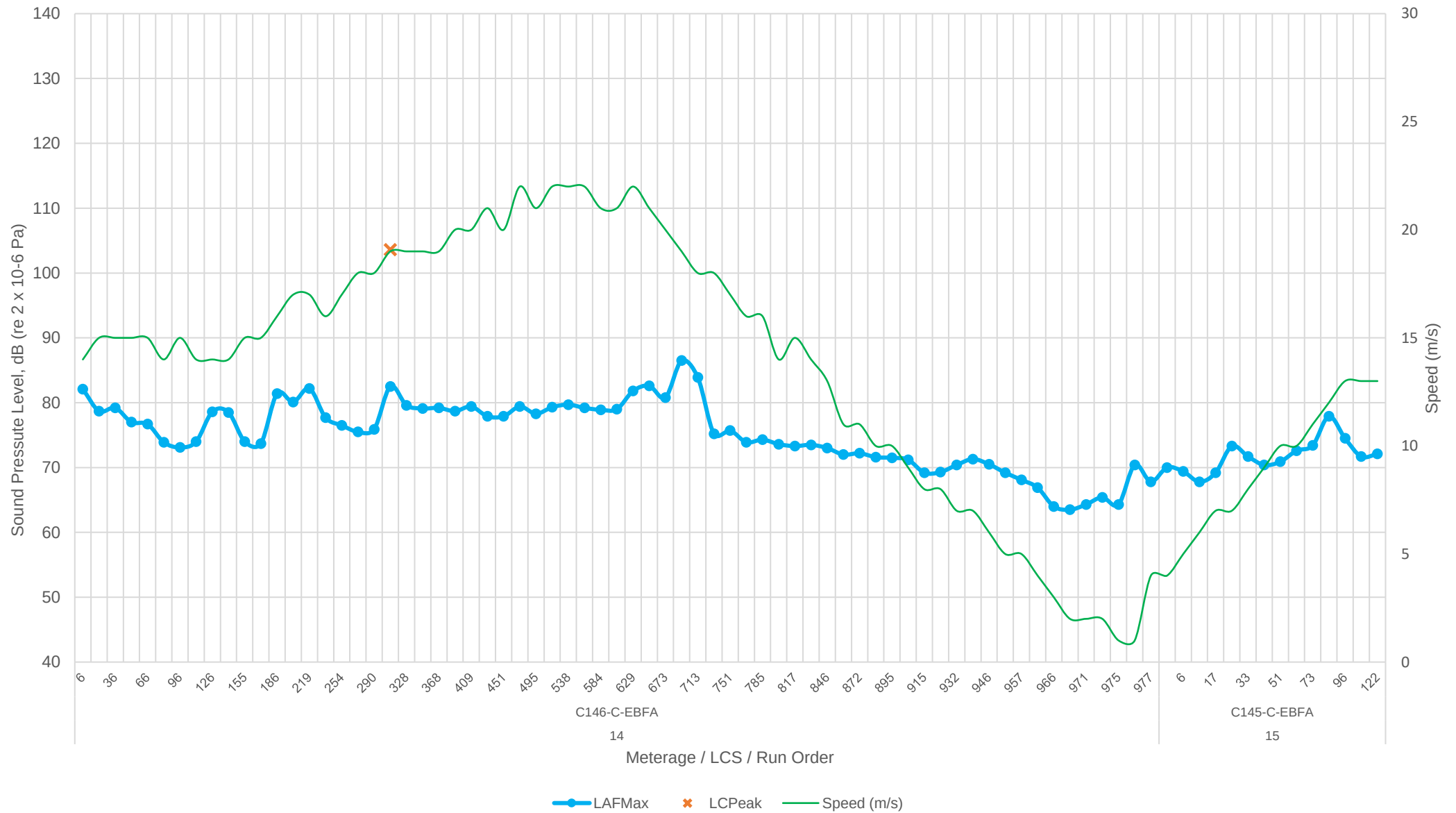


Figure C7 - North Acton (EB) to East Acton (EB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

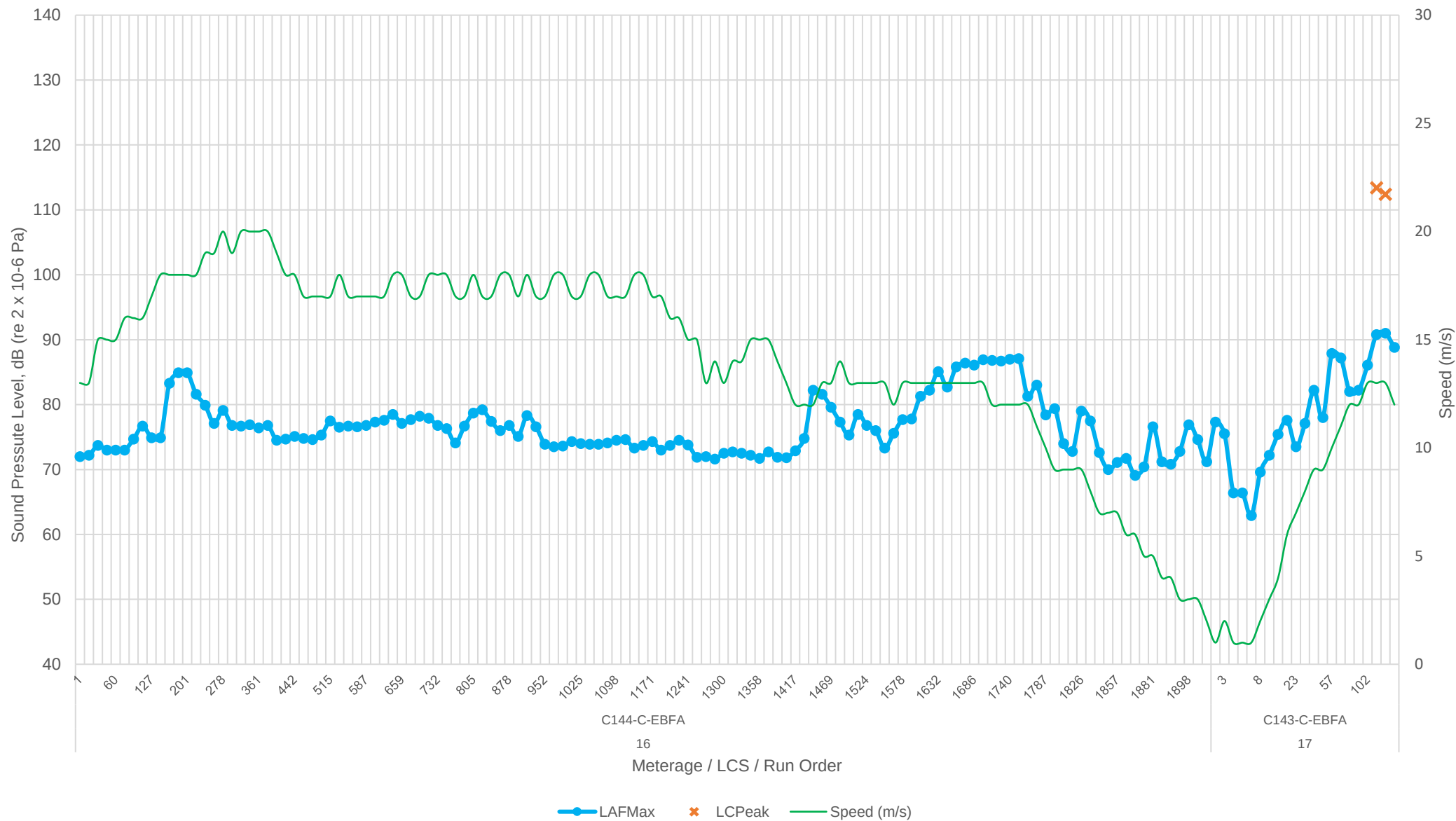
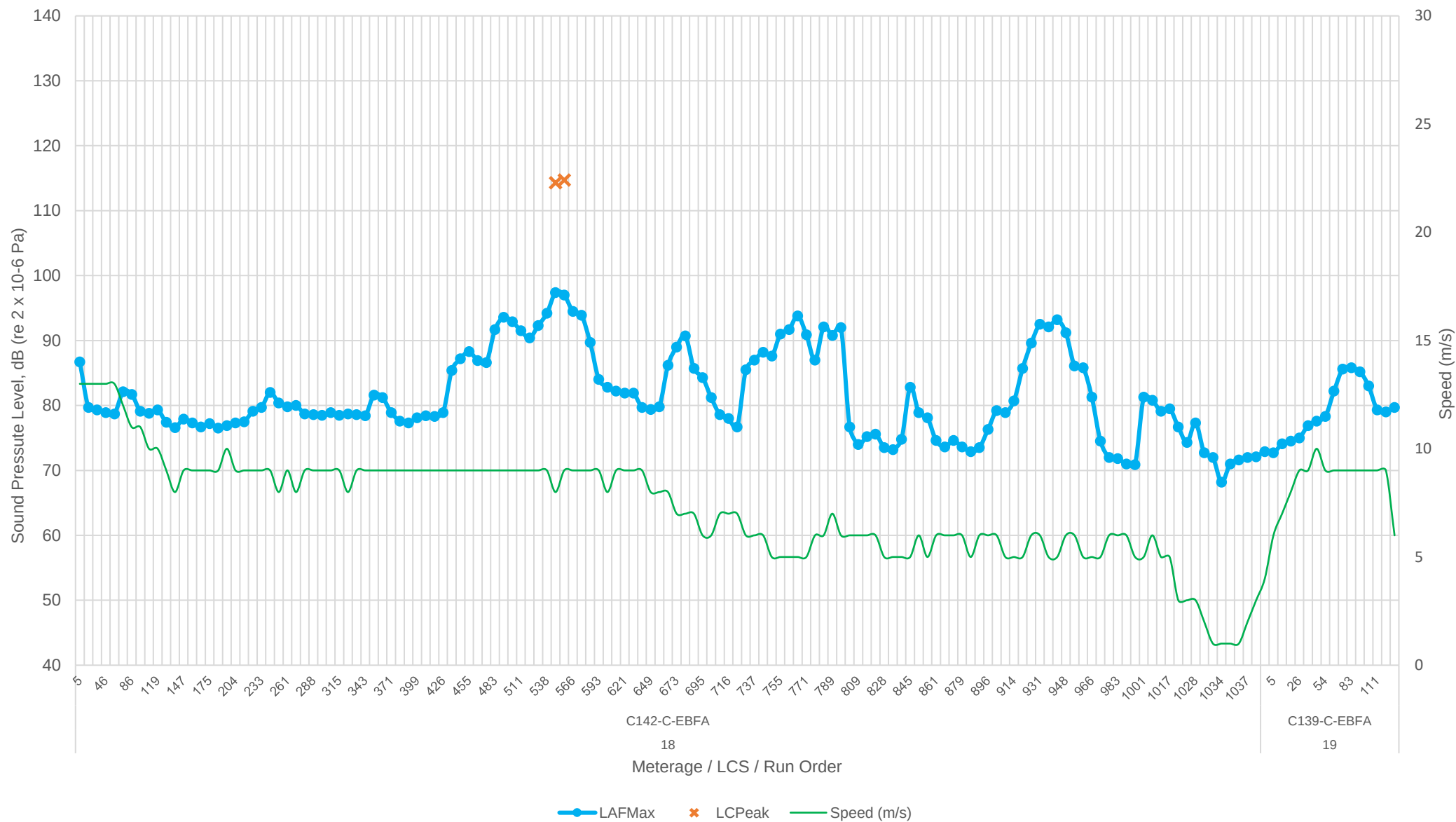


Figure C8 - East Acton (EB) to White City (EB)

# **Acoustic Test Report: Interior Noise Measurement of a Test Vehicle**



**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

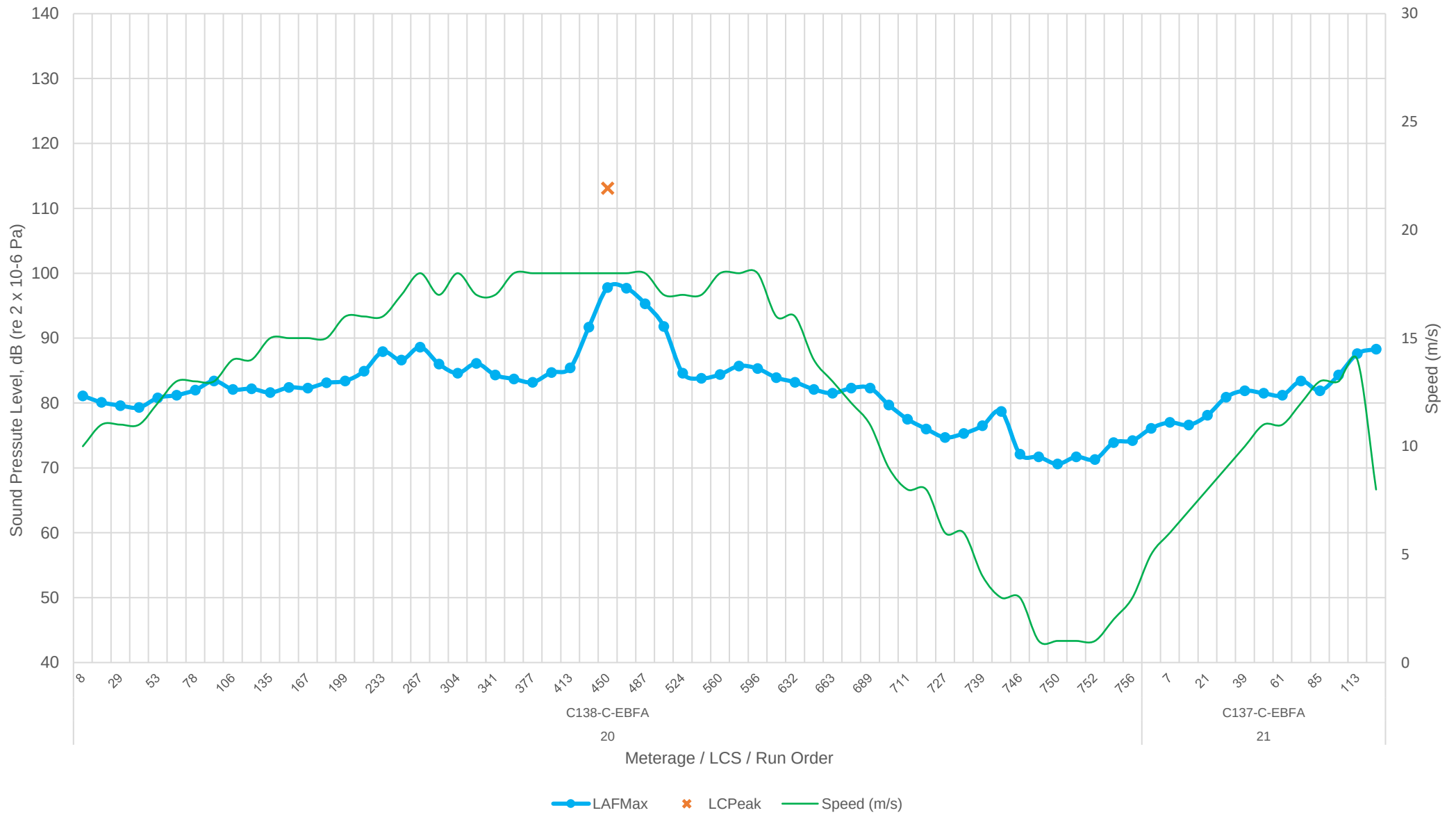


Figure C10 - Shepherd's Bush (EB) to Holland Park (EB)



Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

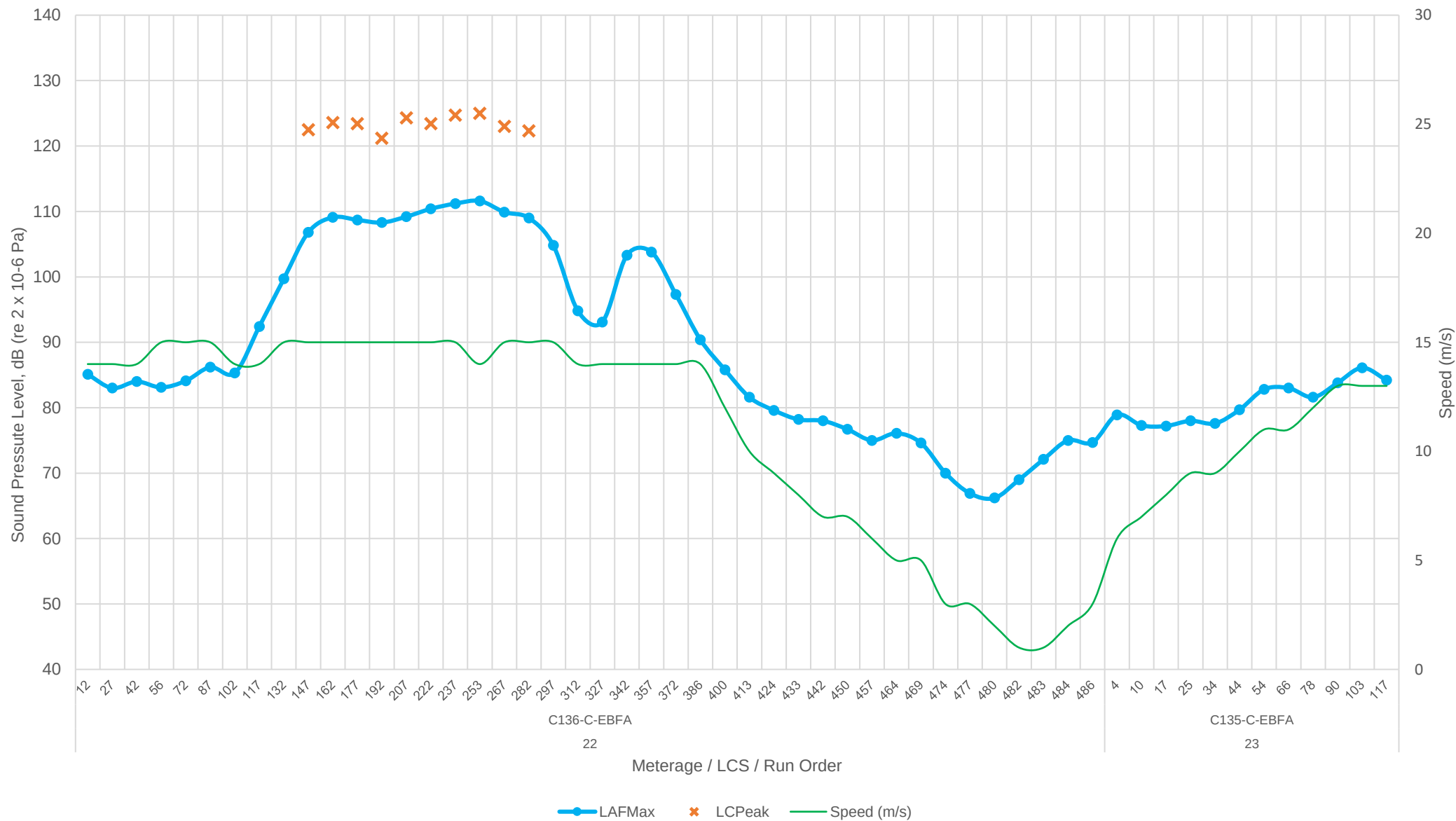


Figure C11 - Holland Park (EB) to Notting Hill Gate (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

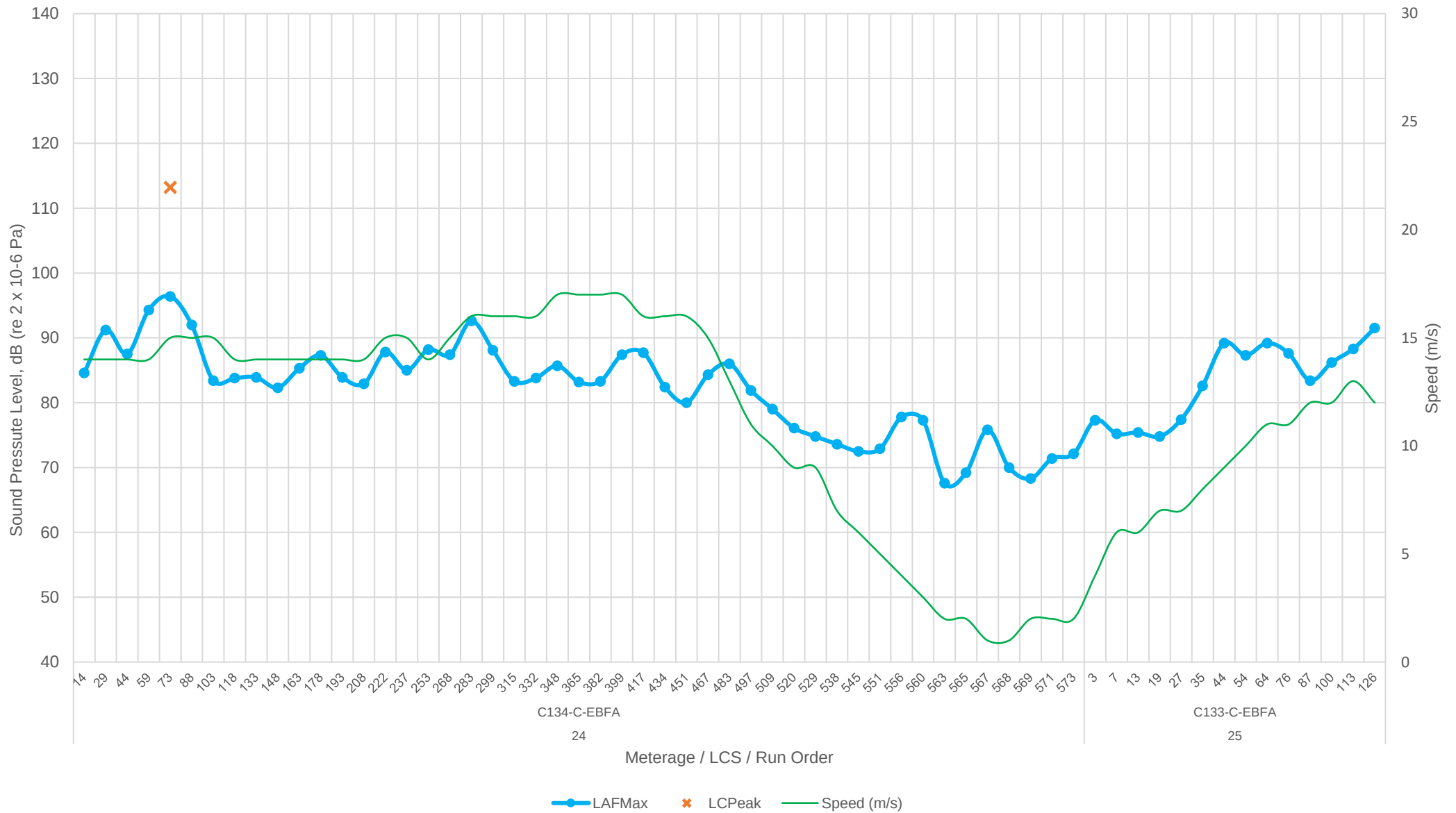


Figure C12 - Notting Hill Gate (EB) to Queensway (EB)

**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

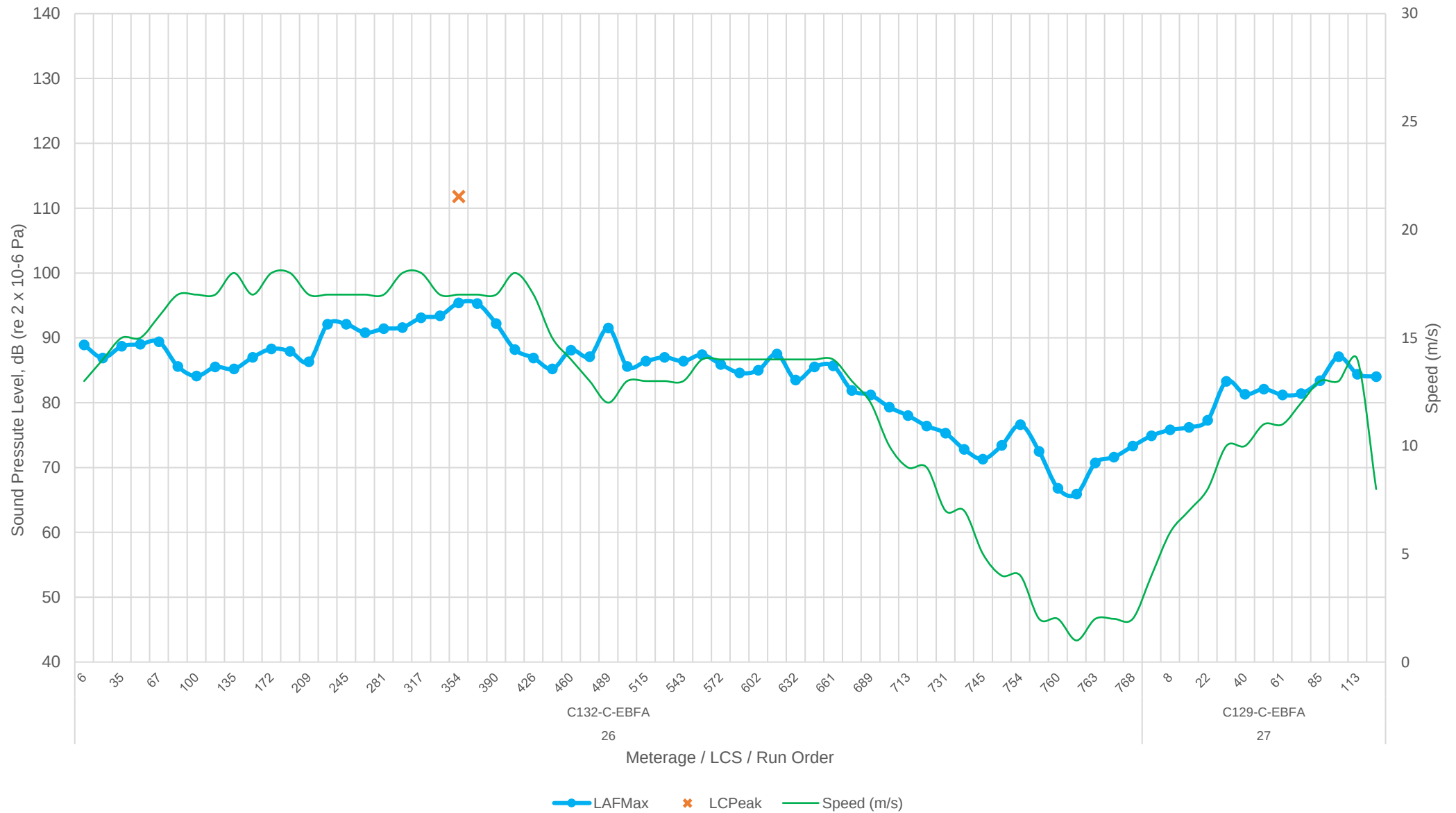


Figure C13 - Queensway (EB) to Lancaster Gate (EB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

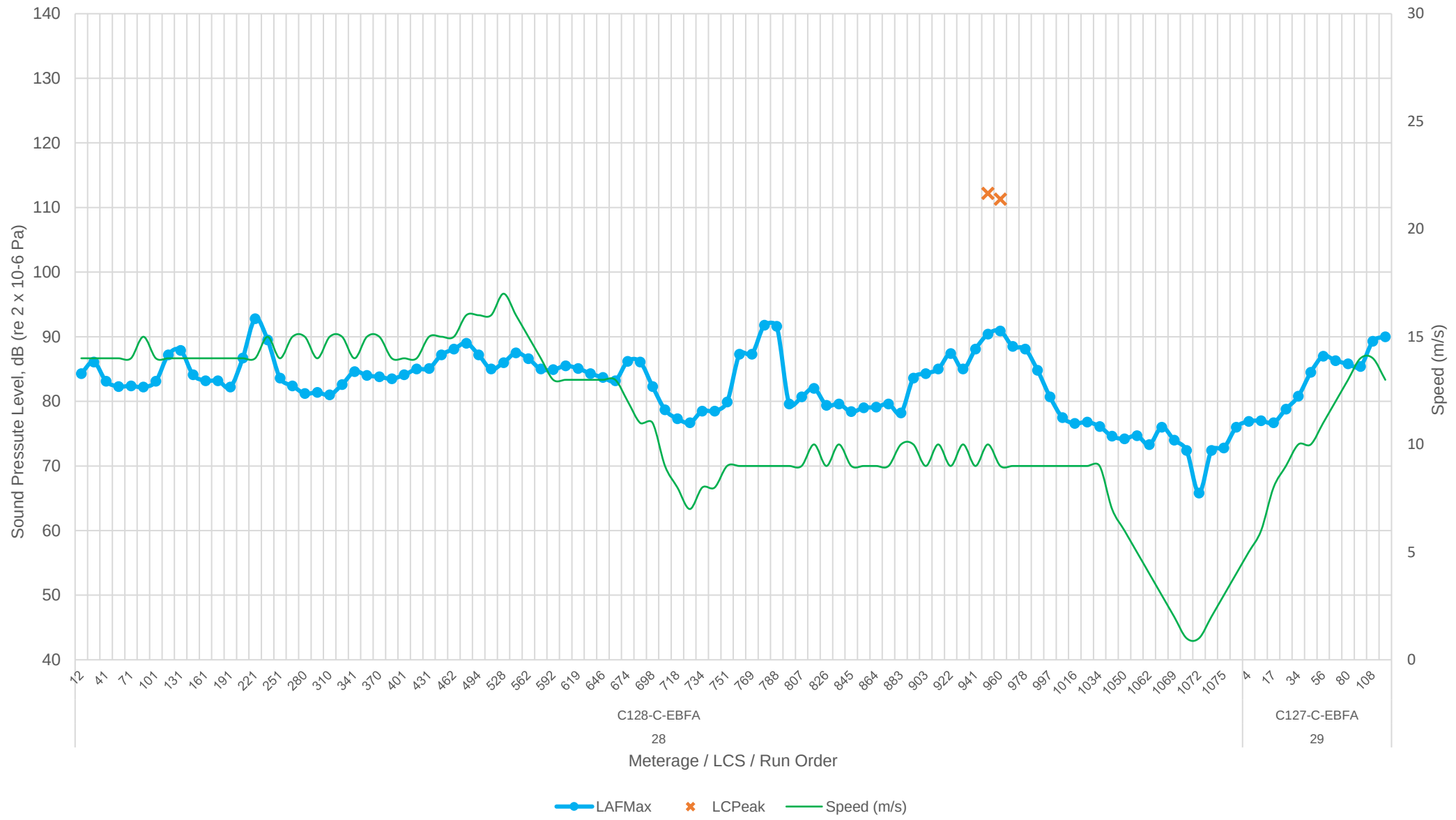


Figure C14 - Lancaster Gate (EB) to Marble Arch (EB)

# **Acoustic Test Report: Interior Noise Measurement of a Test Vehicle**

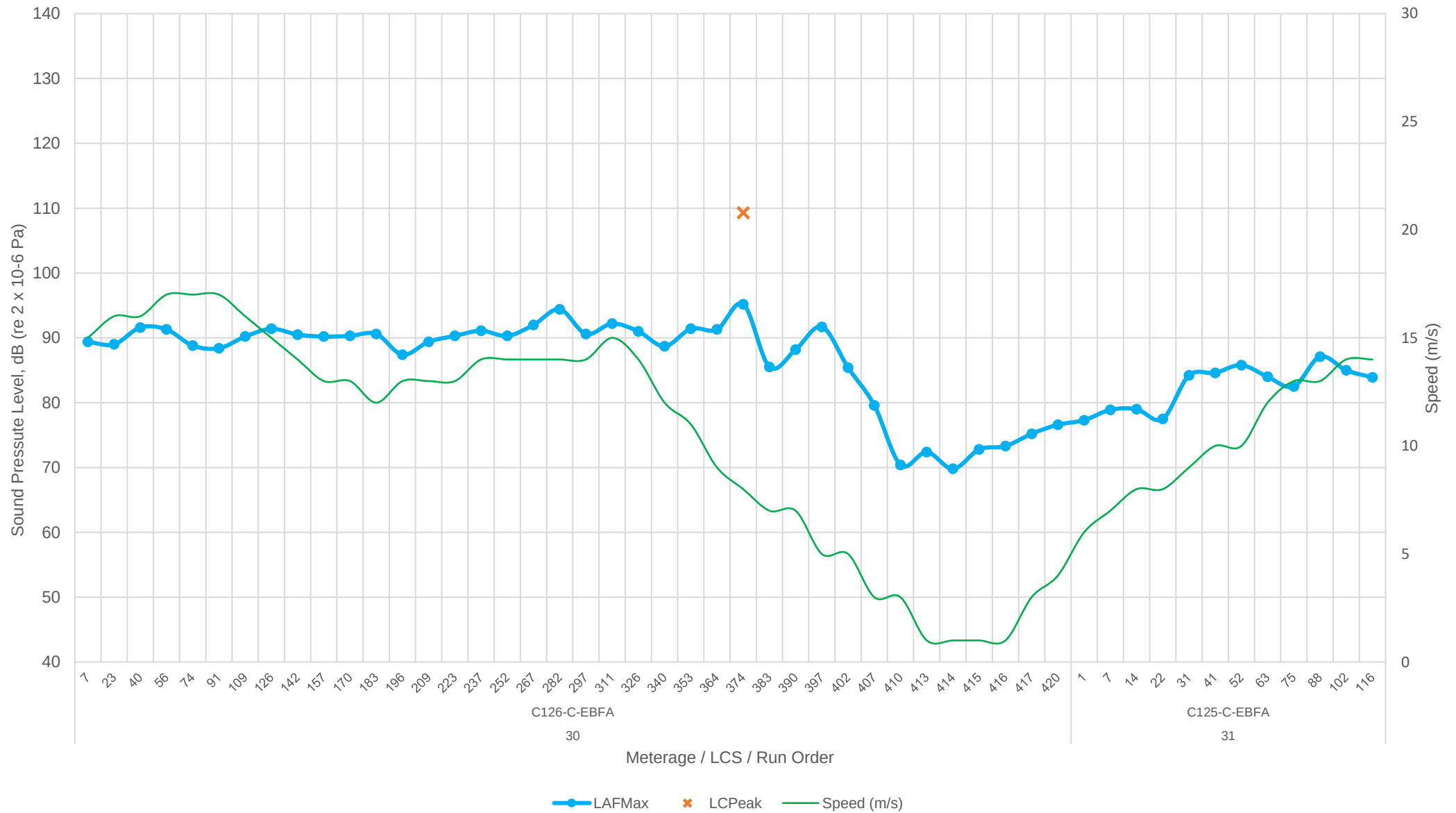


Figure C15 - Marble Arch (EB) to Bond Street (EB)

# **Acoustic Test Report:** **Interior Noise Measurement of a Test Vehicle**

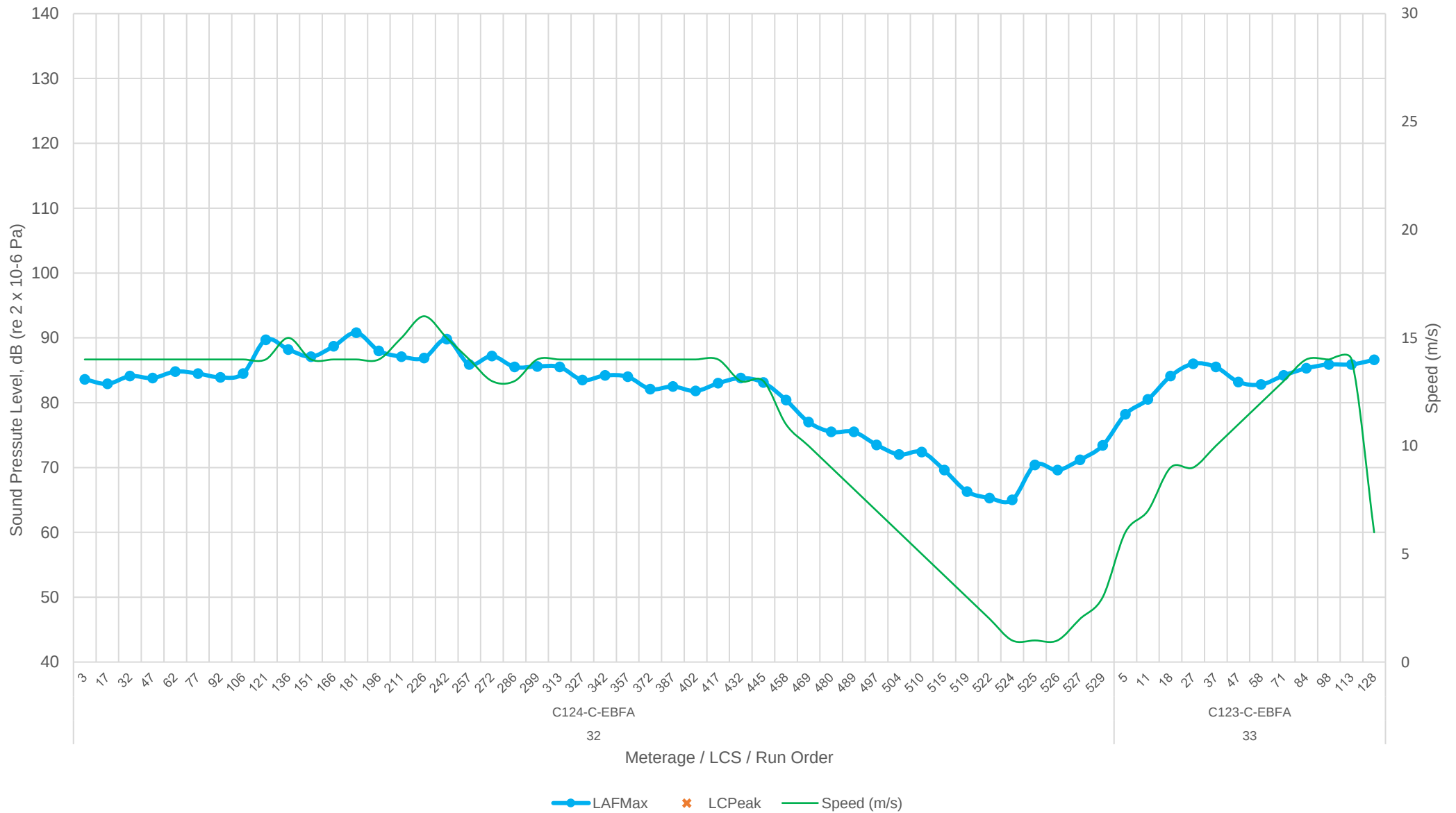


Figure C16 - Bond Street (EB) to Oxford Circus (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

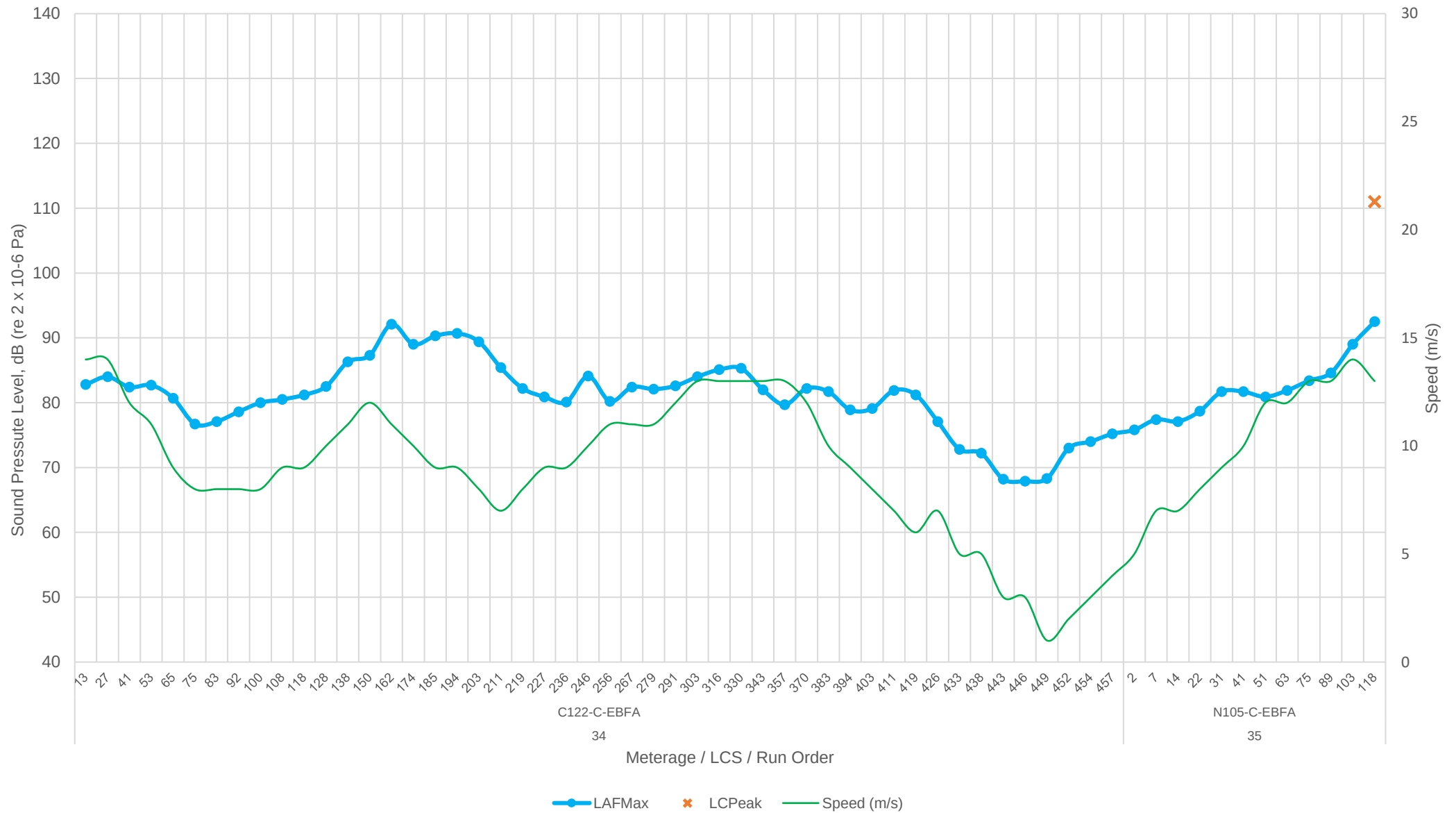


Figure C17 - Oxford Circus (EB) to Tottenham Court Road (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

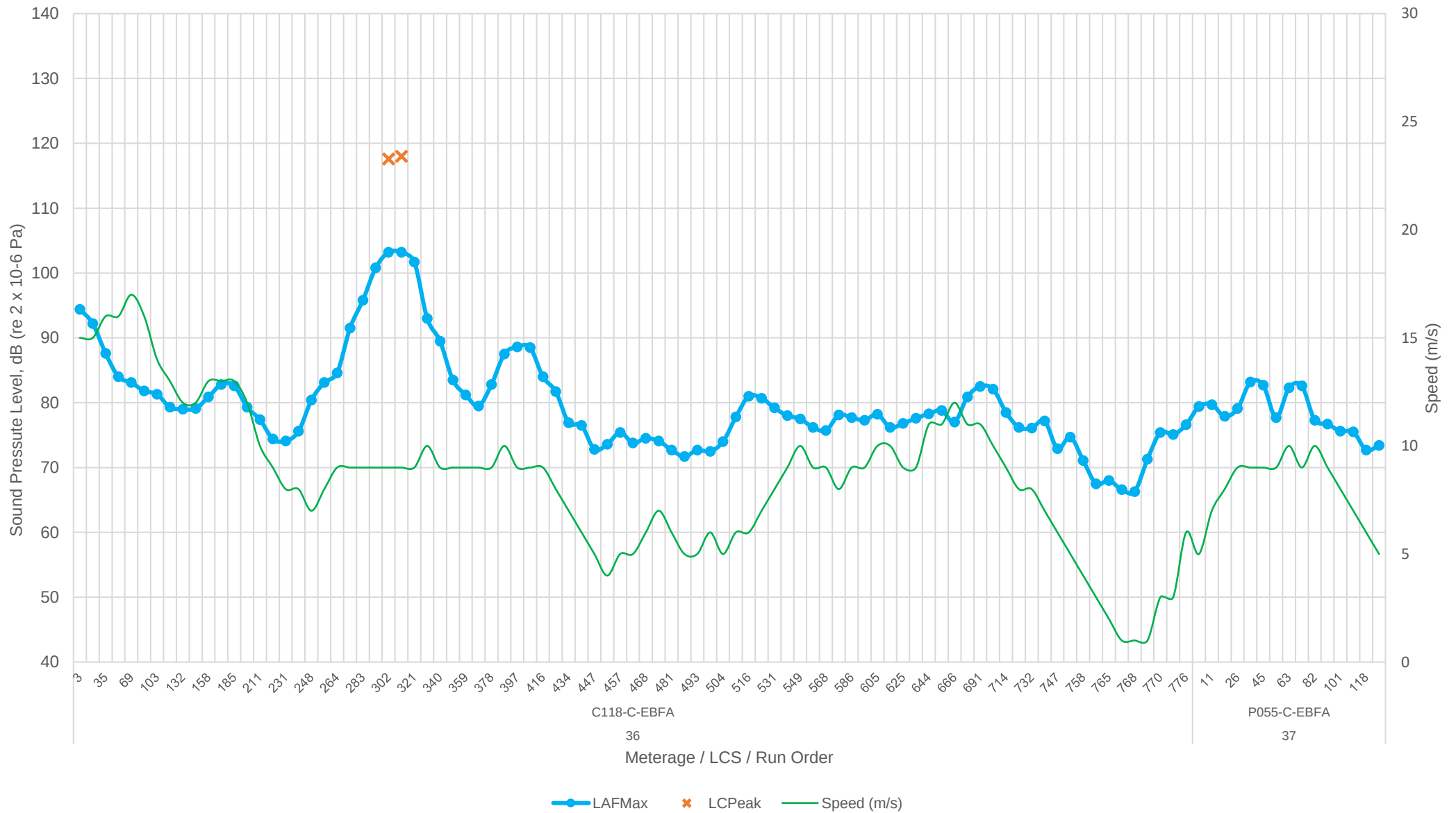


Figure C18 - Tottenham Court Road (EB) to Holborn (EB)



Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

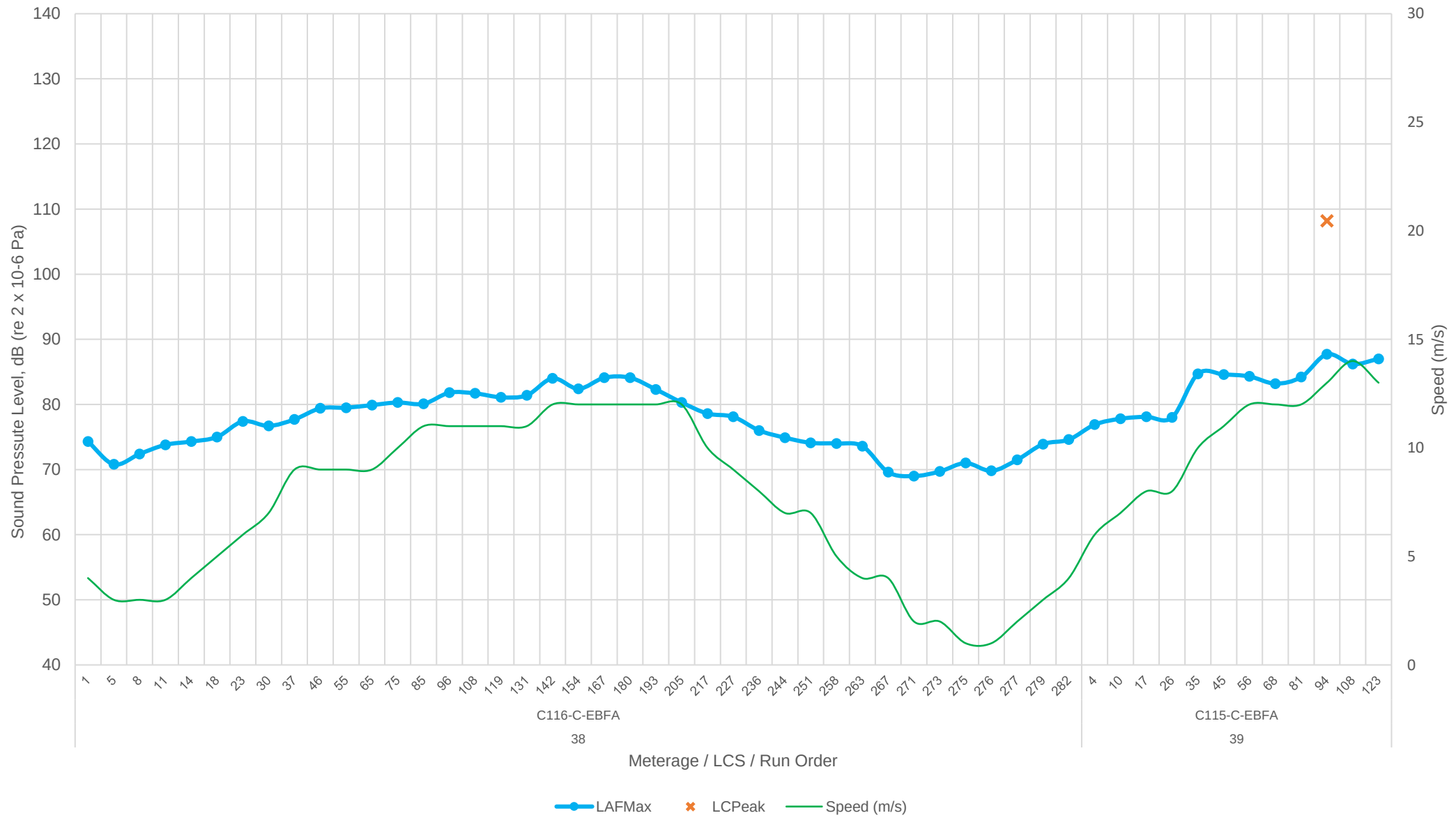


Figure C19 - Holborn (EB) to Chancery Lane (EB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

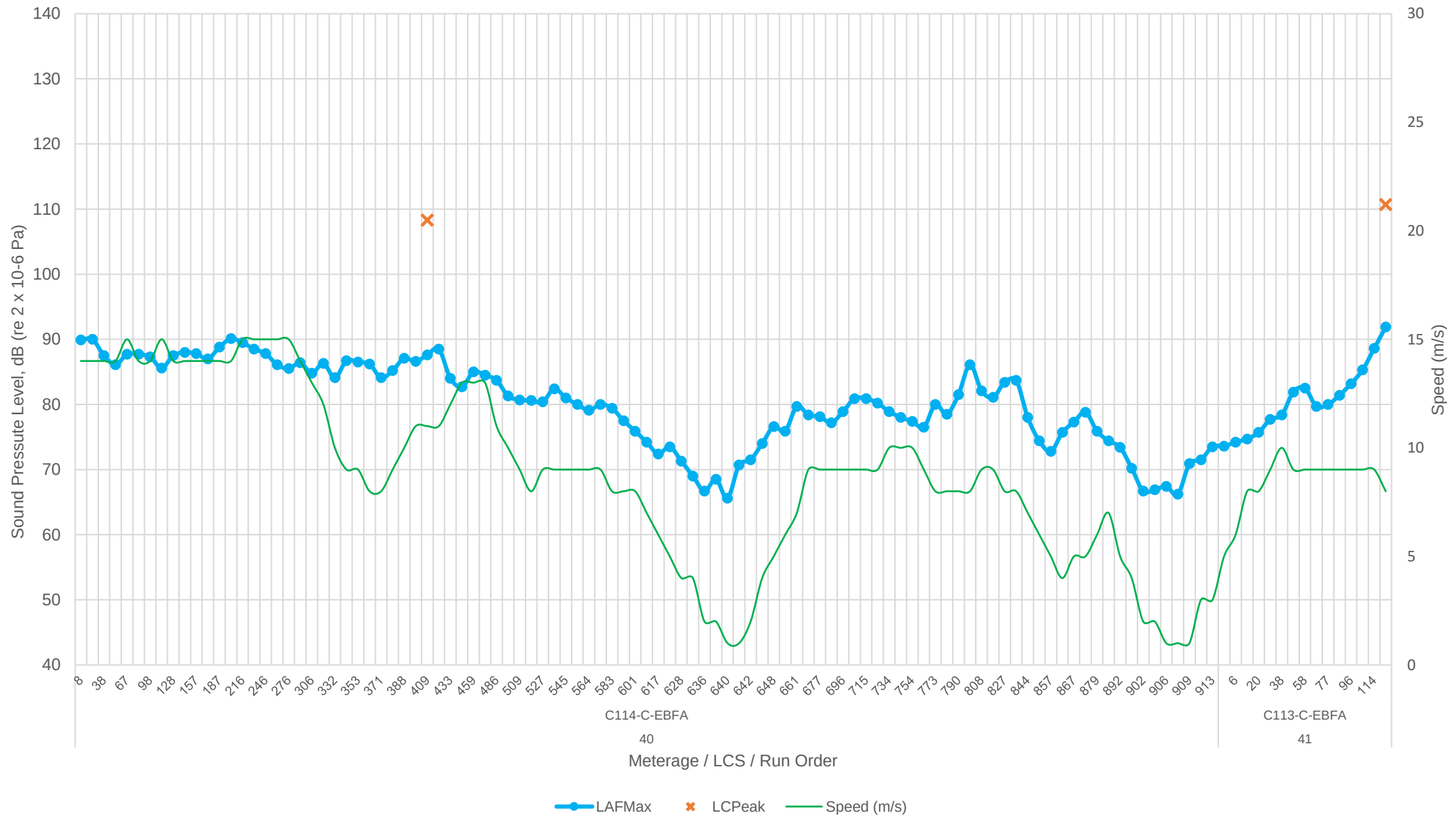
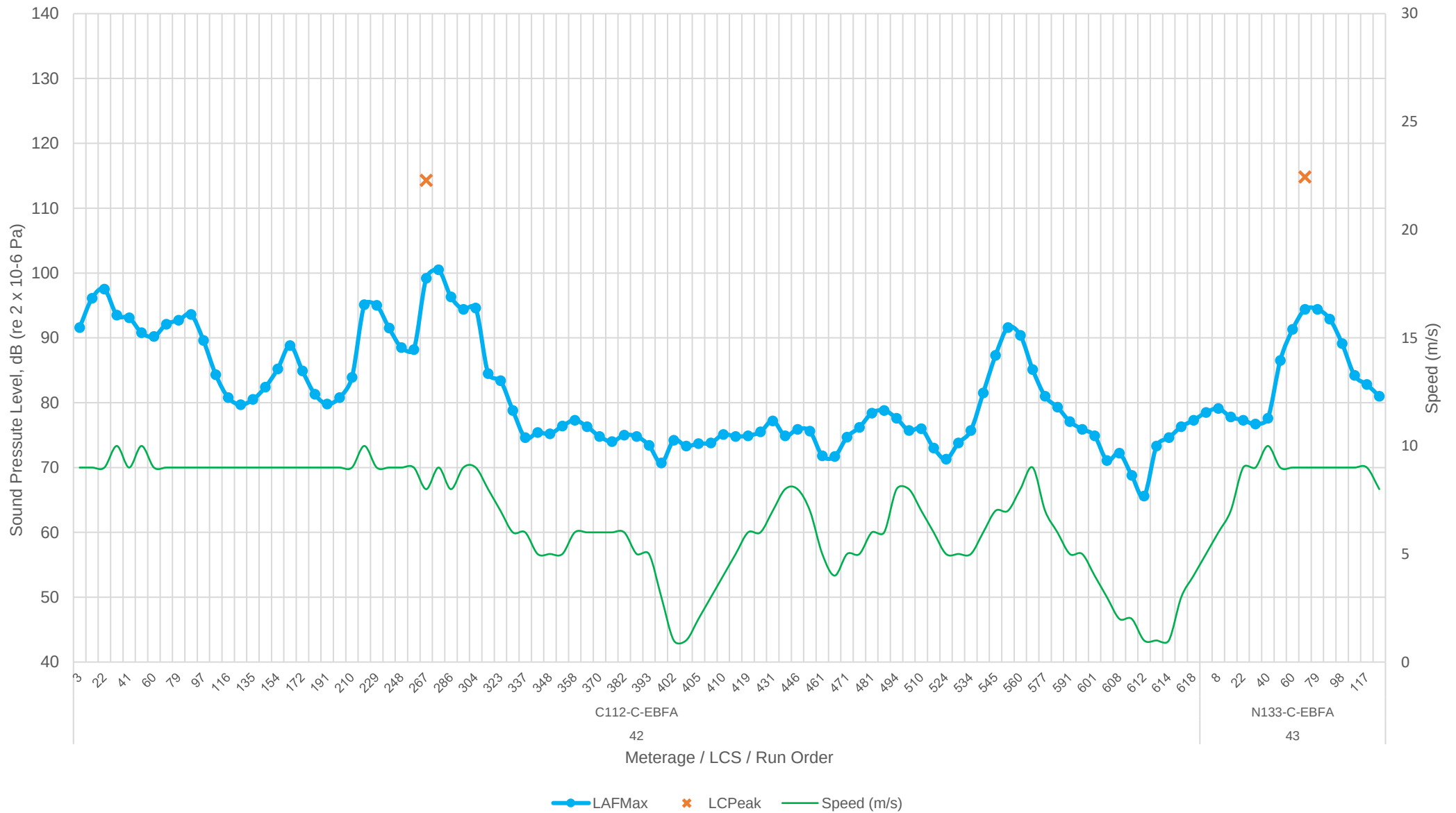


Figure C20 - Chancery Lane (EB) to St. Paul's (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle



**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

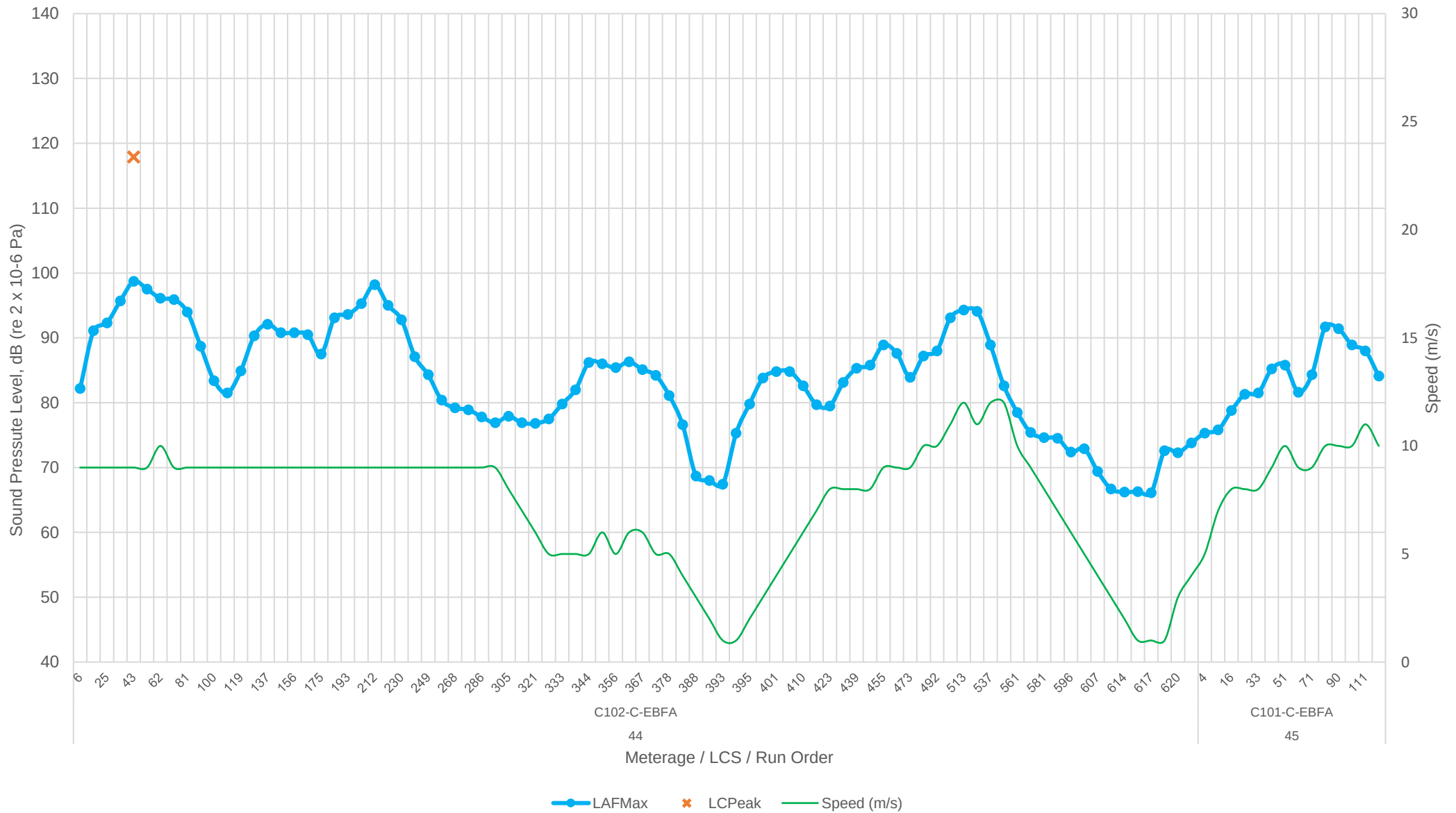


Figure C22 - Bank (EB) to Liverpool Street (EB)

**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

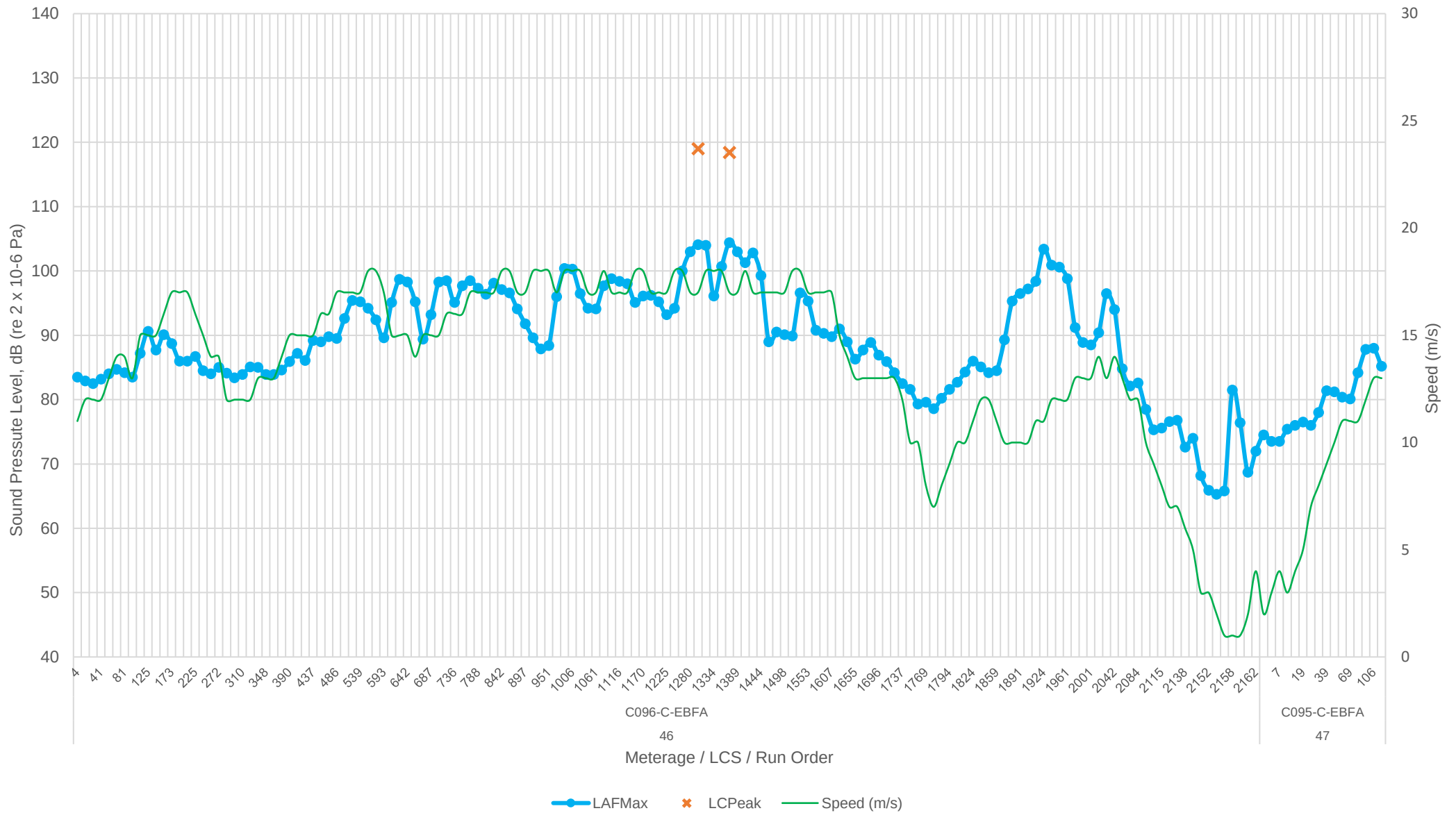


Figure C23 - Liverpool Street (EB) to Bethnal Green (EB)

**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

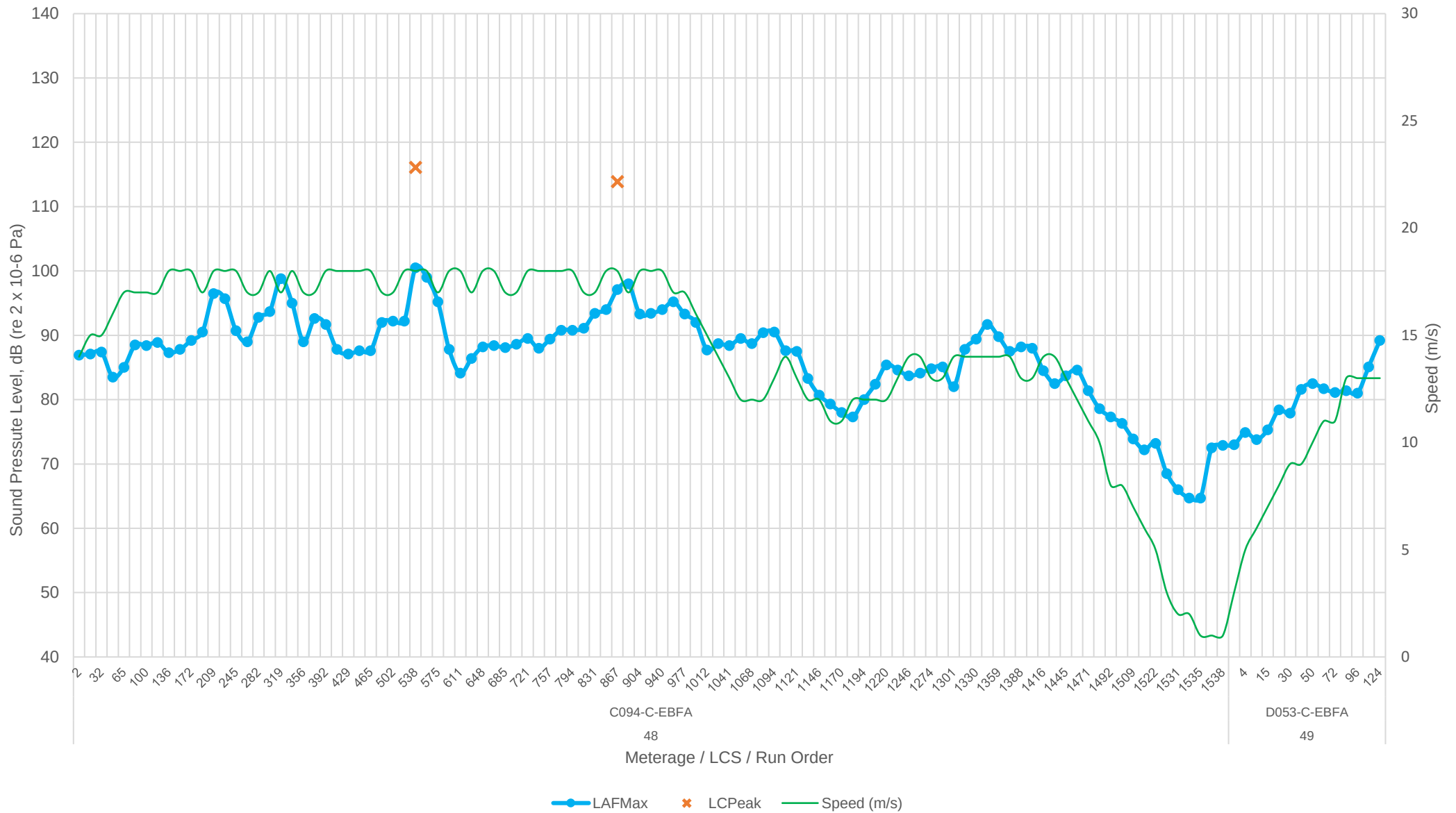


Figure C24 - Bethnal Green (EB) to Mile End (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

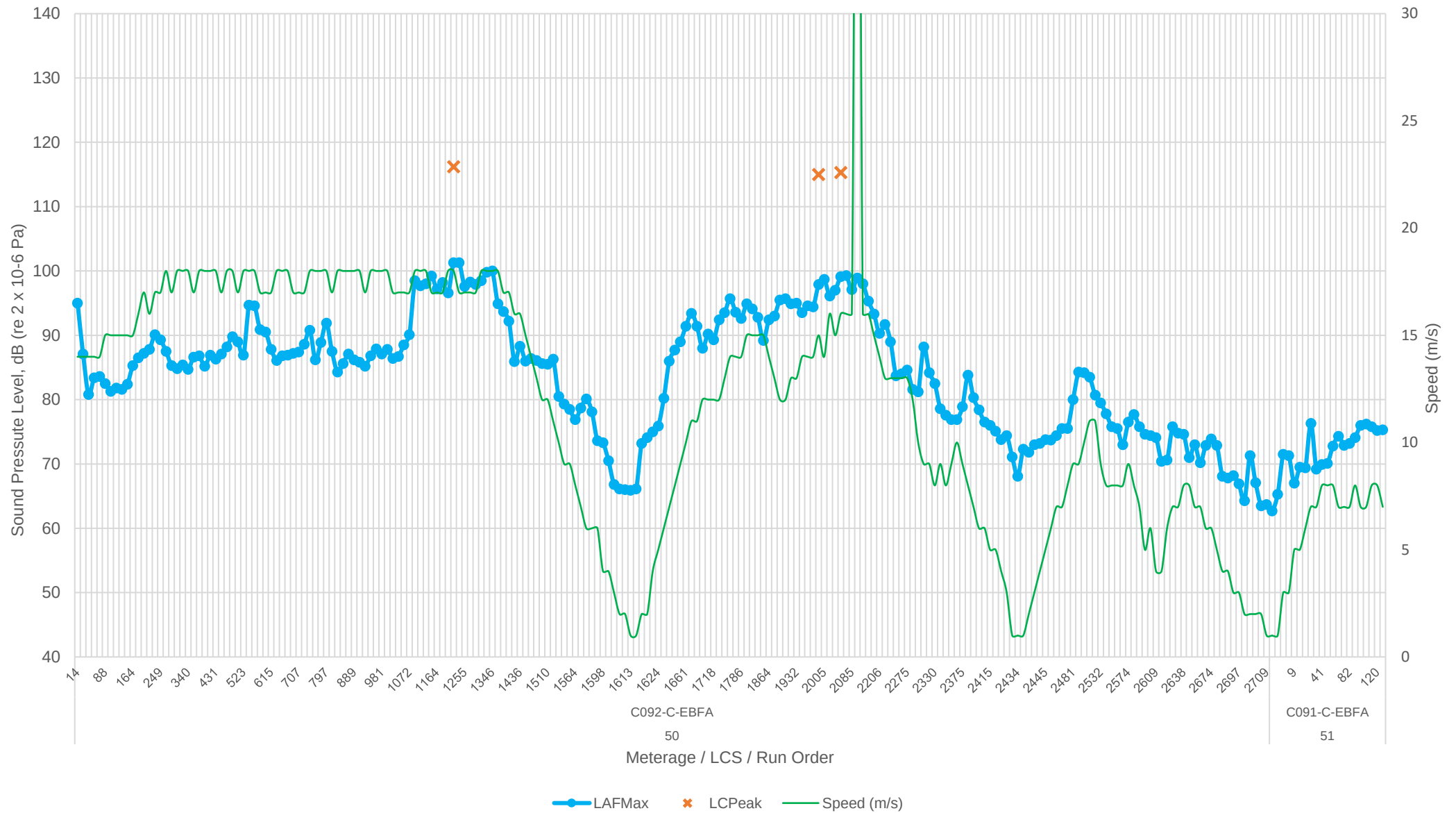


Figure C25 - Mile End (EB) to Stratford (EB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

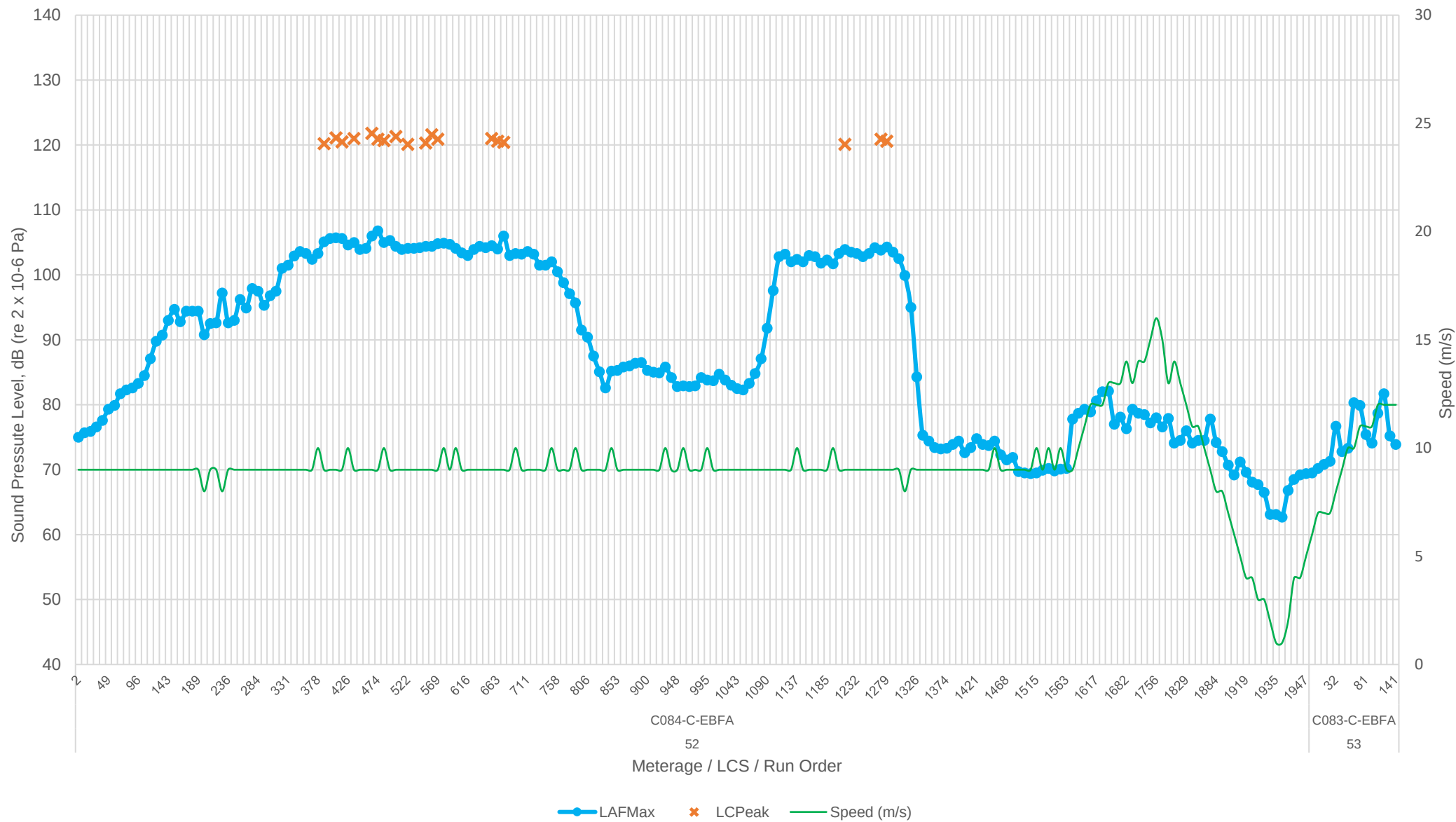


Figure C26 - Stratford (EB) to Leyton (EB)



Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

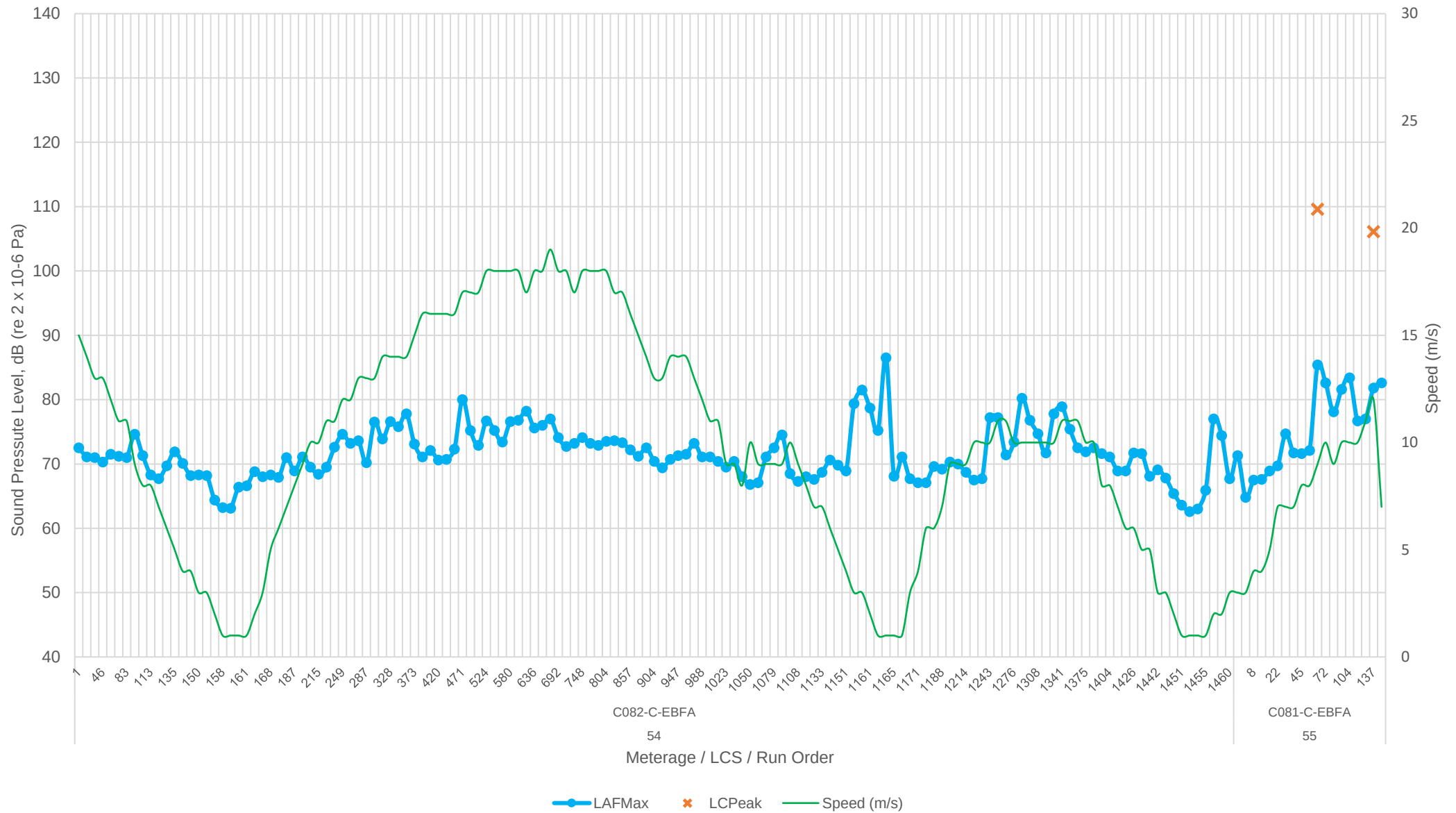


Figure C27 - Leyton (EB) to Leytonstone (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

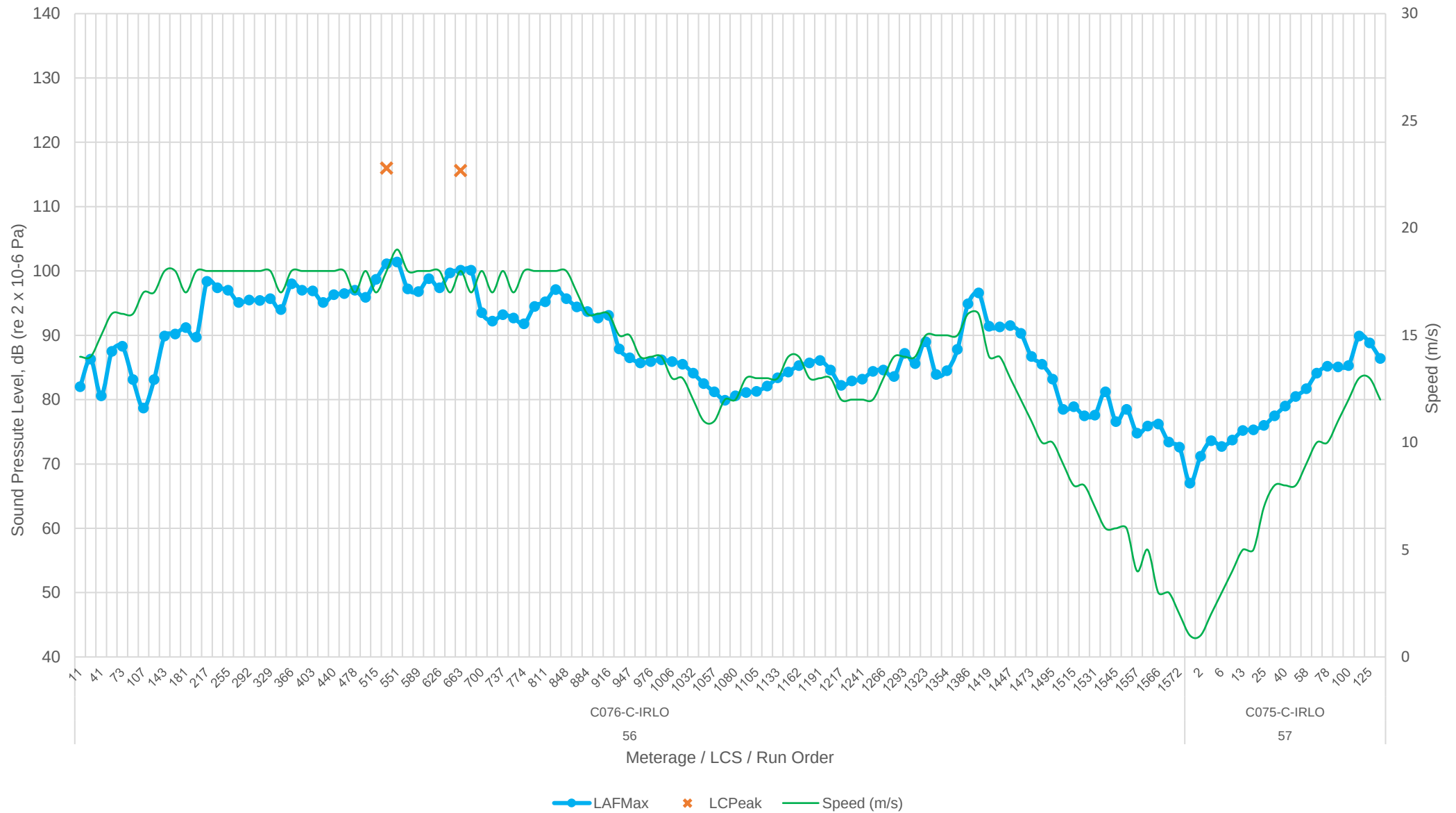


Figure C28 - Leytonstone (EB) to Wanstead (IR)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

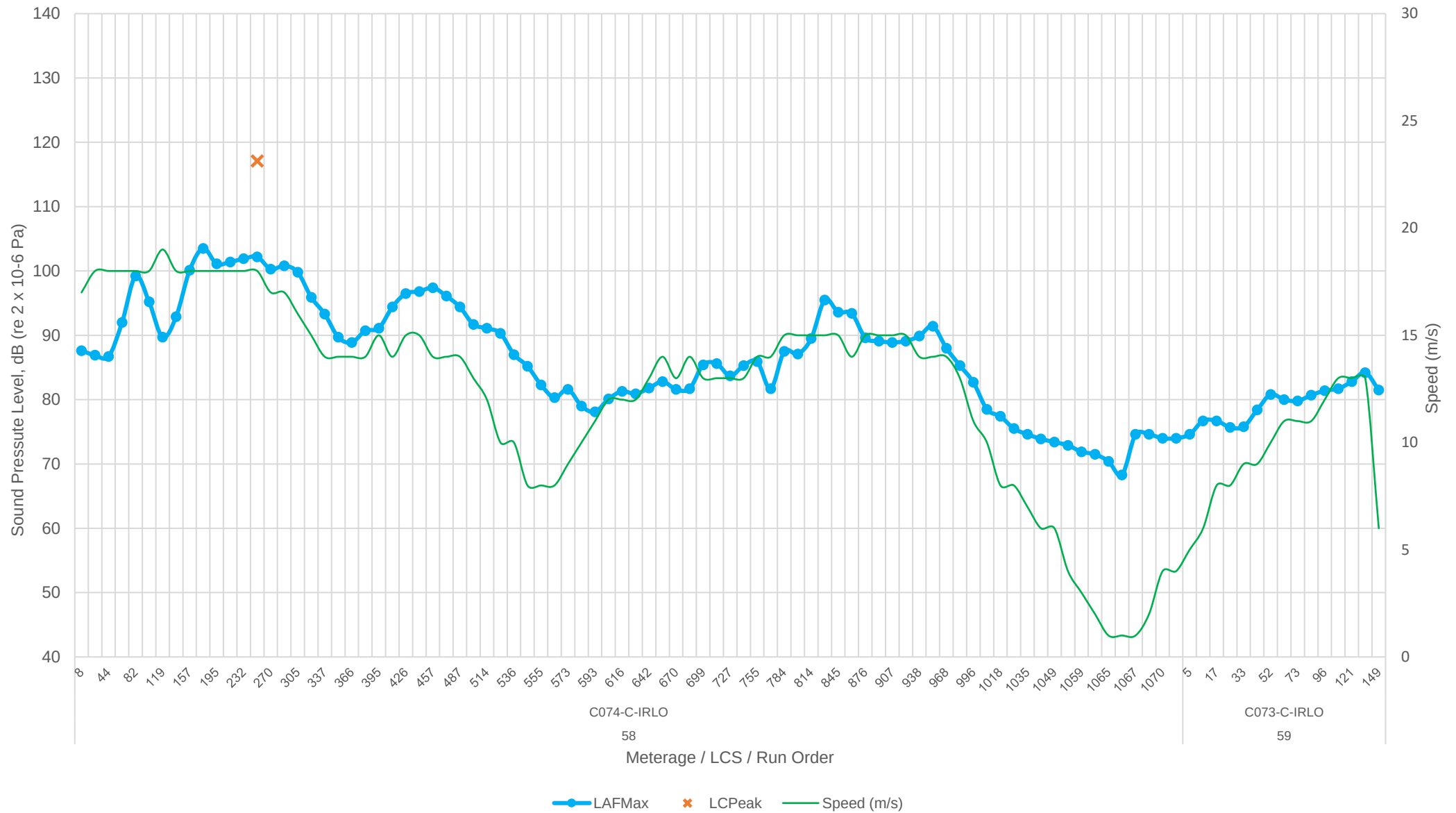


Figure C29 - Wanstead (IR) to Redbridge (IR)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

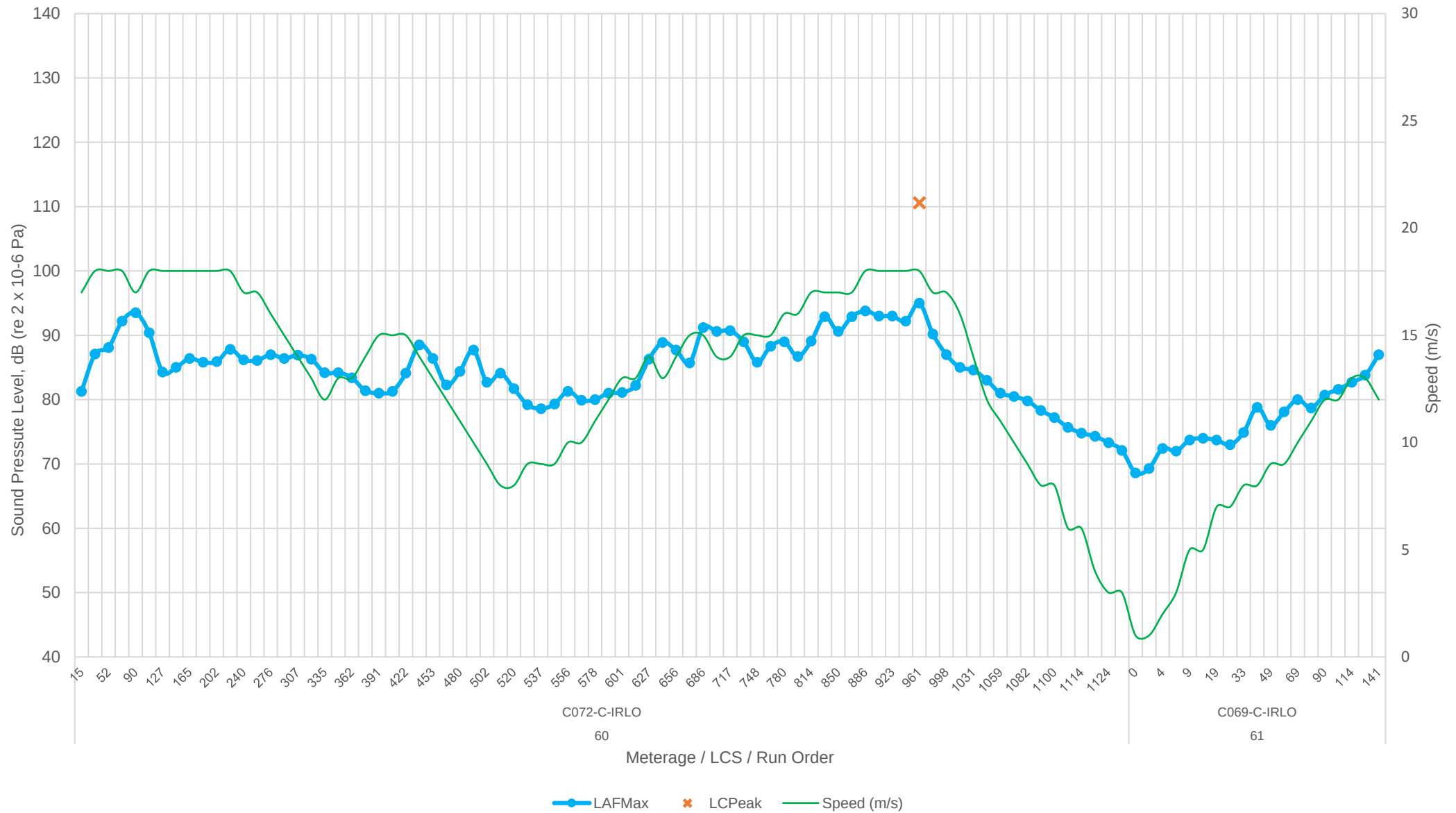


Figure C30 - Redbridge (IR) to Gants Hill (IR)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

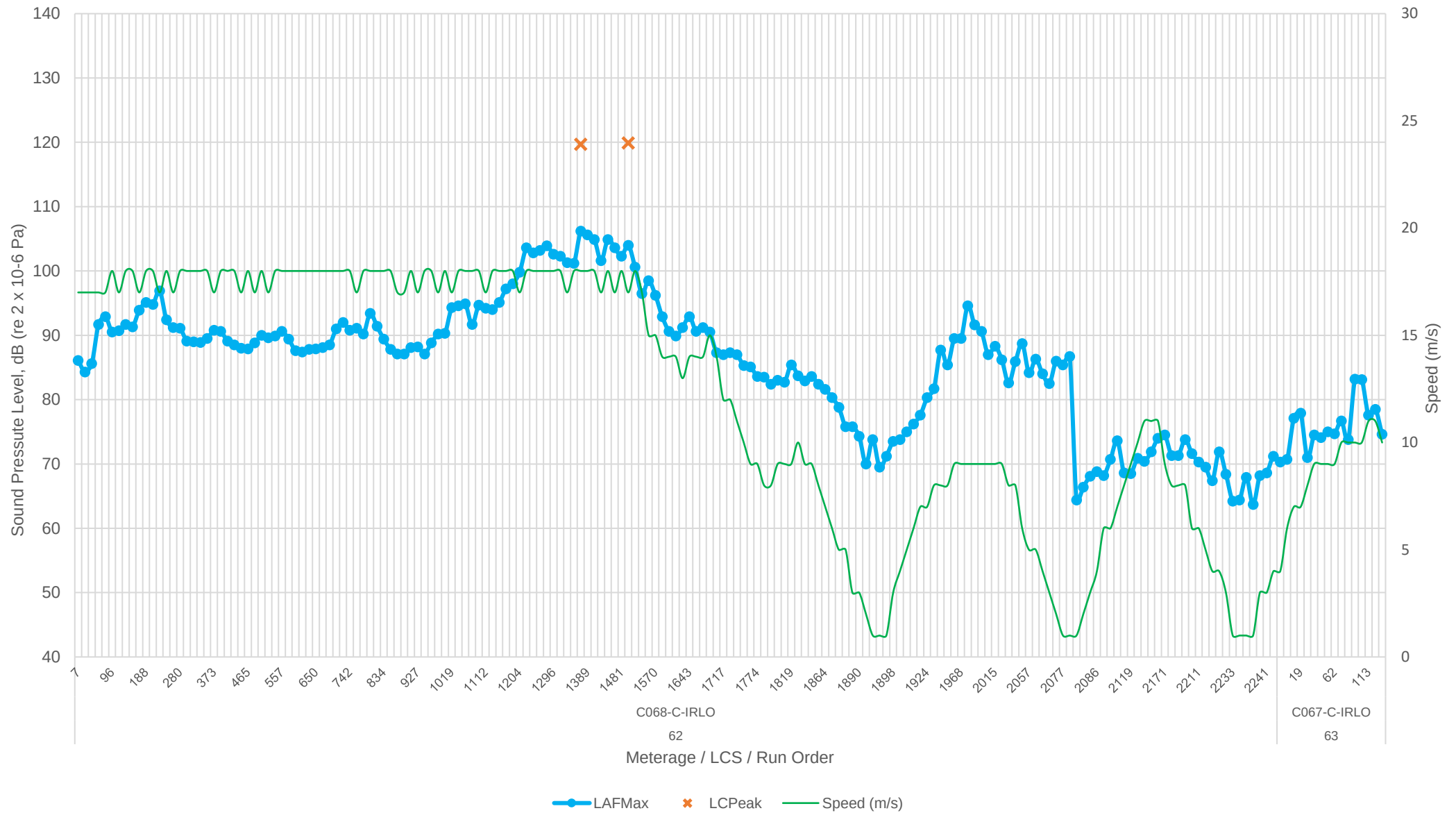


Figure C31 - Gants Hill (IR) to Newbury Park (IR)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

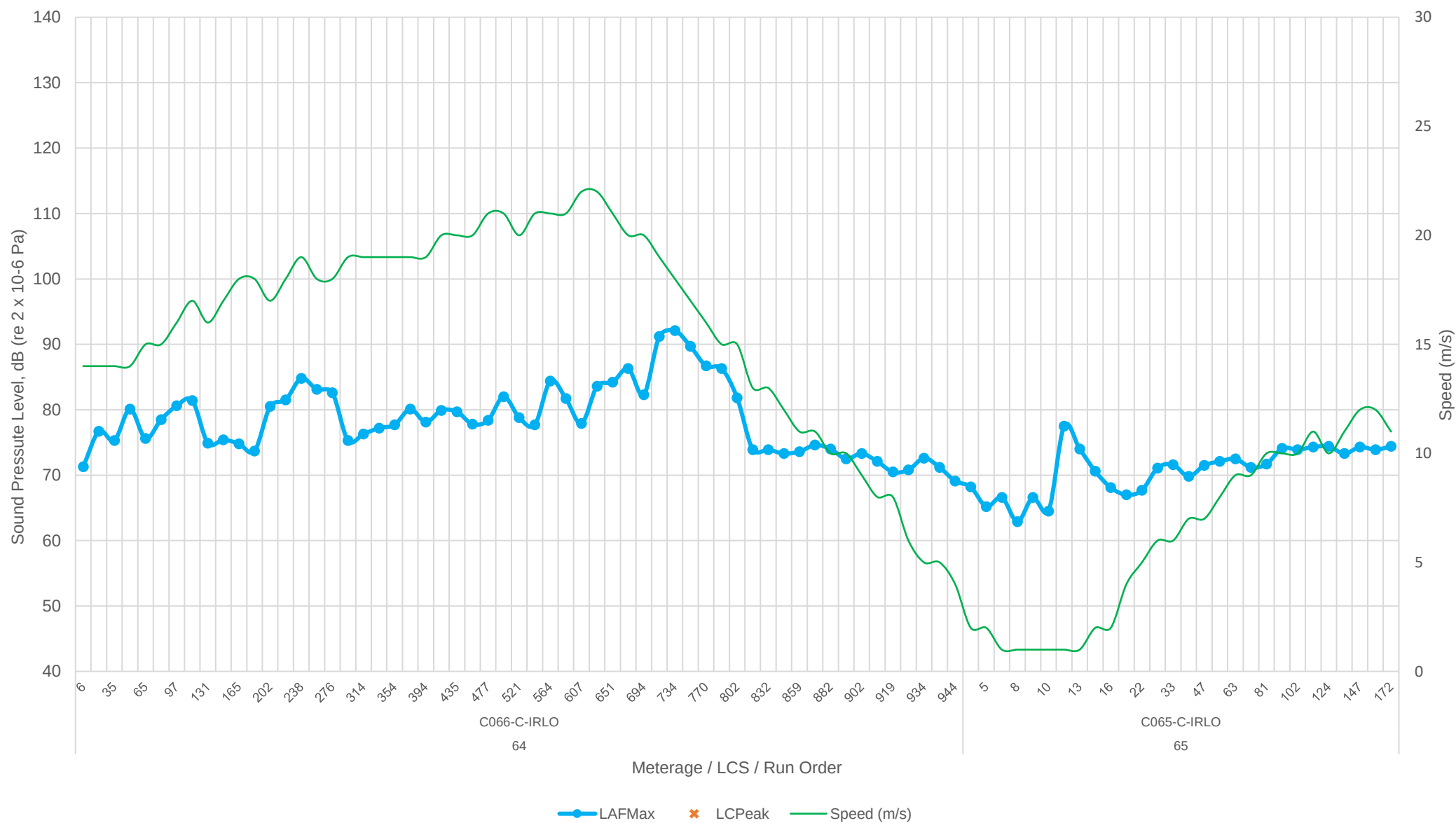
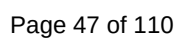


Figure C32 - Newbury Park (IR) to Barkingside (IR)



Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

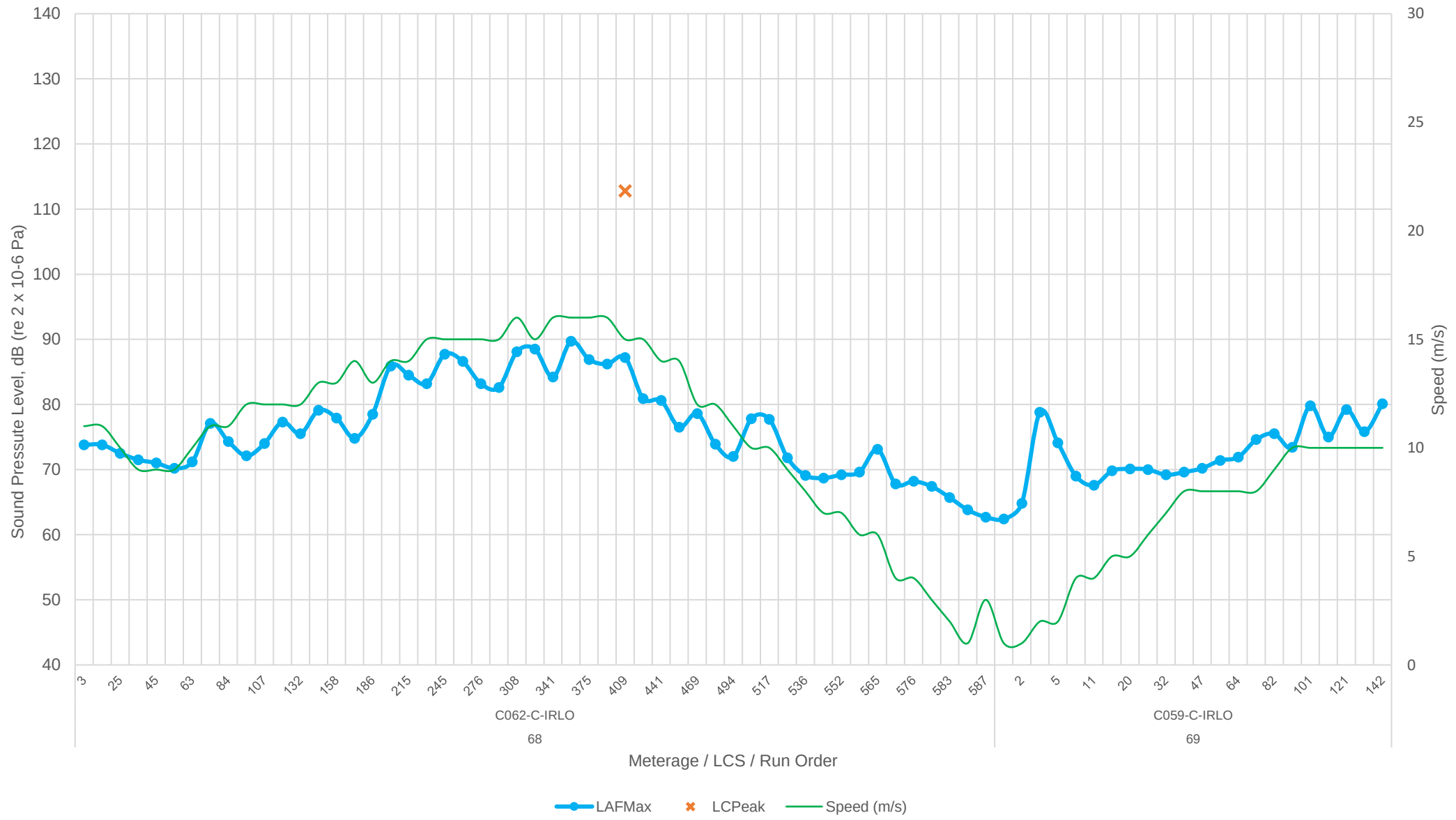


Figure C34 - Fairlop (IR) to Hainault (IR)



**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

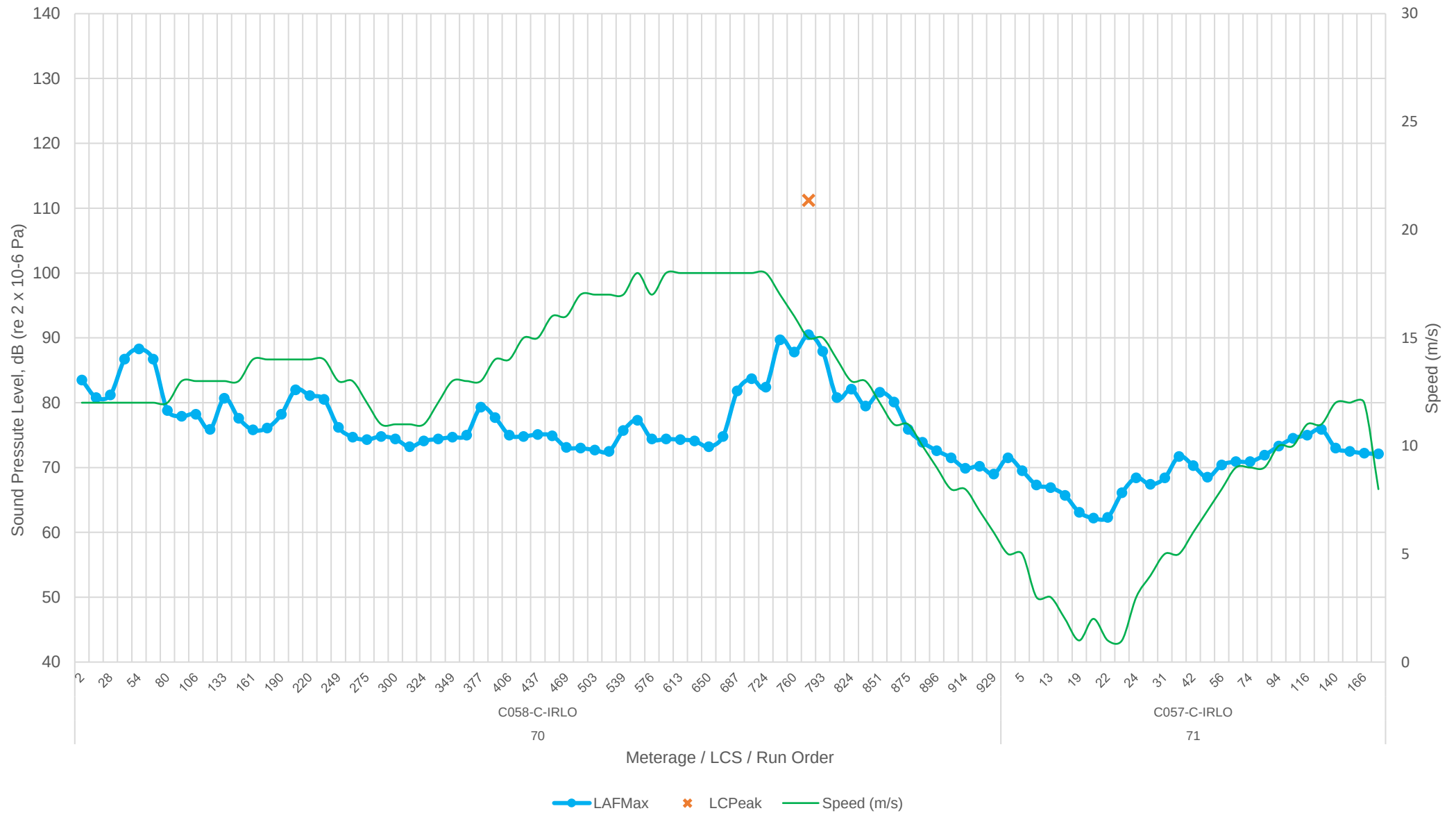


Figure C35 - Hainault (IR) to Grange Hill (IR)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

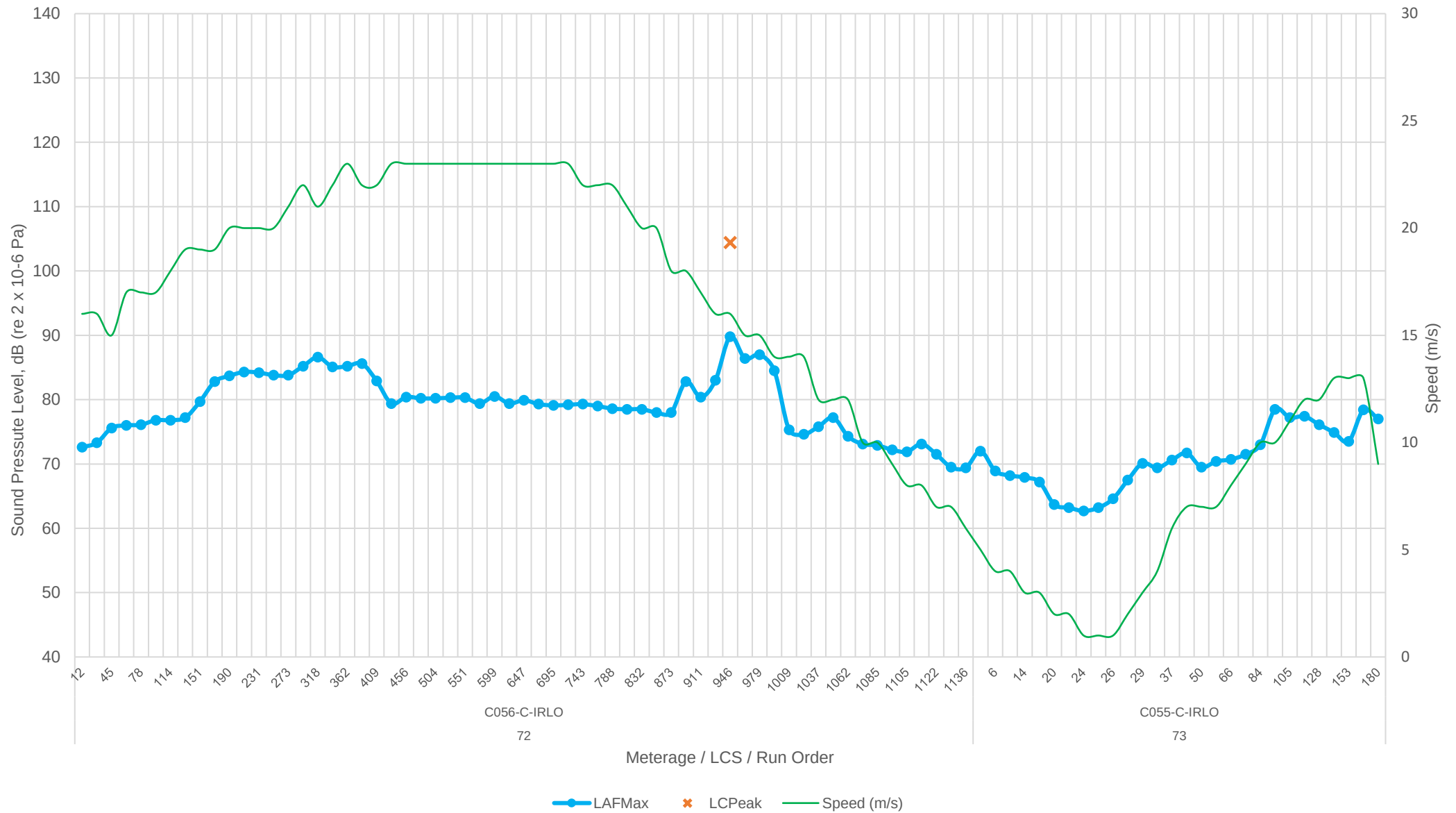


Figure C36 - Grange Hill (IR) to Chigwell (IR)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

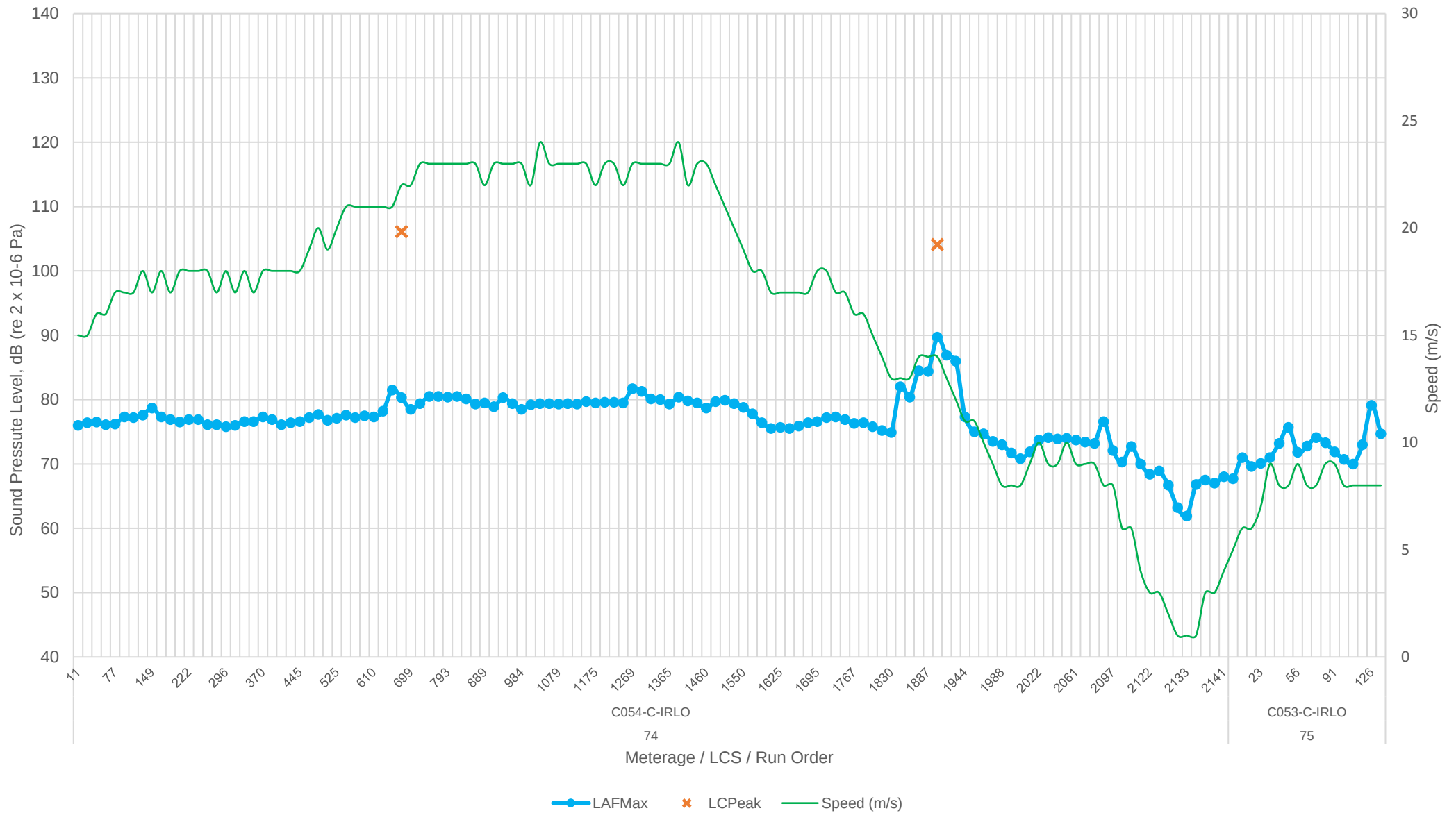


Figure C37 - Chigwell (IR) to Roding Valley (IR)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

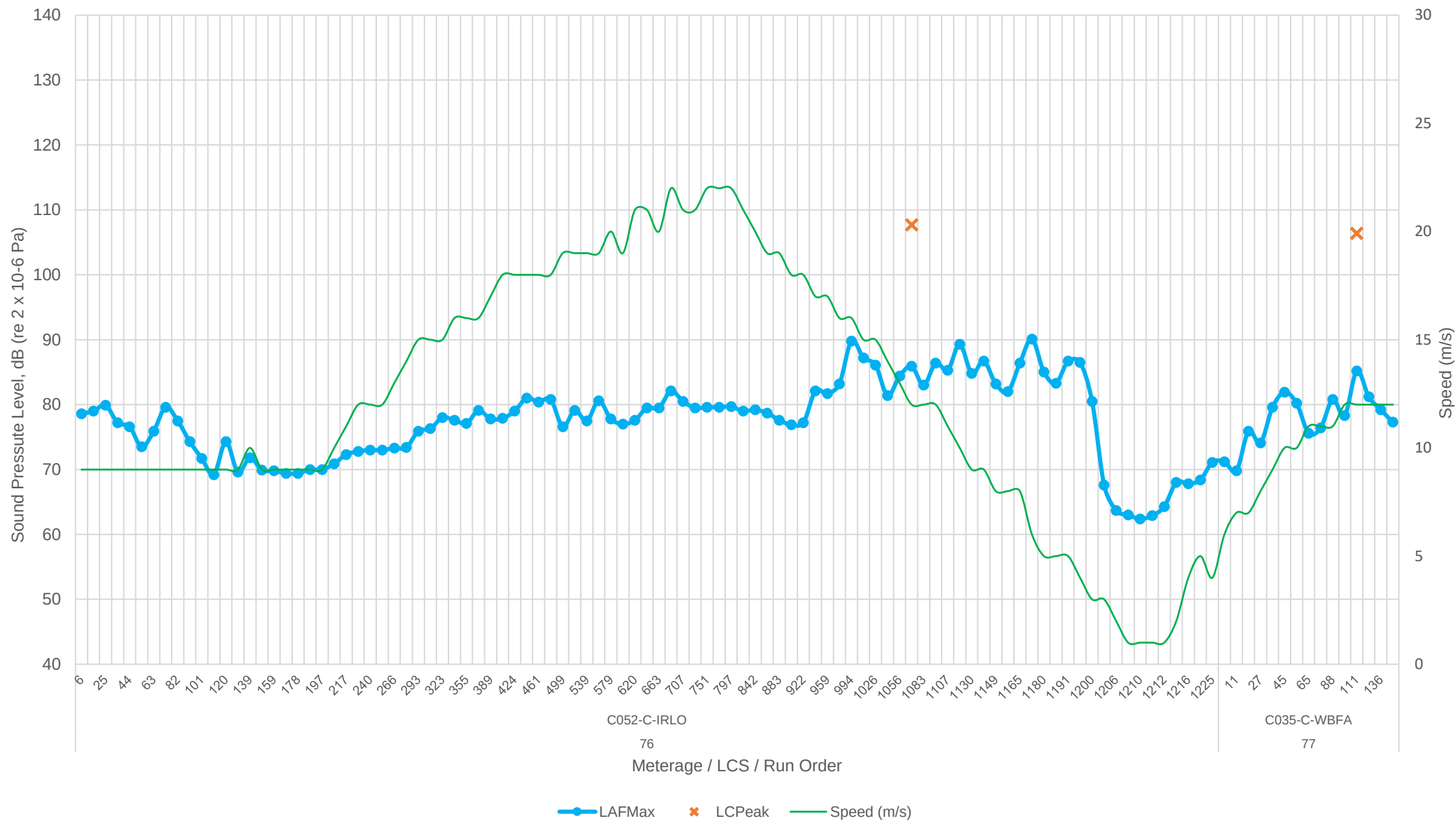


Figure C38 - Roding Valley (IR) to Woodford (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

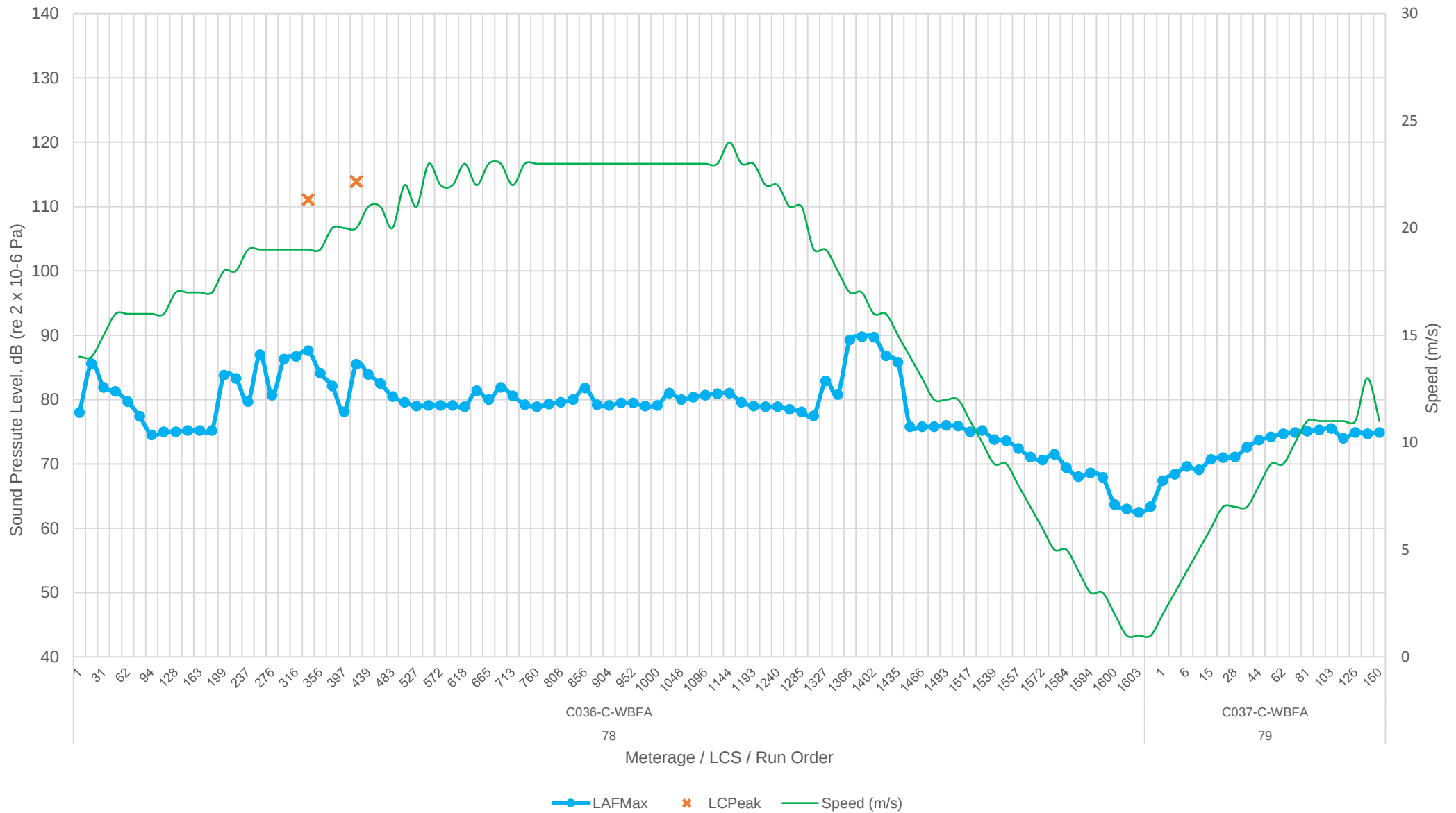


Figure C39 - Woodford (WB) to South Woodford (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

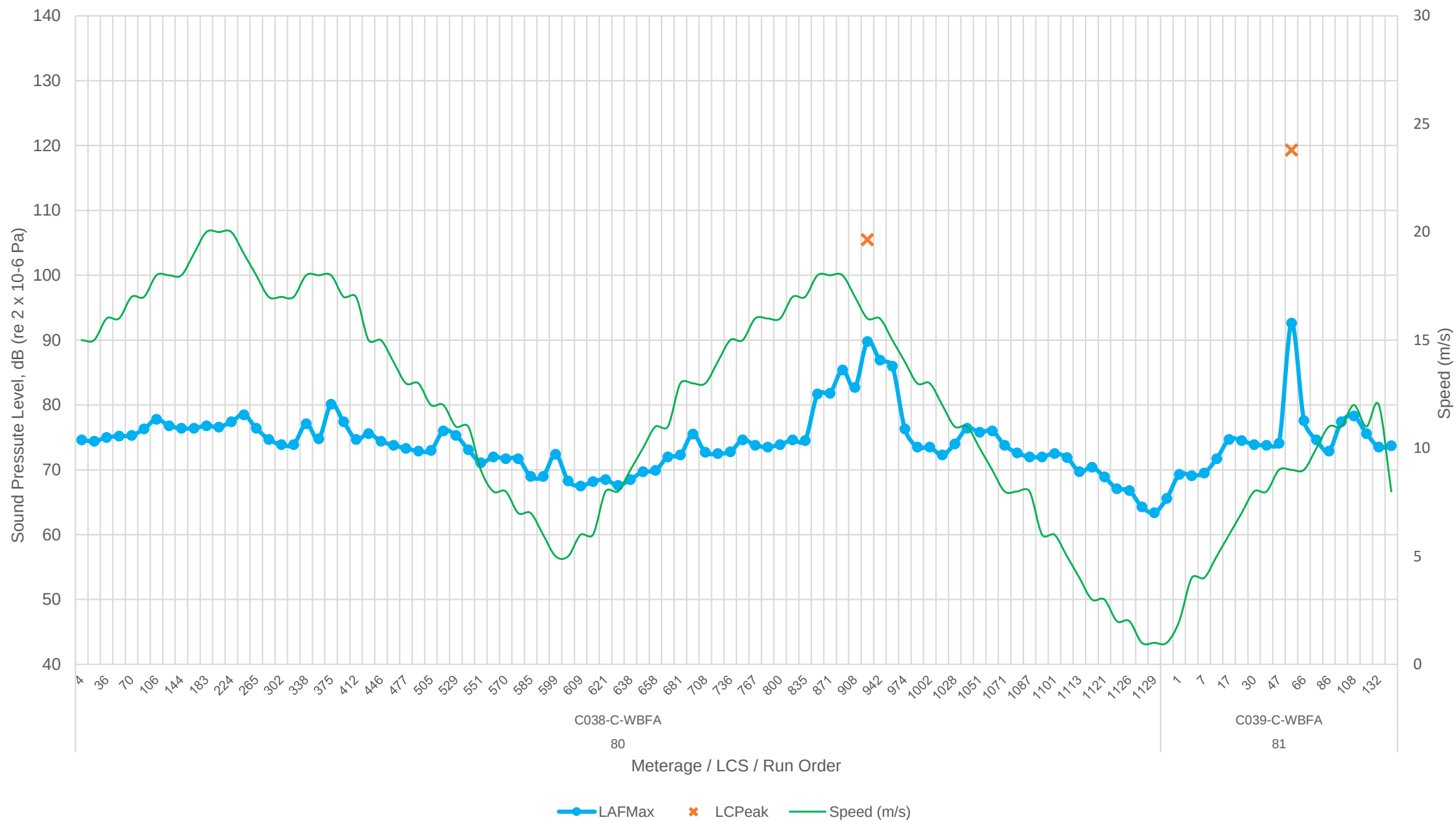


Figure C40 - South Woodford (WB) to Snaresbrook (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

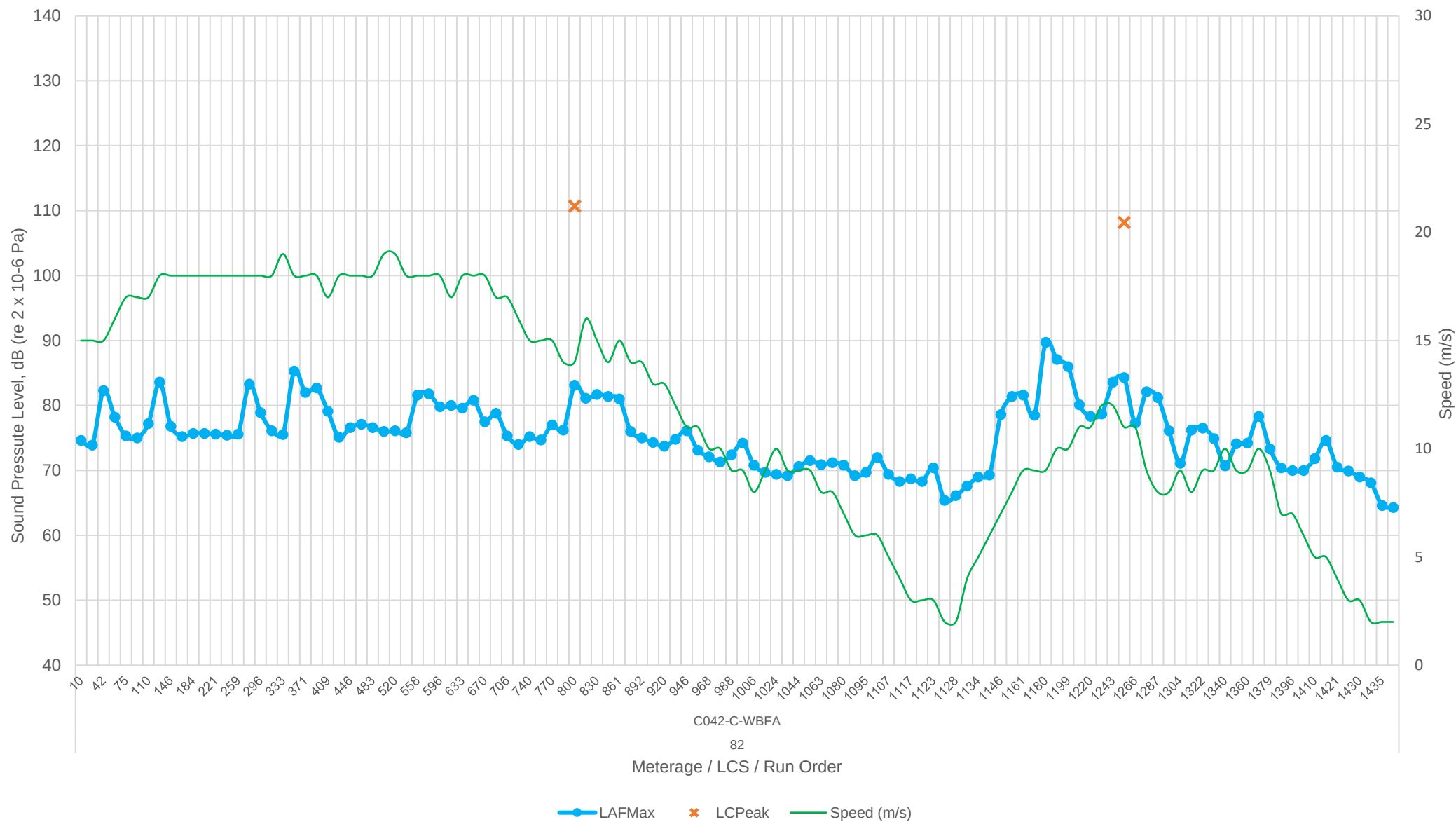


Figure C41 - Snaresbrook (WB) to Leytonstone (WB)

**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

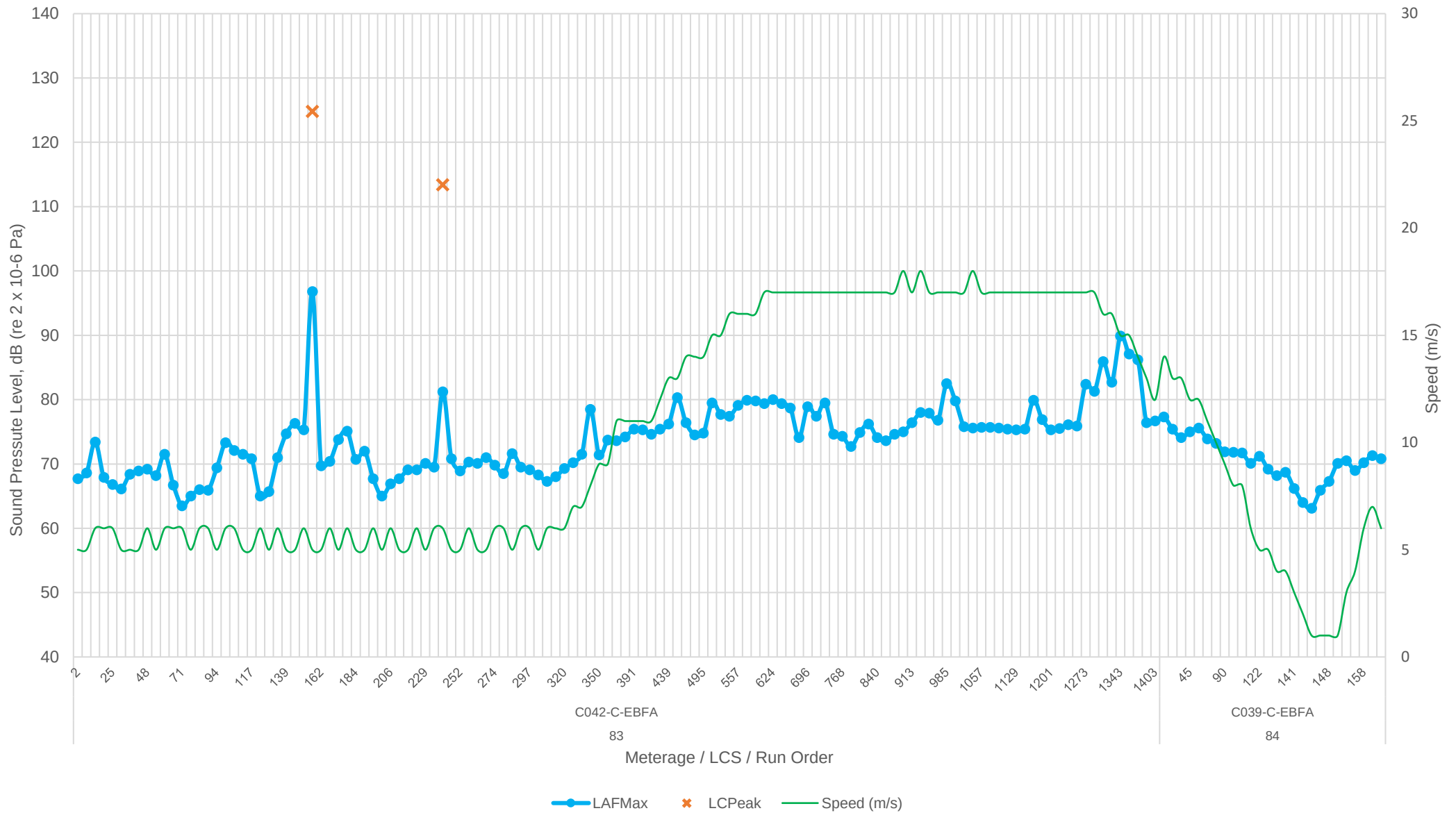


Figure C42 - Leytonstone (WB) to Snaresbrook (EB)



# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

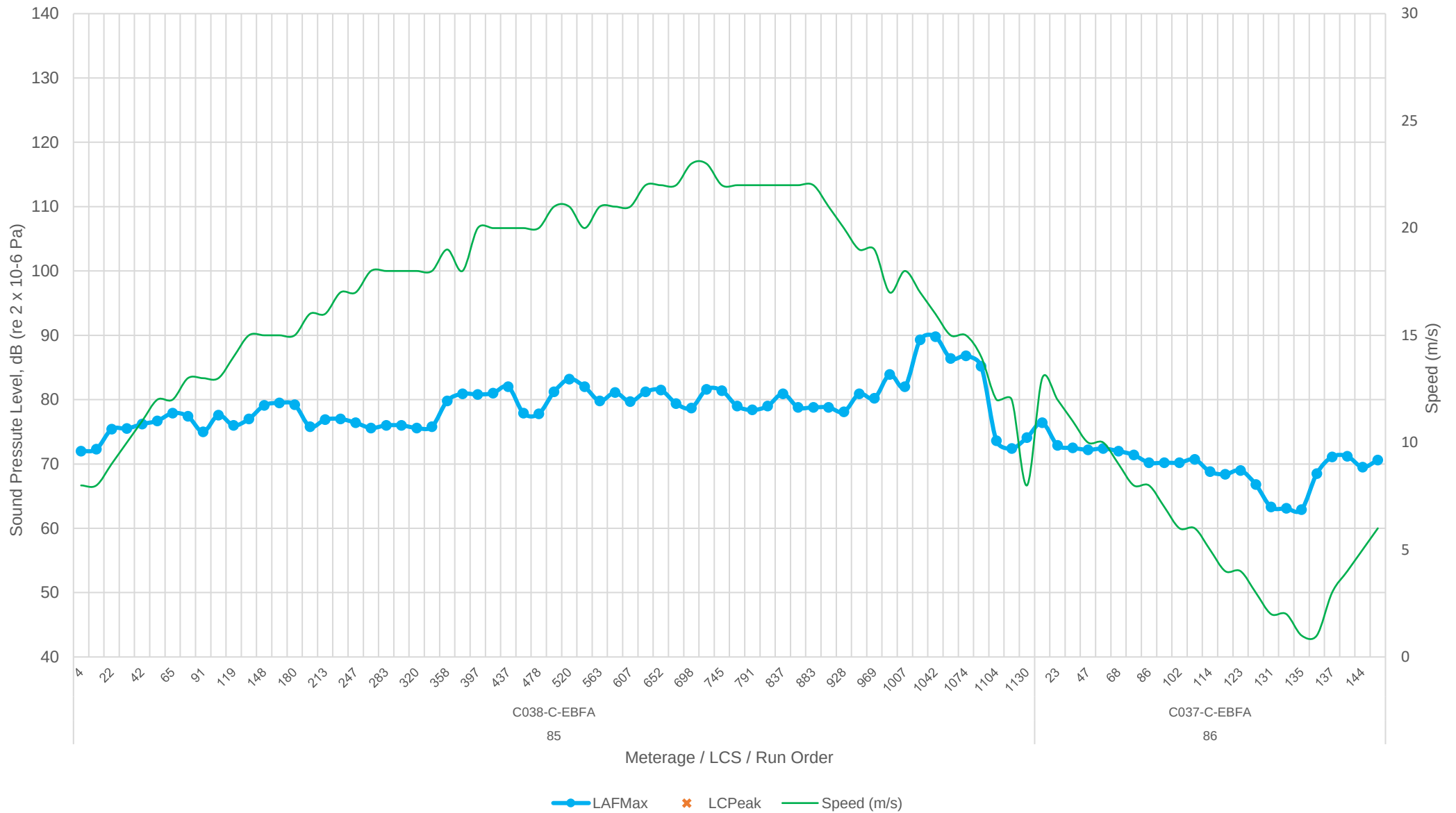


Figure C43 - Snaresbrook (EB) to South Woodford (EB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

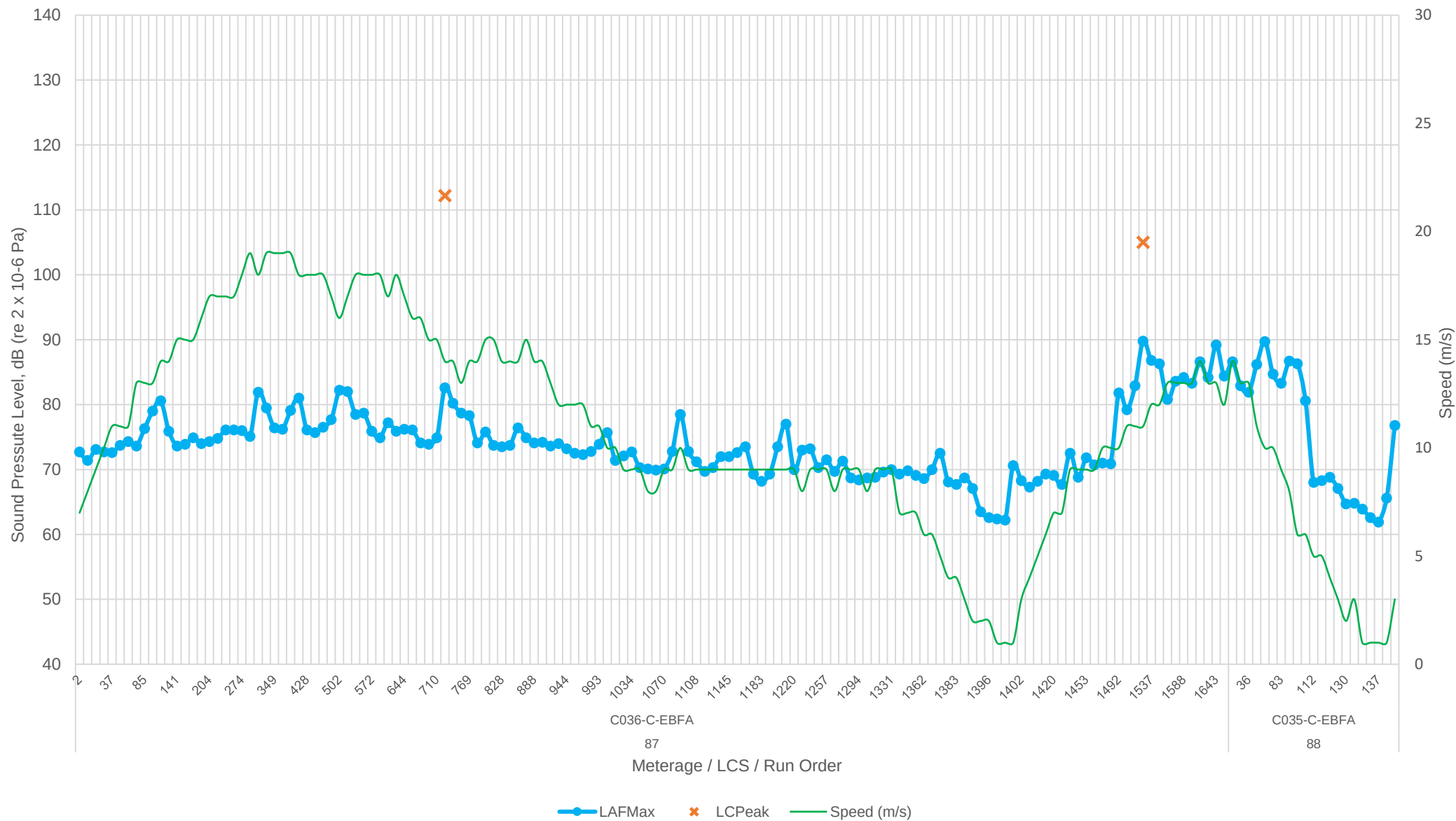


Figure C44 - South Woodford (EB) to Woodford (EB)

**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

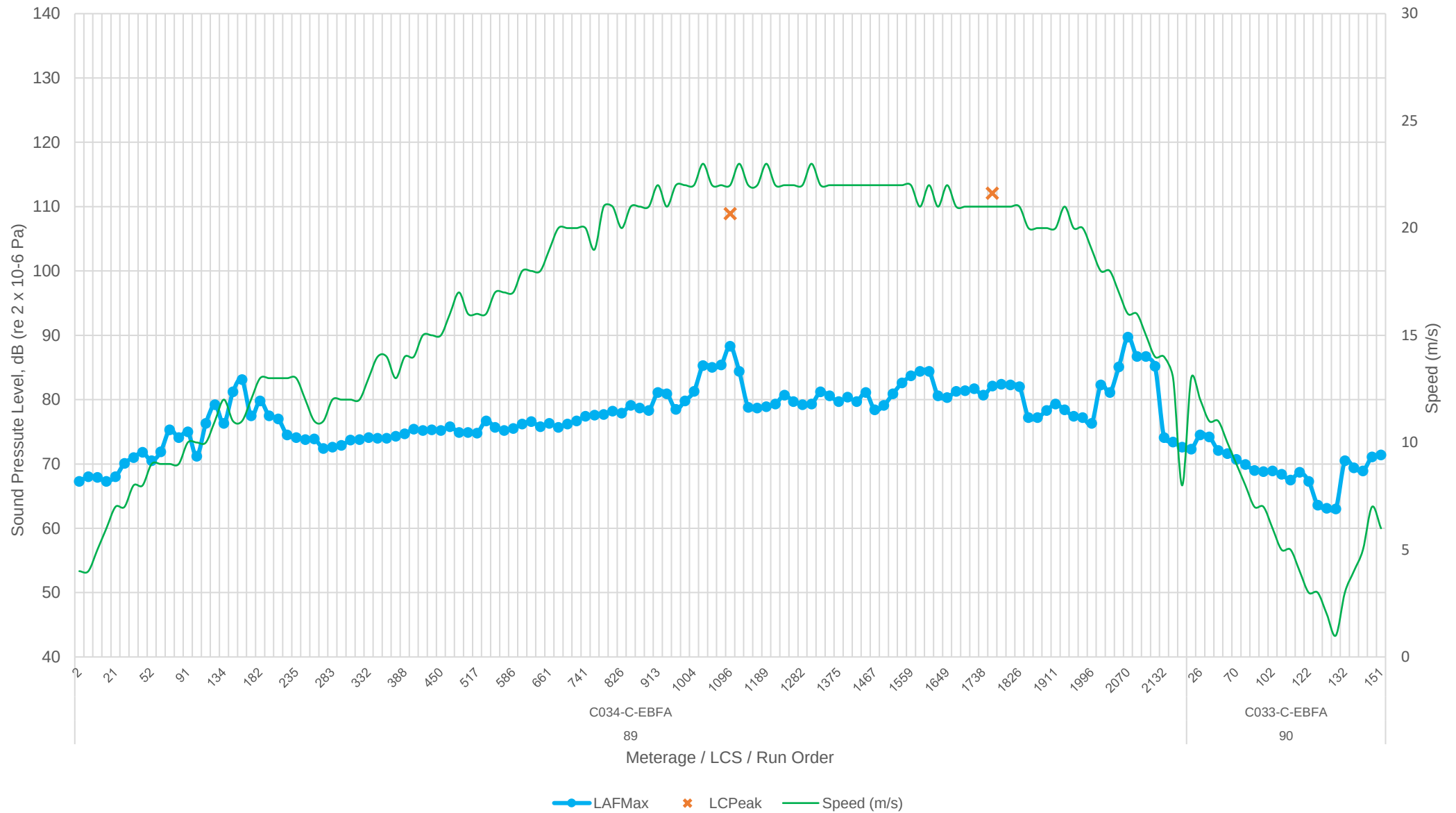


Figure C45 - Woodford (EB) to Buckhurst Hill (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

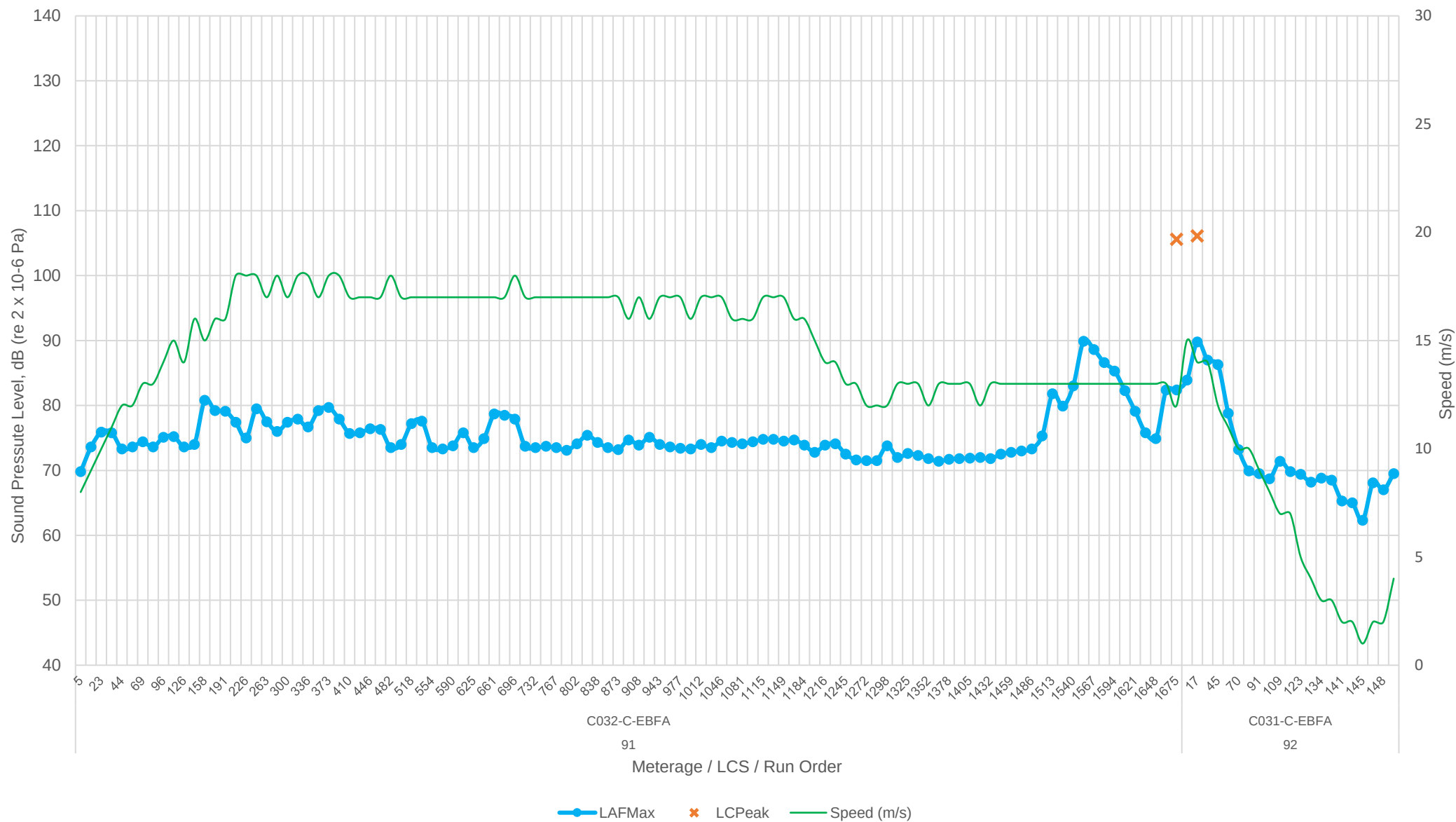


Figure C46 - Buckhurst Hill (EB) to Loughton (EB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

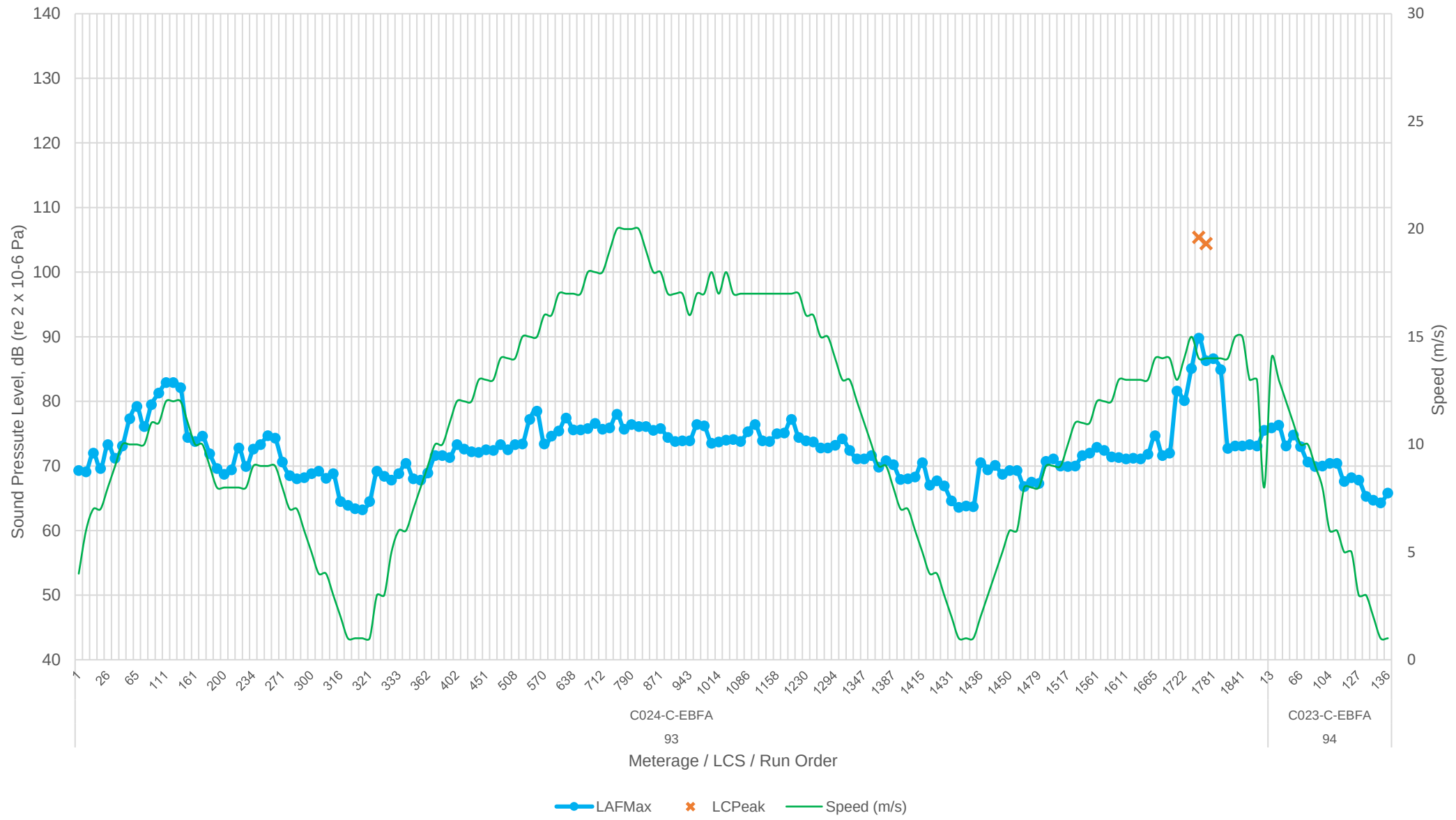


Figure C47 - Loughton (EB) to Debden (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

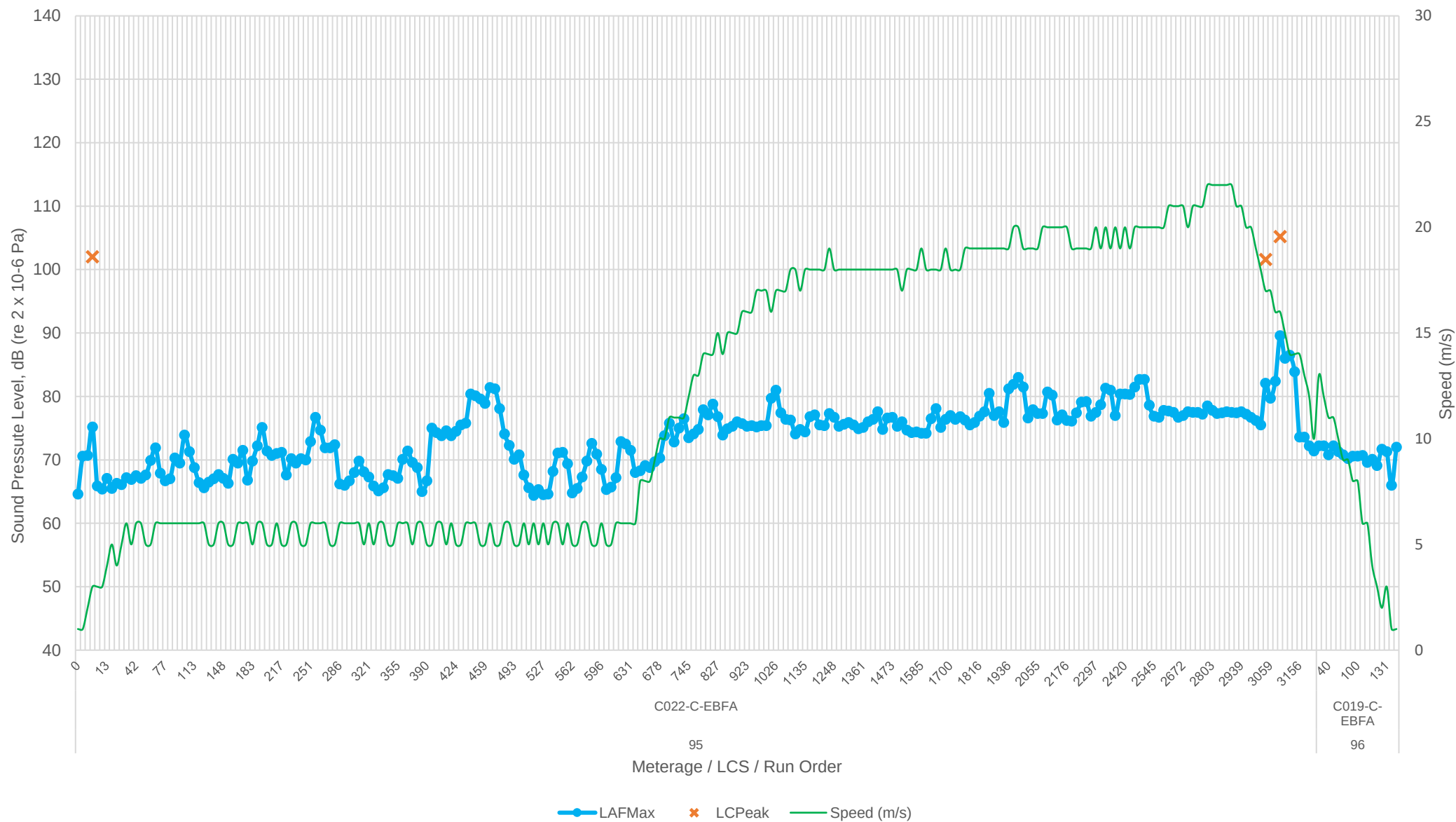


Figure C48 - Debden (EB) to Theydon Bois (EB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

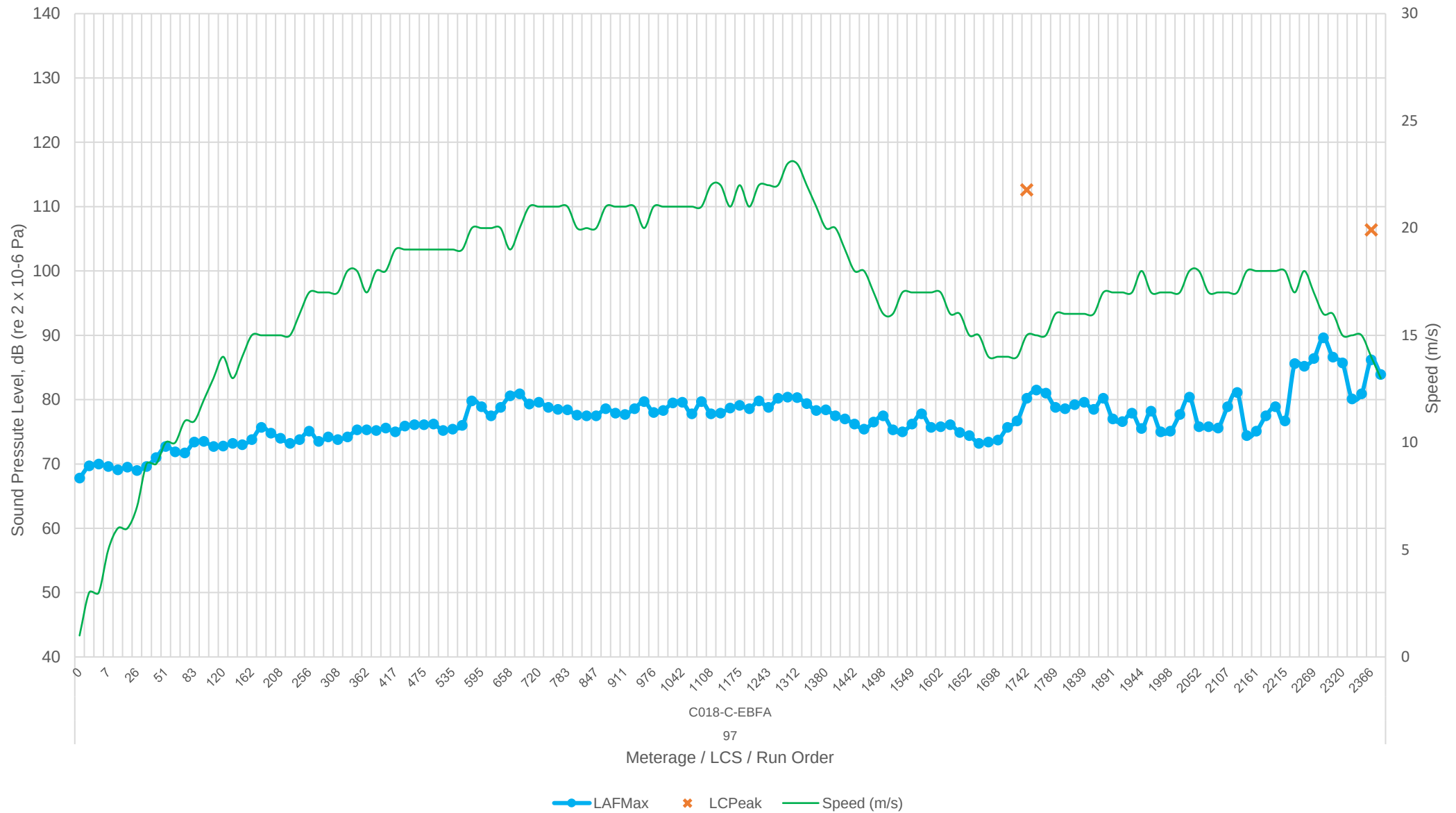


Figure C49 - Theydon Bois (EB) to Epping (Terminating)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

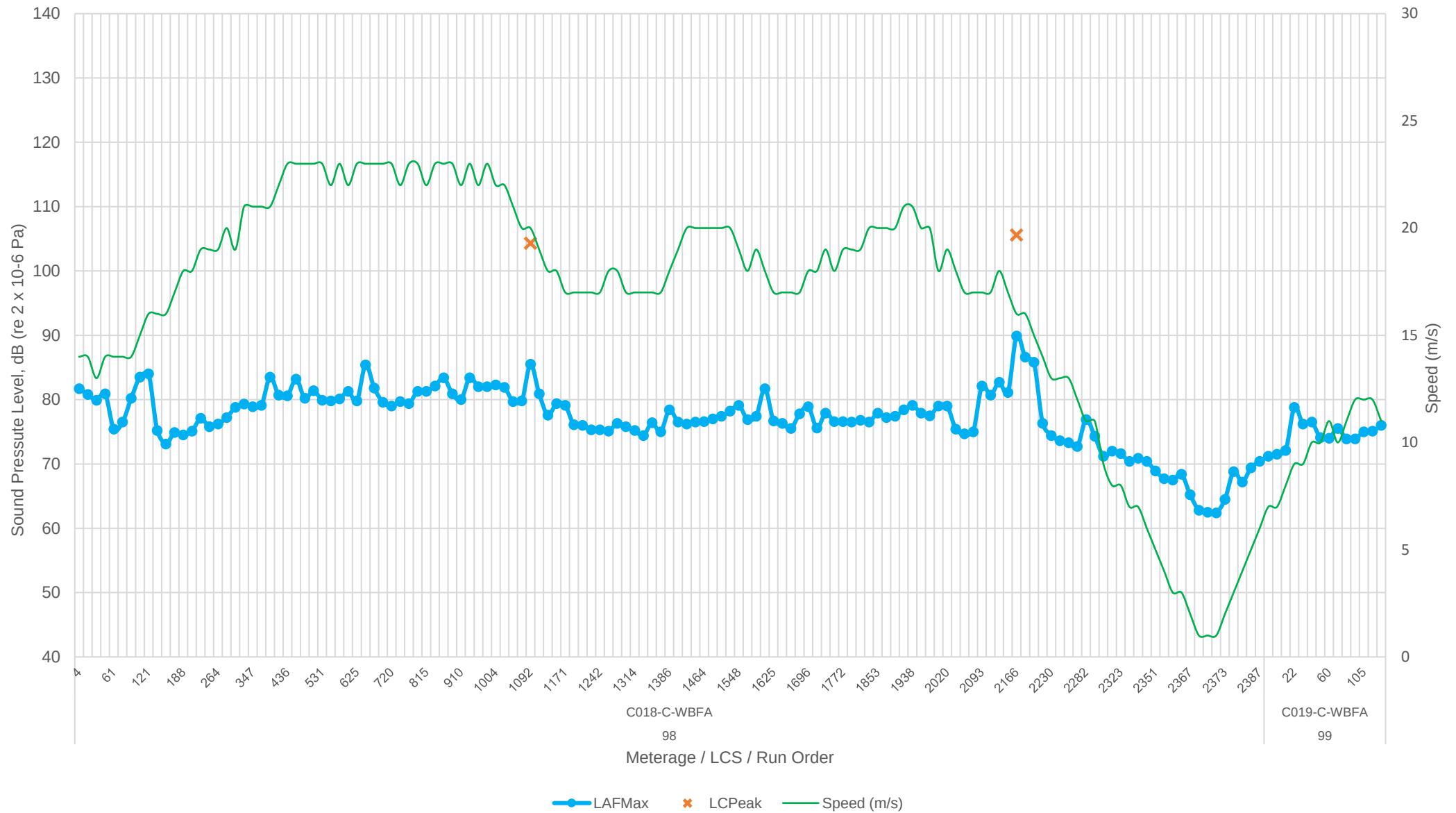


Figure C50 - Epping (Terminating) to Theydon Bois (WB)



Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

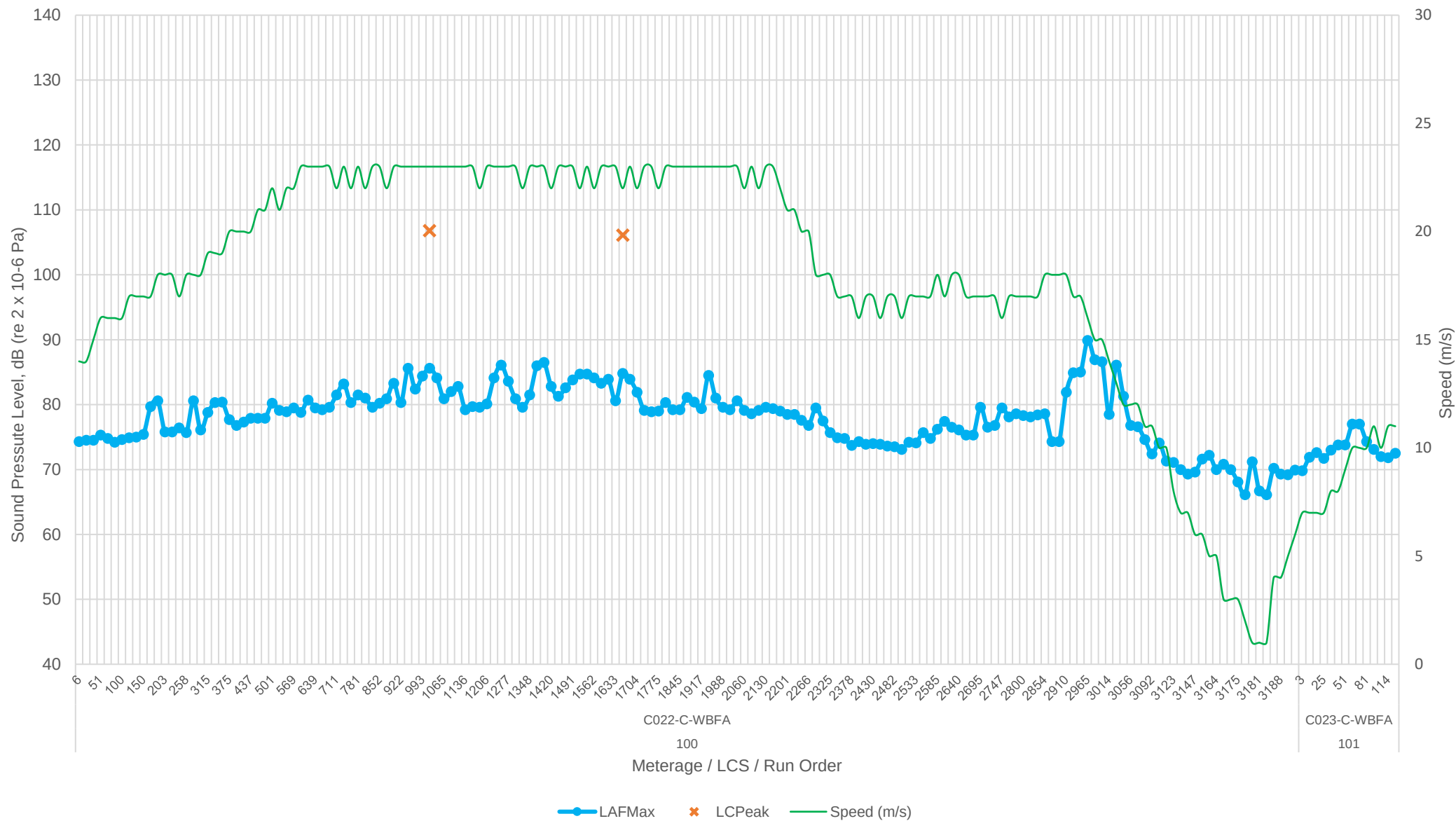


Figure C51 - Theydon Bois (WB) to Debden (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

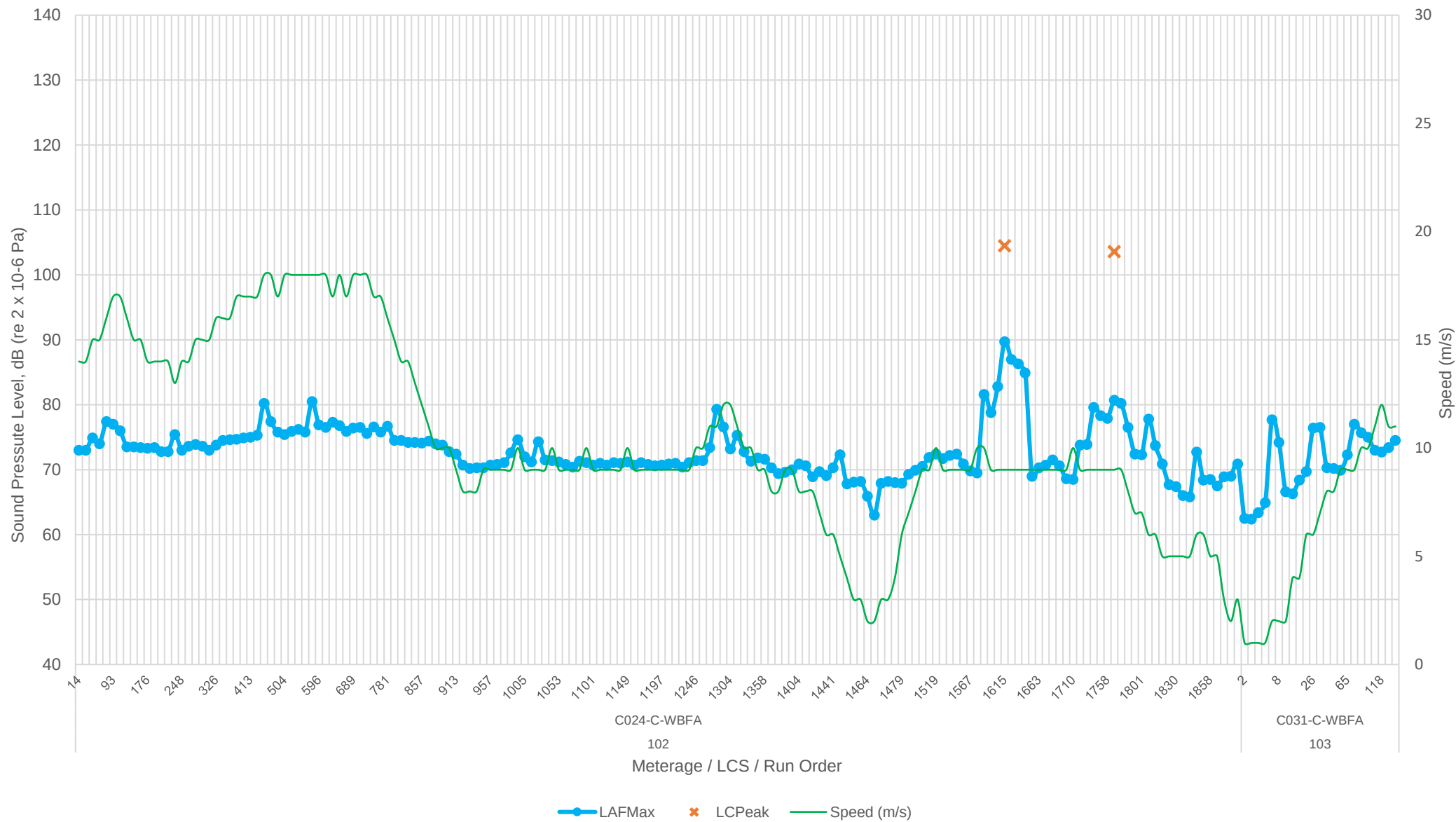


Figure C52 - Debden (WB) to Loughton (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

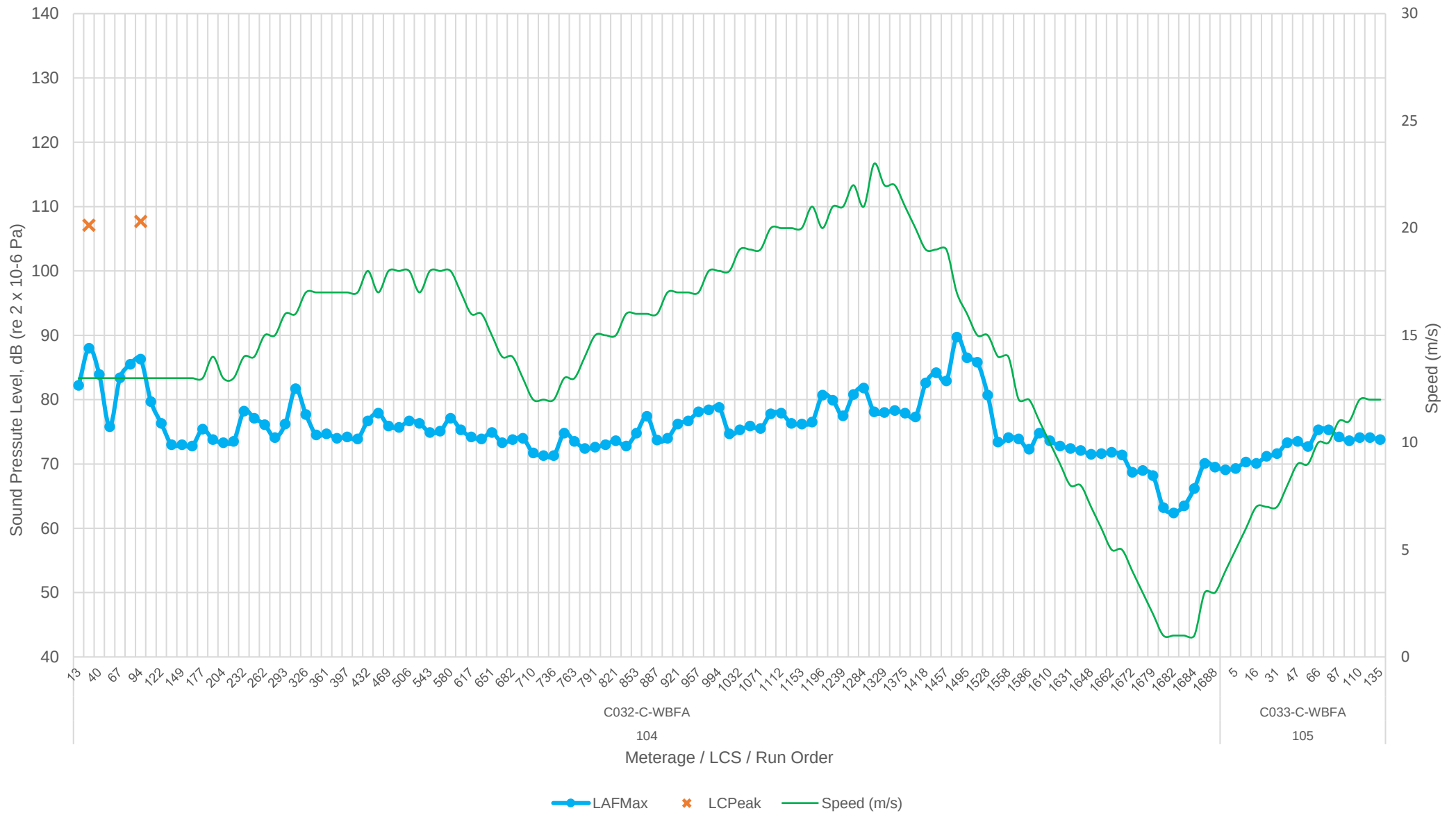


Figure C53 - Loughton (WB) to Buckhurst Hill (WB)

**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

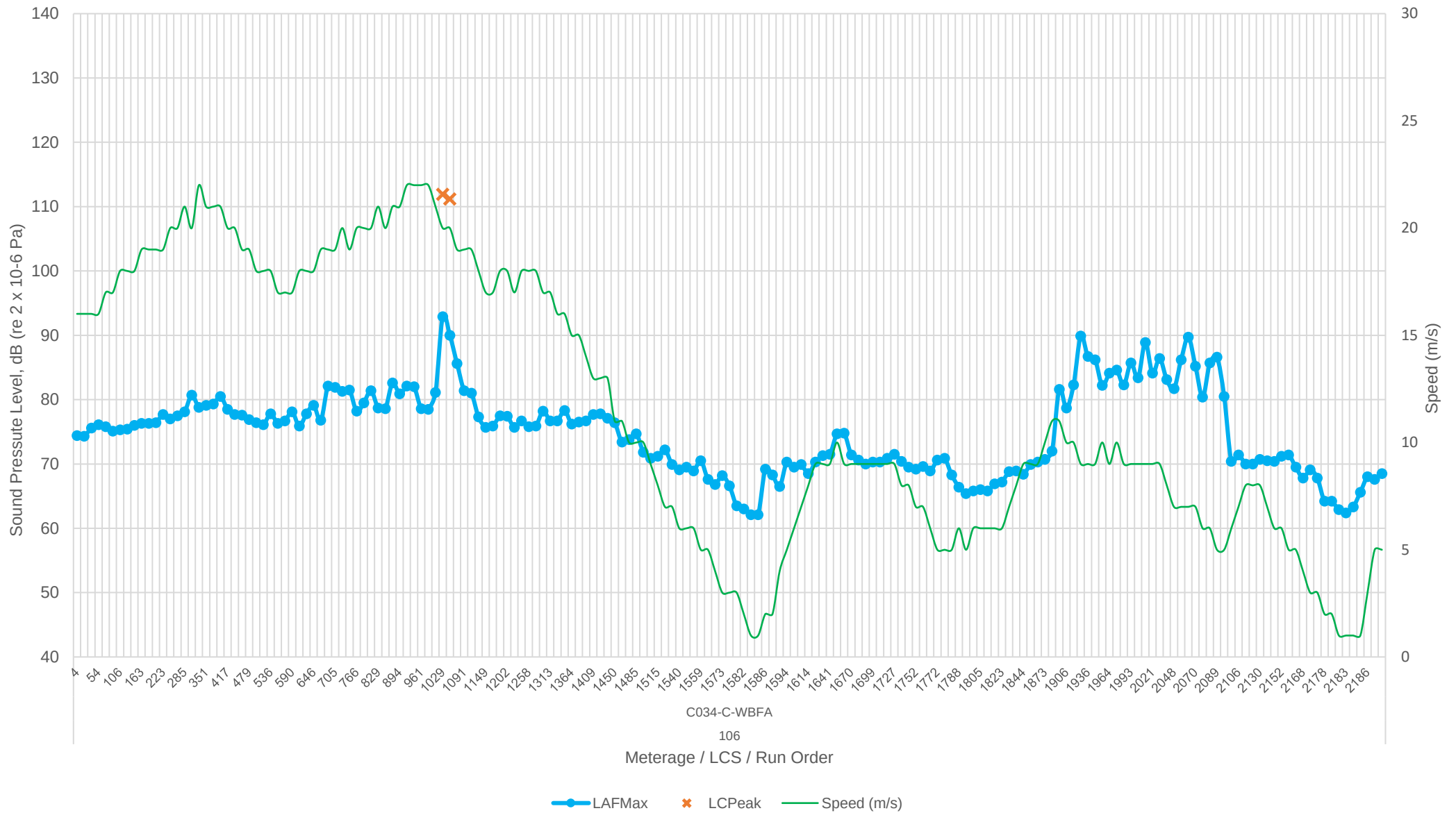


Figure C54 - Buckhurst Hill (WB) to Woodford (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

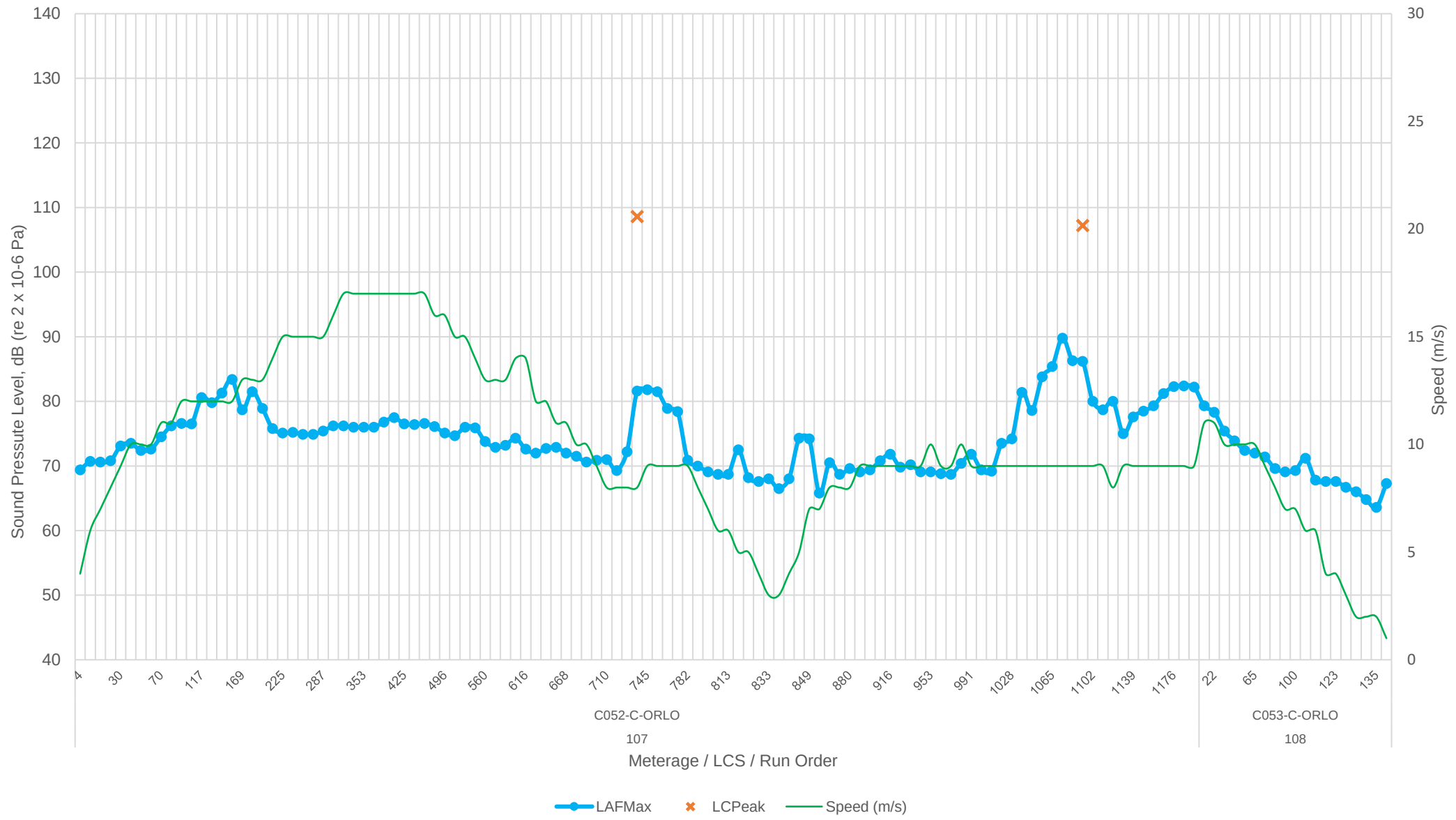
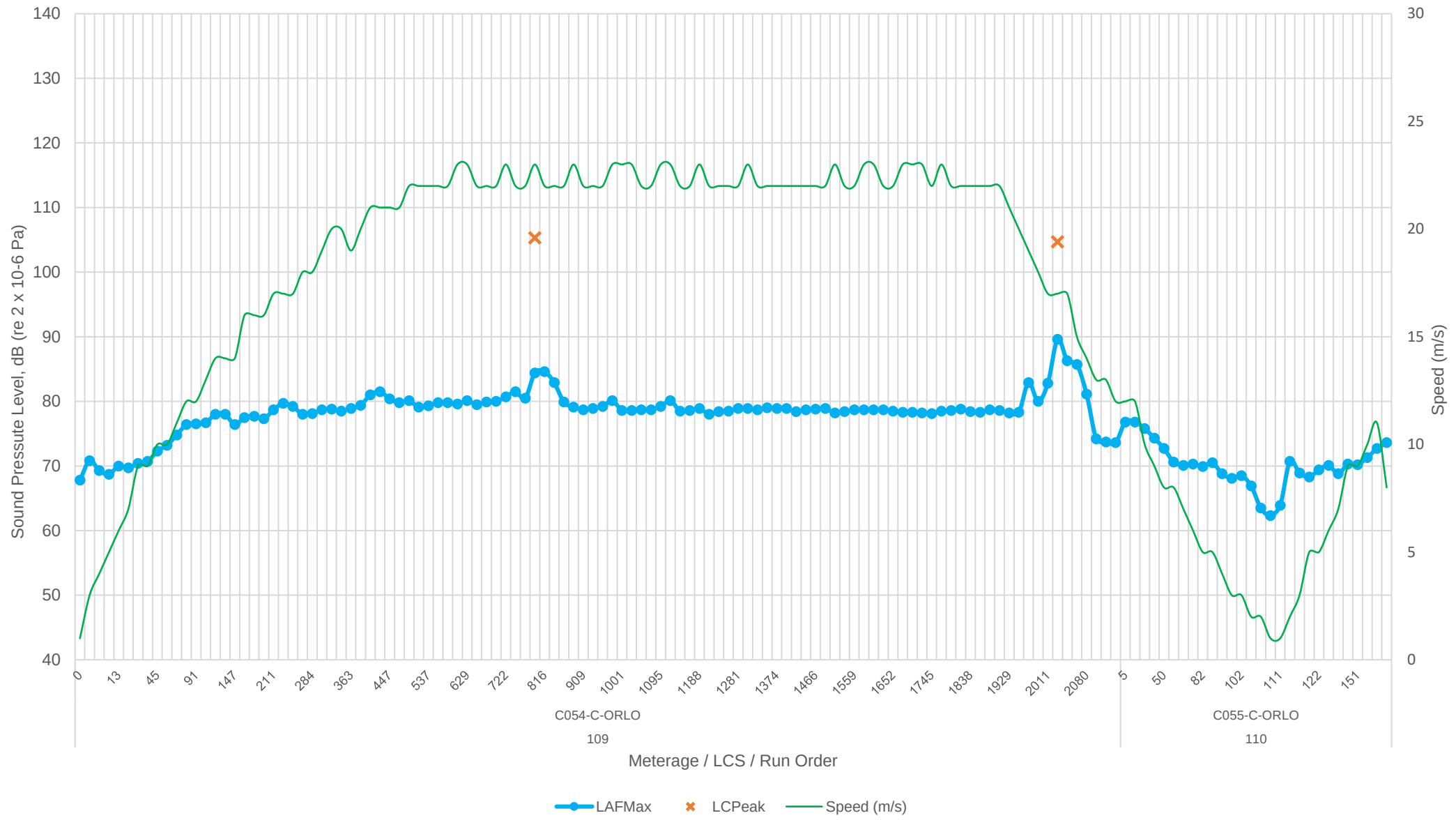


Figure C55 - Woodford (EB) to Roding Valley (OR)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle



Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

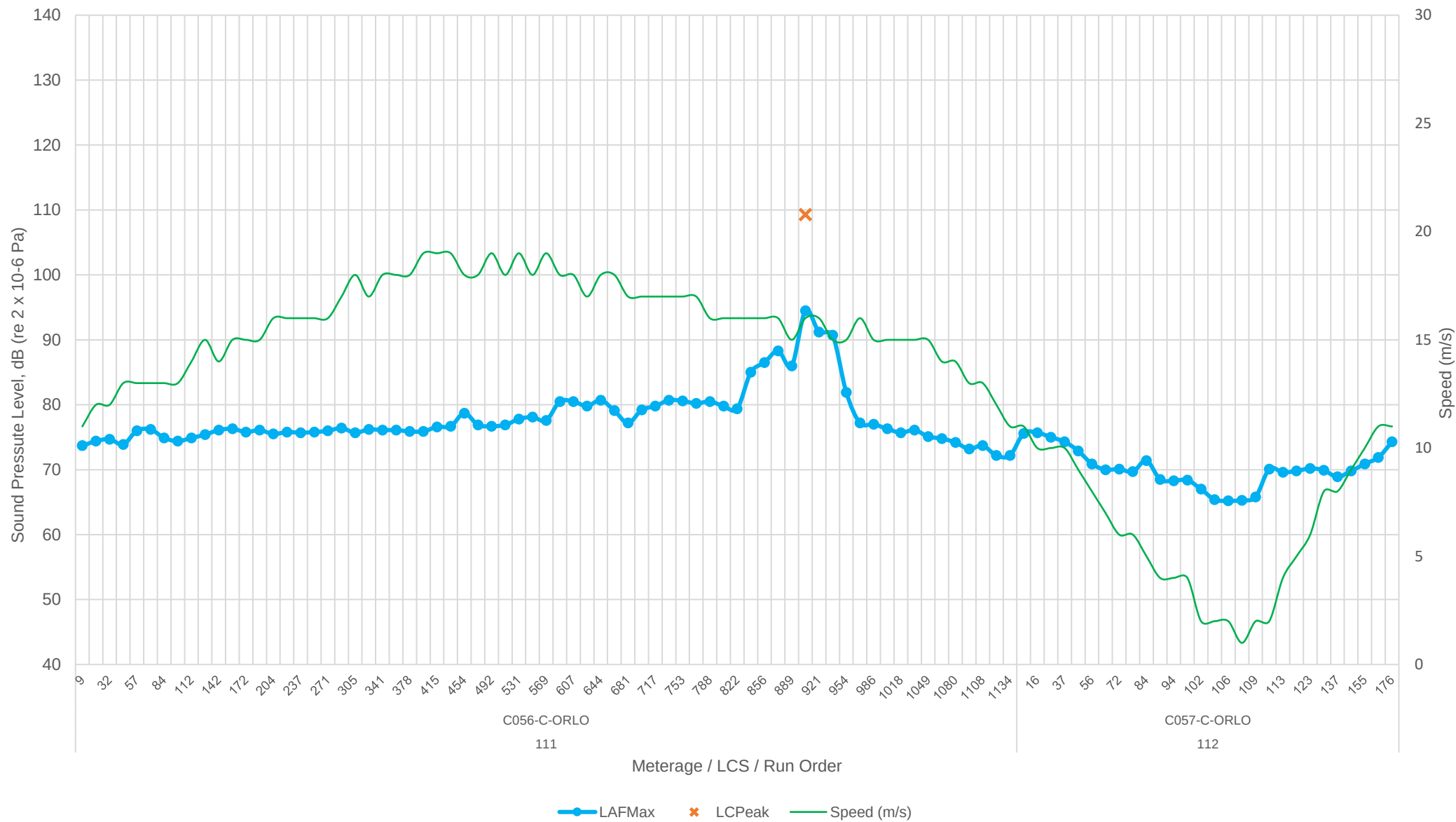


Figure C57 - Chigwell (OR) to Grange Hill (OR)

**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

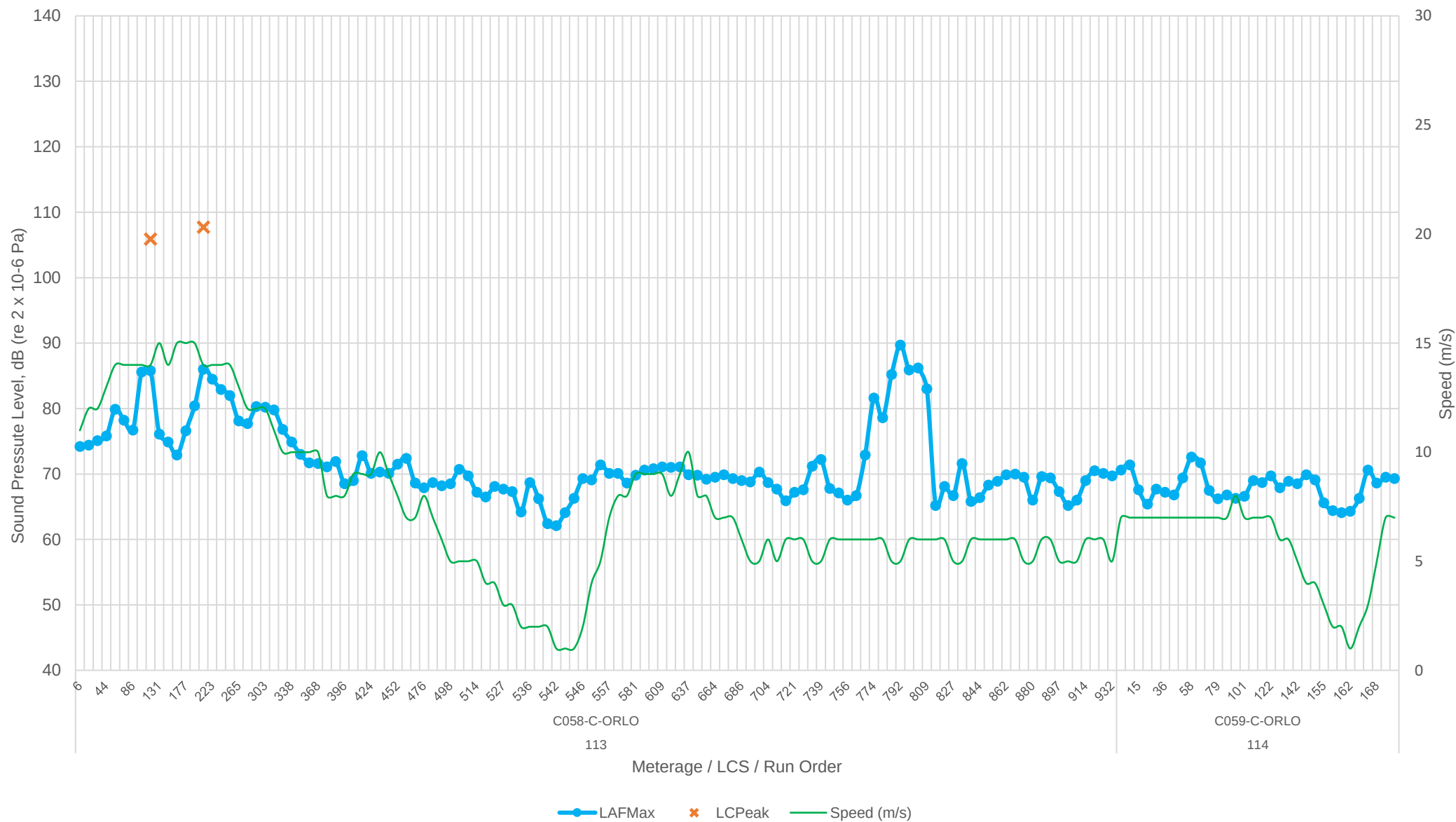


Figure C58 - Grange Hill (OR) to Hainault (OR)



Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

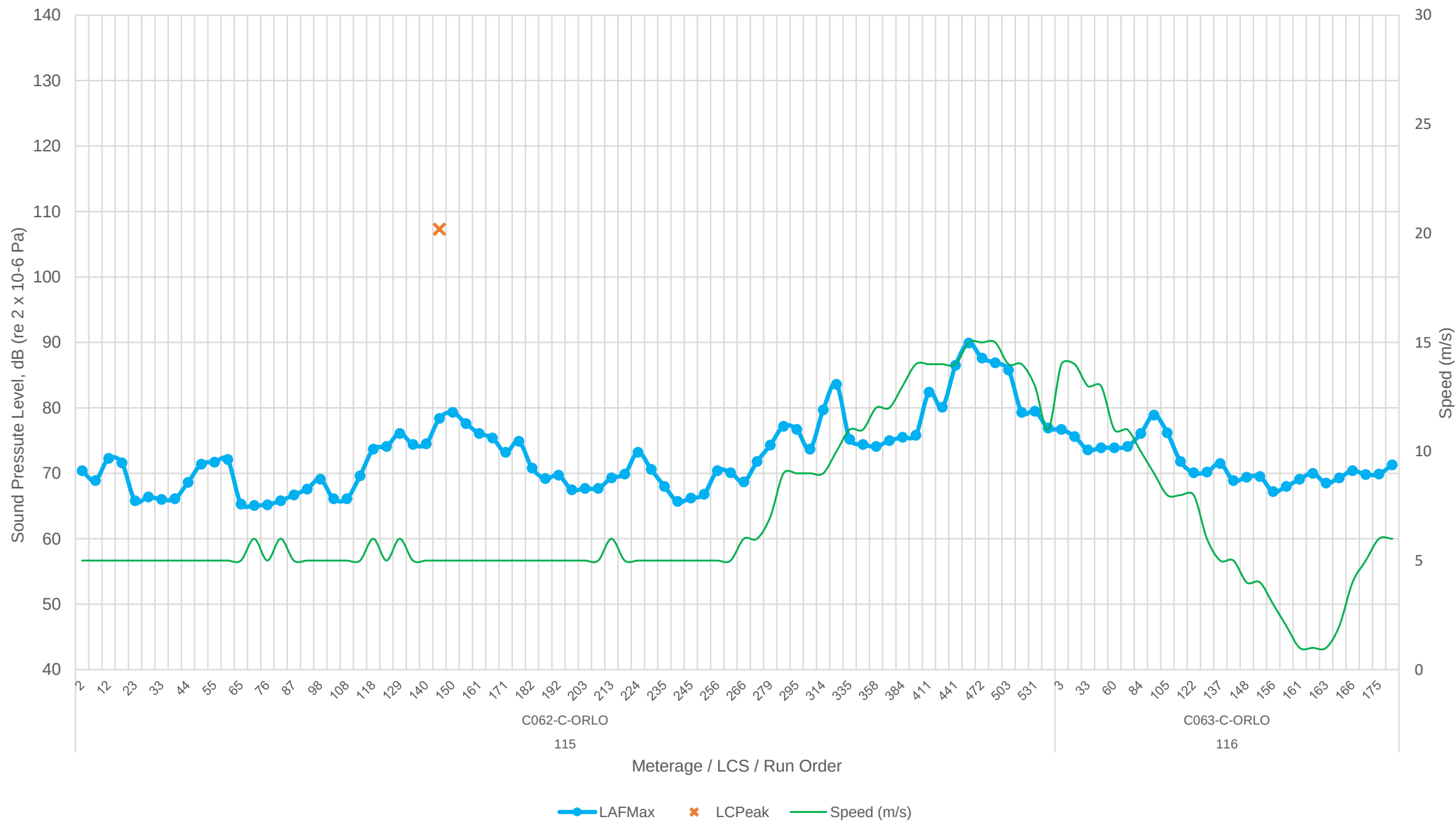


Figure C59 - Hainault (OR) to Fairlop (OR)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

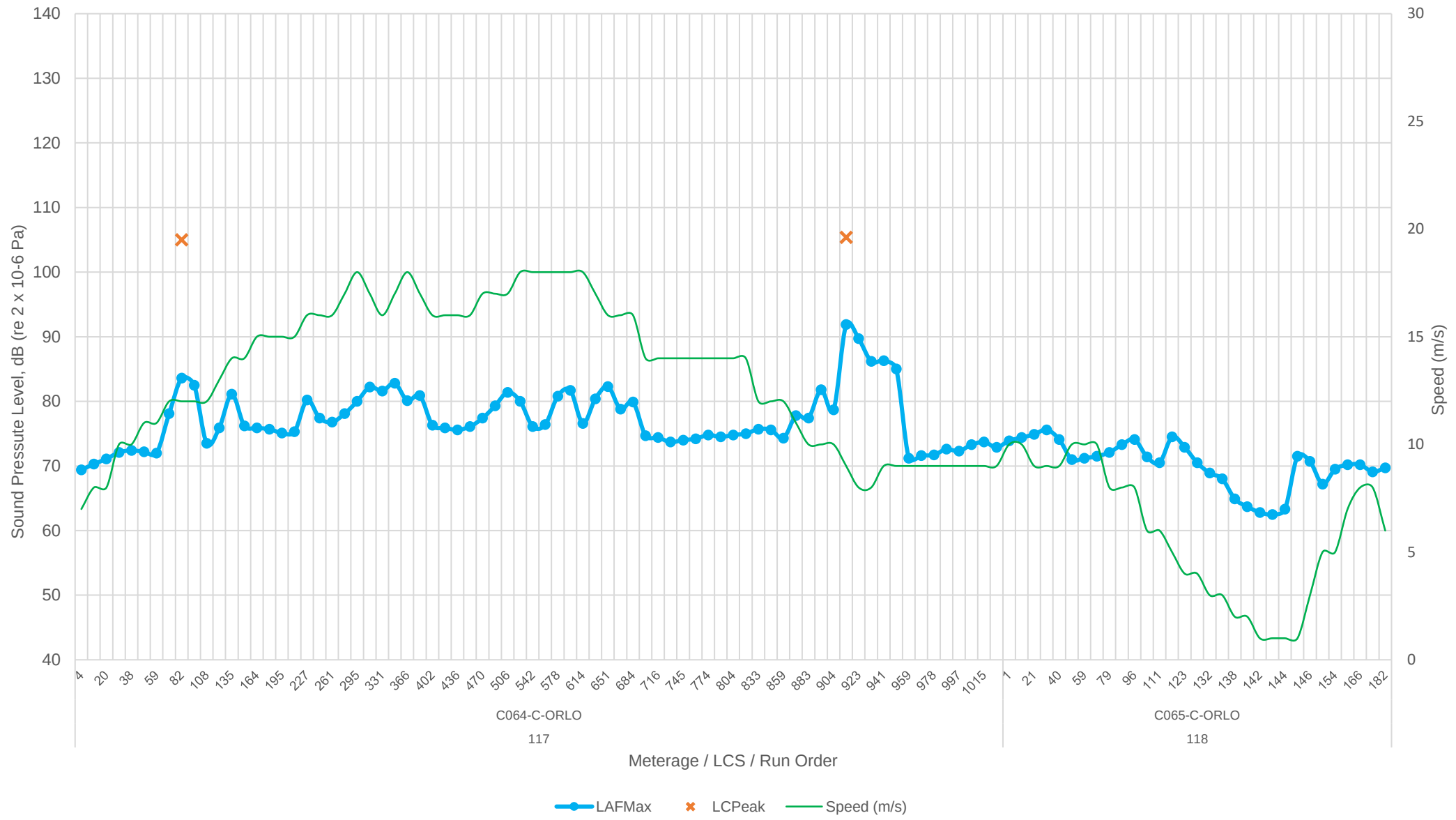


Figure C60 - Fairlop (OR) to Barkingside (OR)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

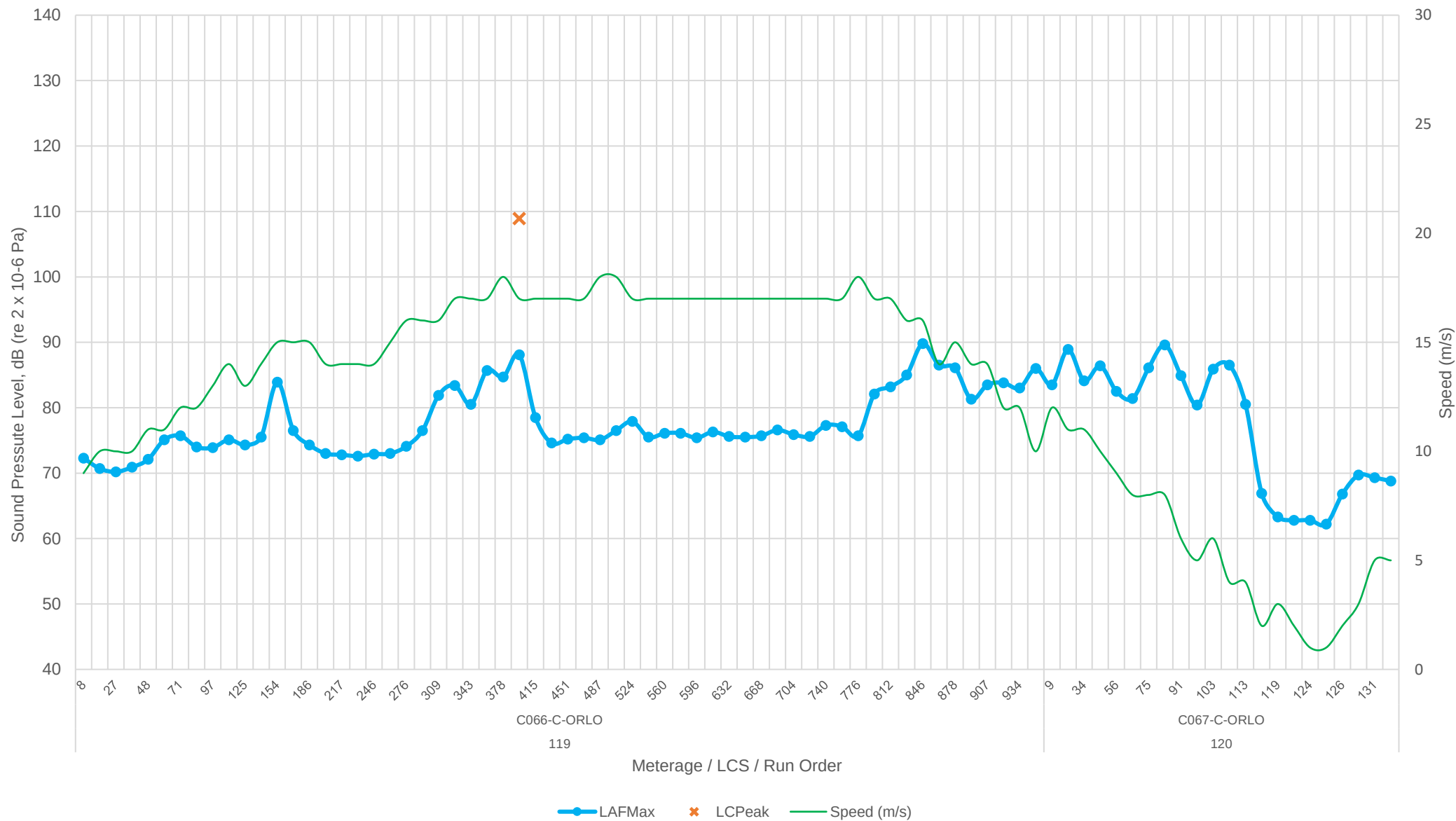


Figure C61 - Barkingside (OR) to Newbury Park (OR)

**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

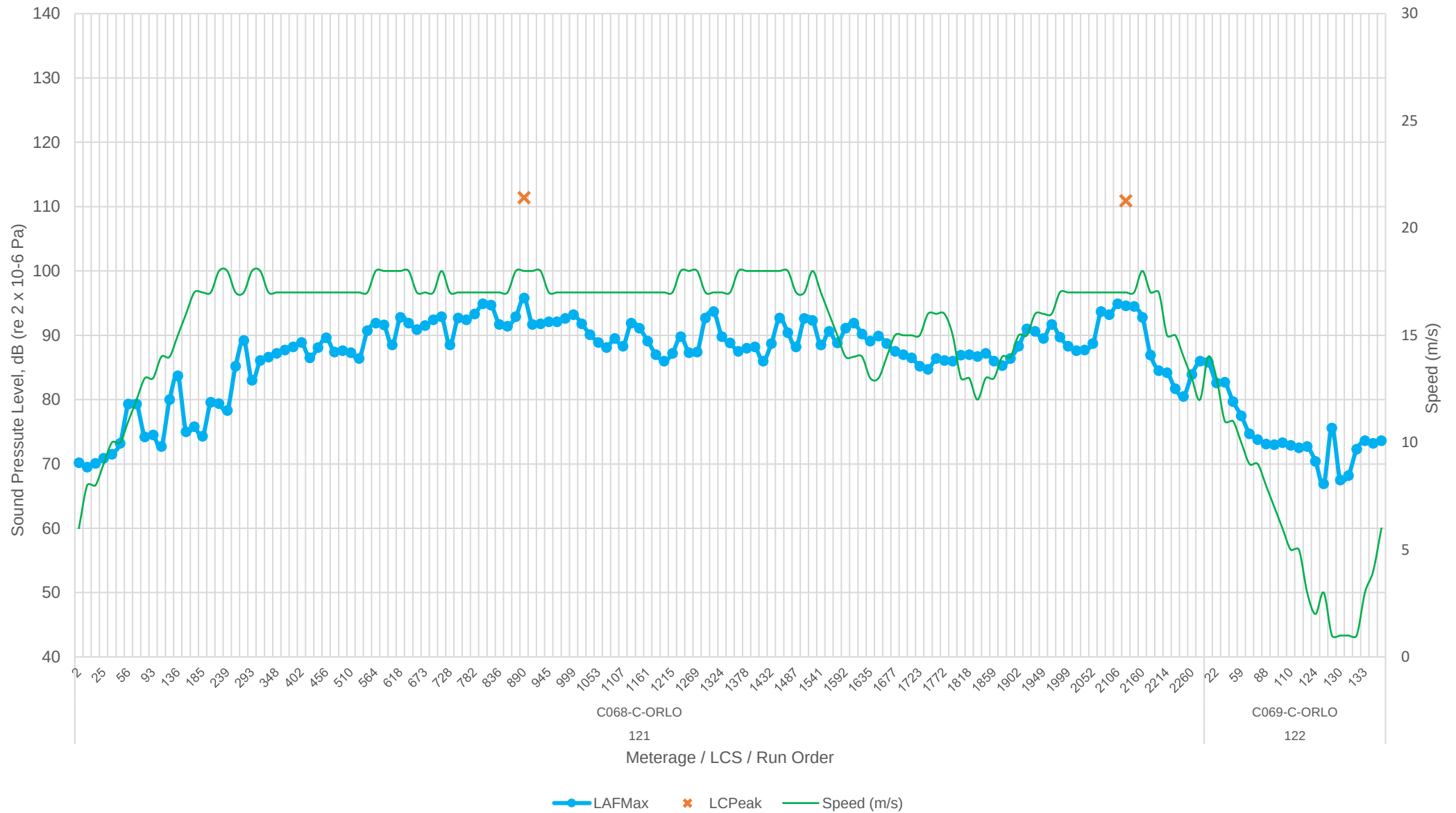


Figure C62 - Newbury Park (OR) to Gants Hill (OR)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

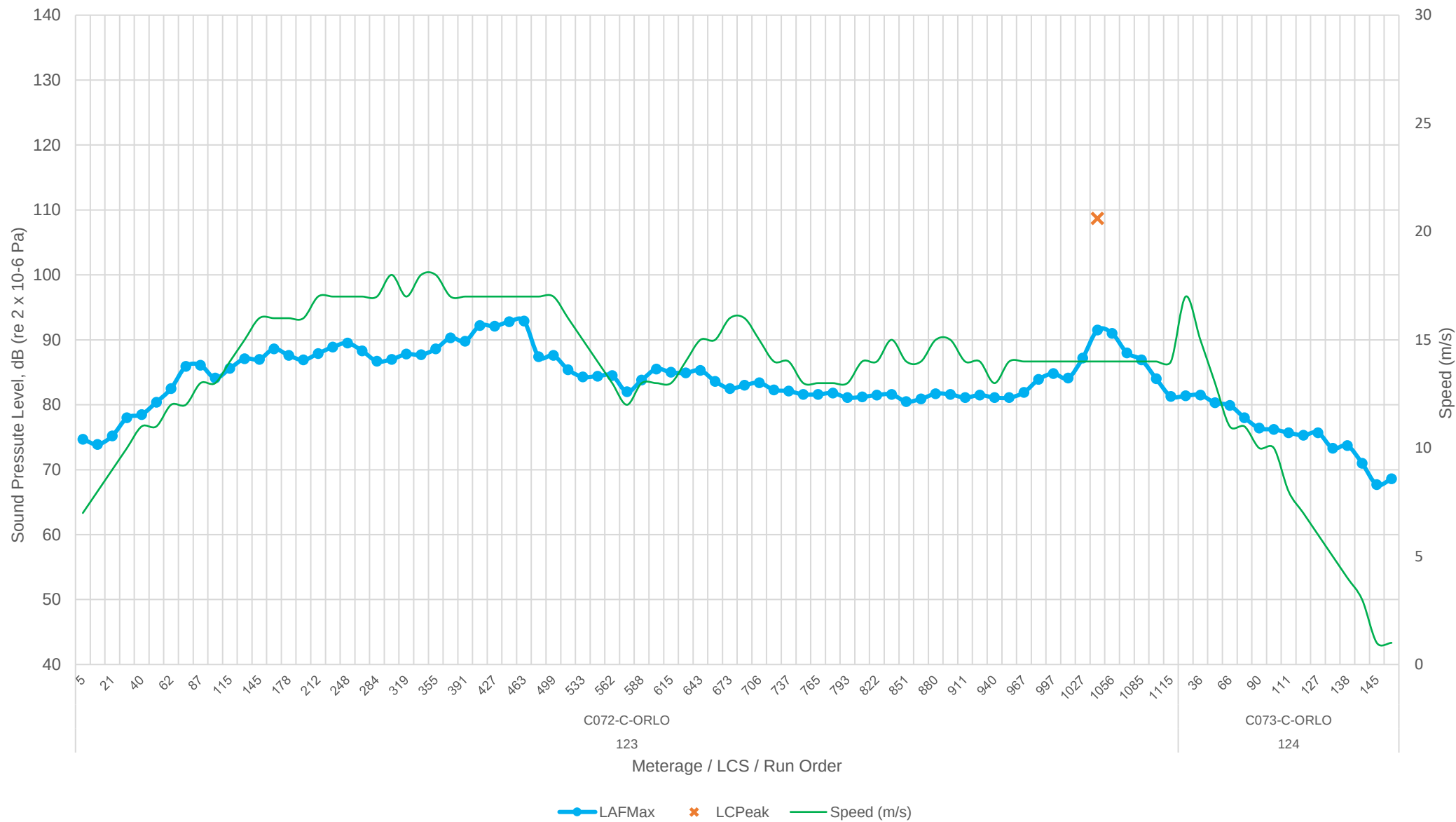


Figure C63 - Gants Hill (OR) to Redbridge (OR)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

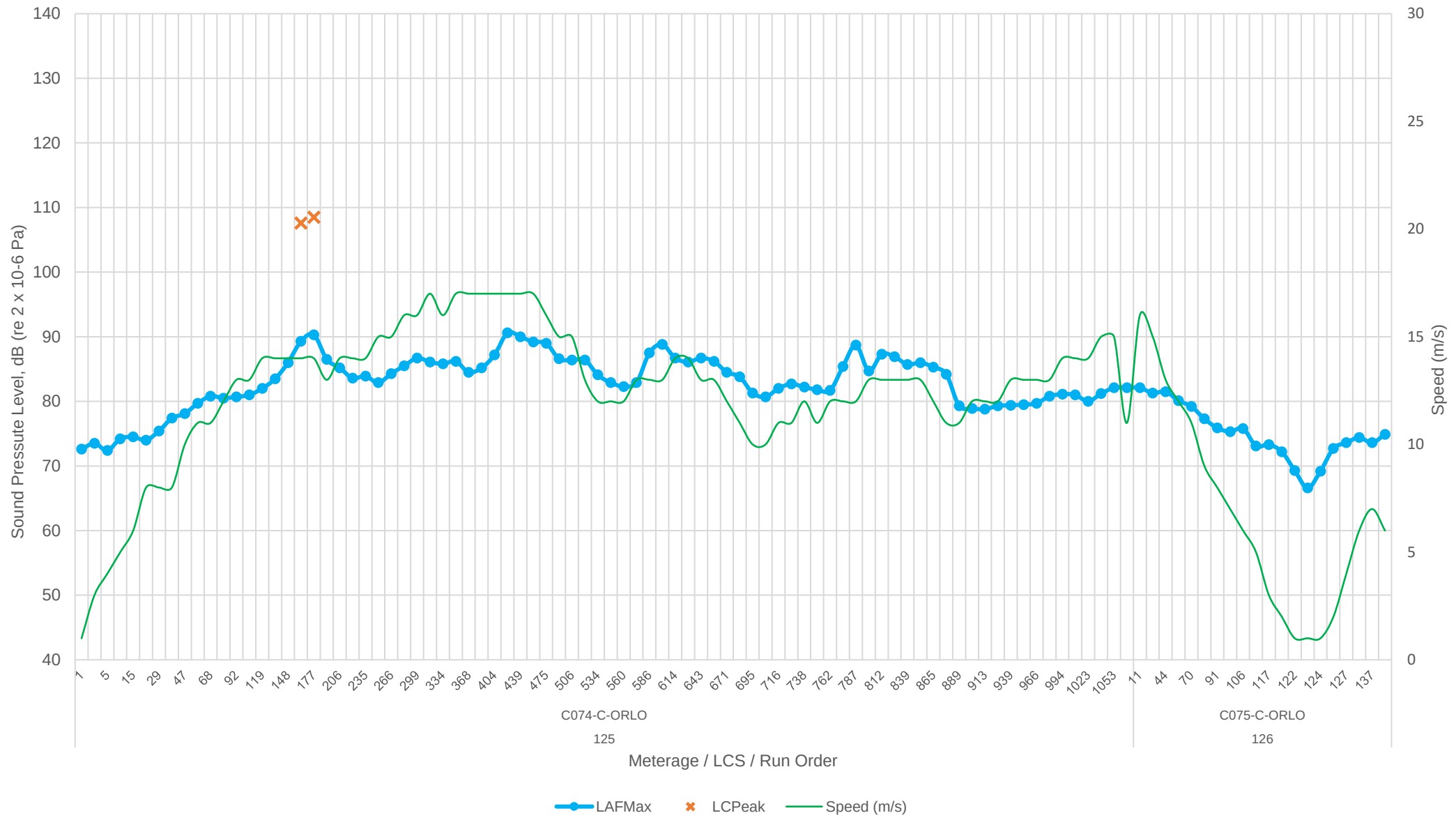


Figure C64 - Redbridge (OR) to Wanstead (OR)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

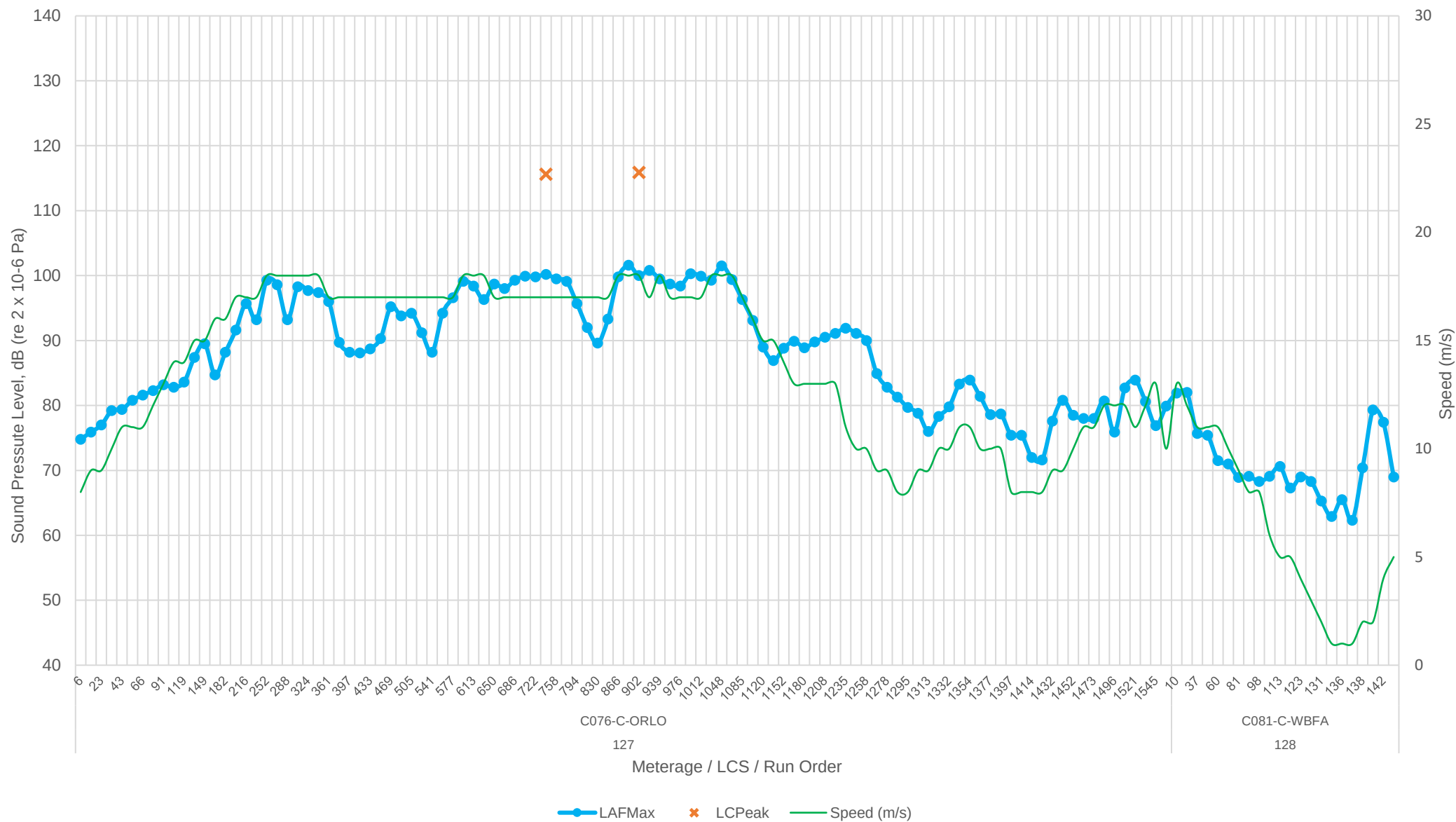
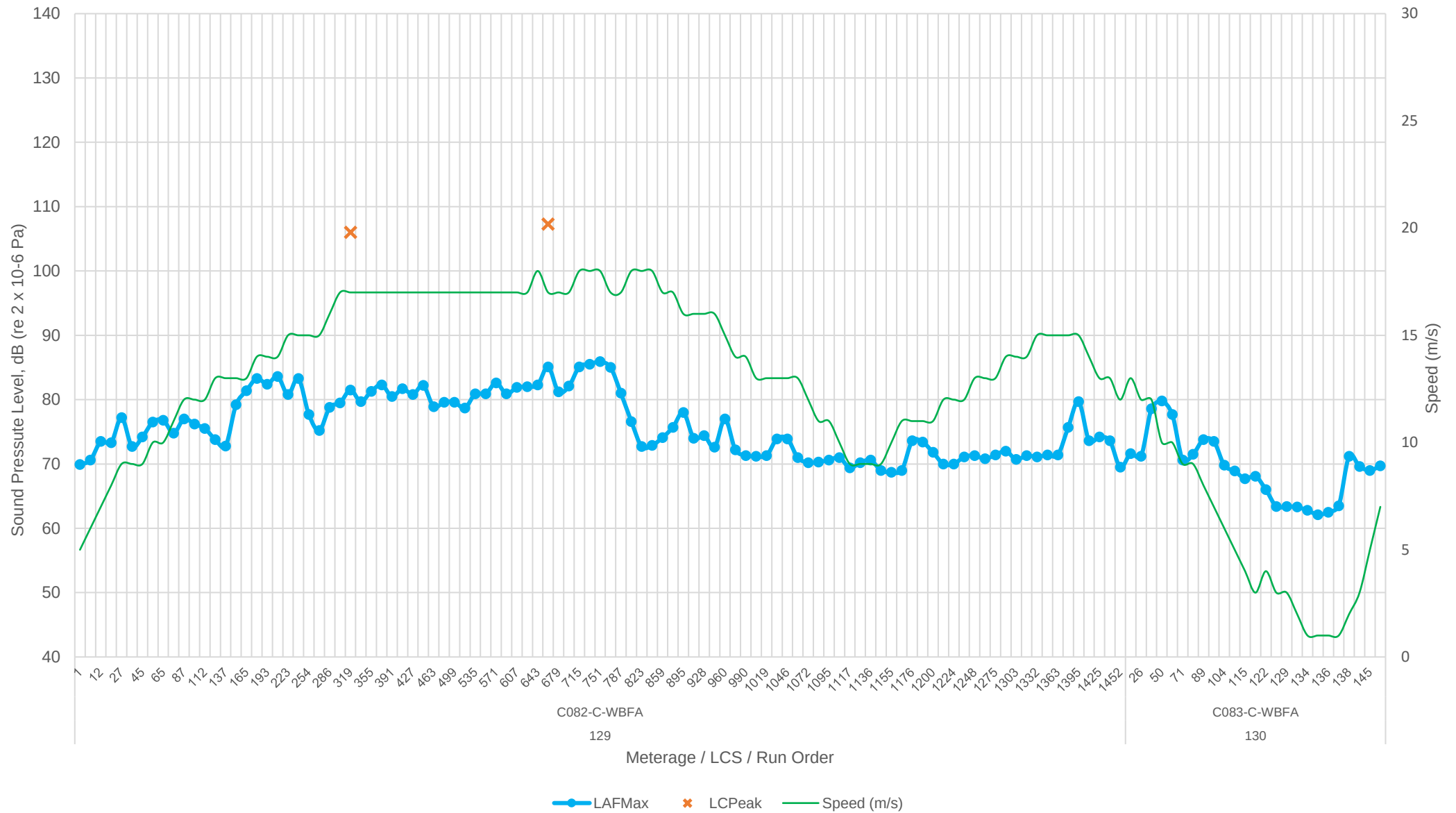


Figure C65 - Wanstead (OR) to Leytonstone (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle





**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

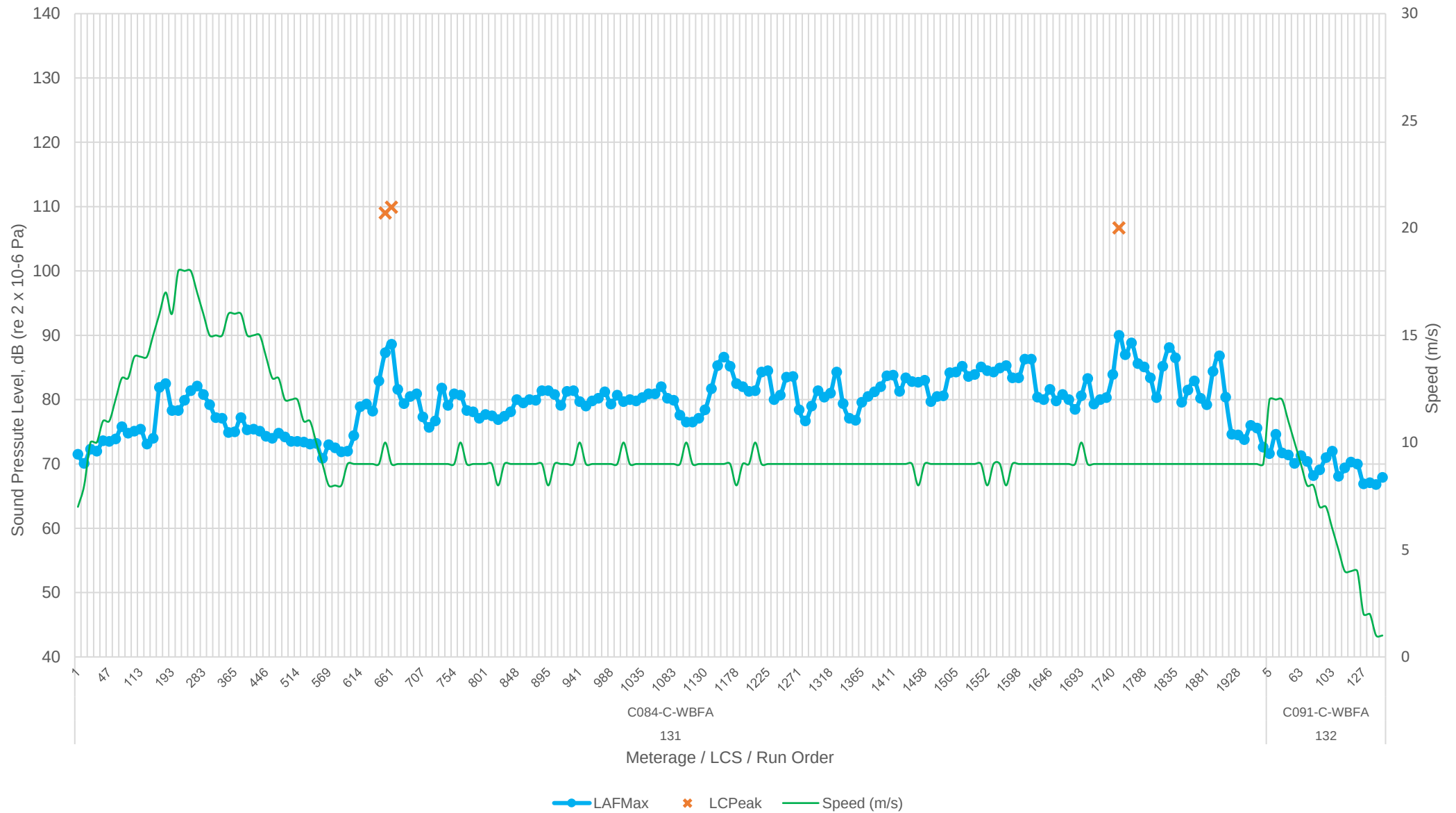


Figure C67 - Leyton (WB) to Stratford (WB)

**Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle**

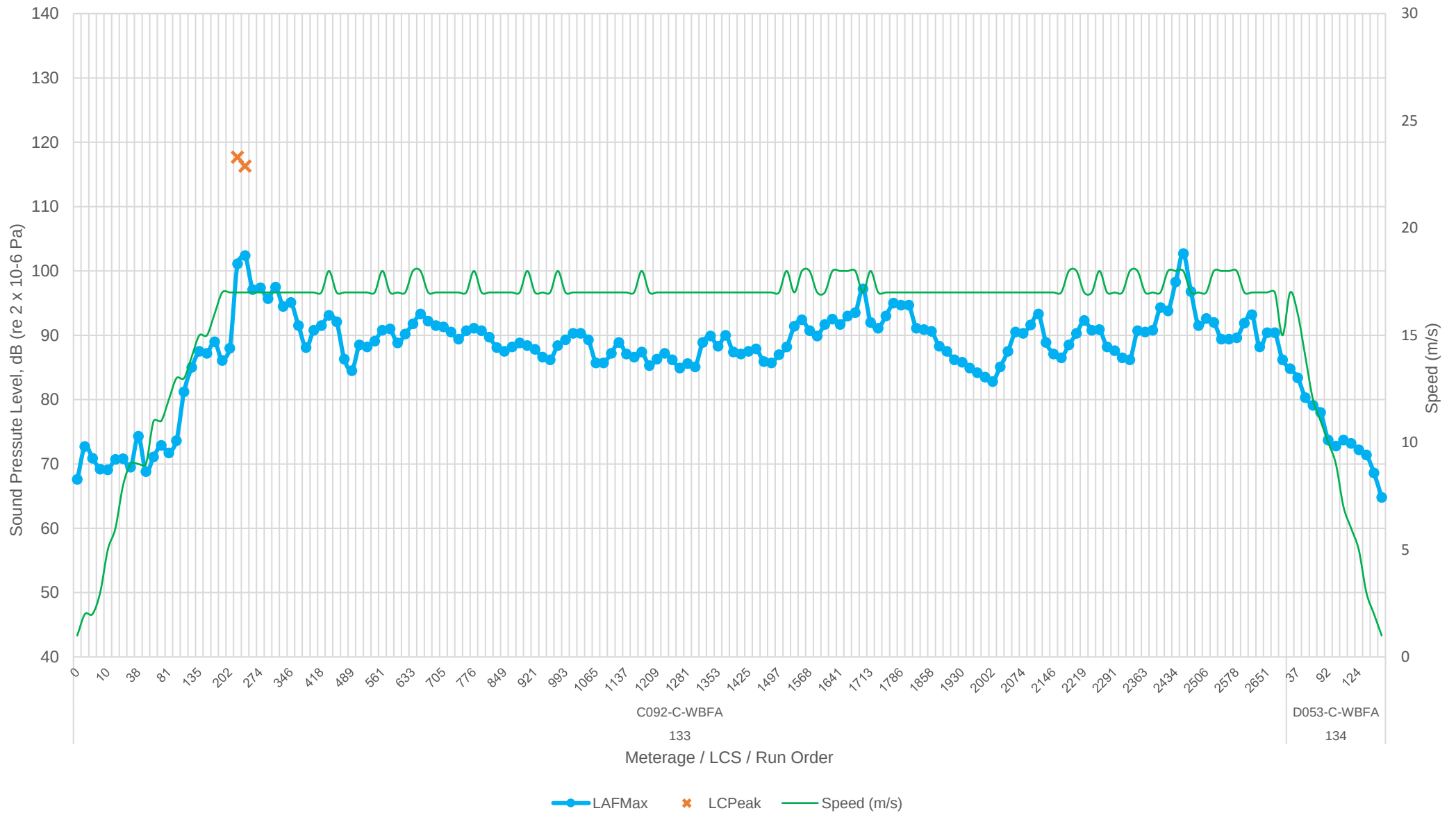


Figure C68 - Stratford (WB) to Mile End (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

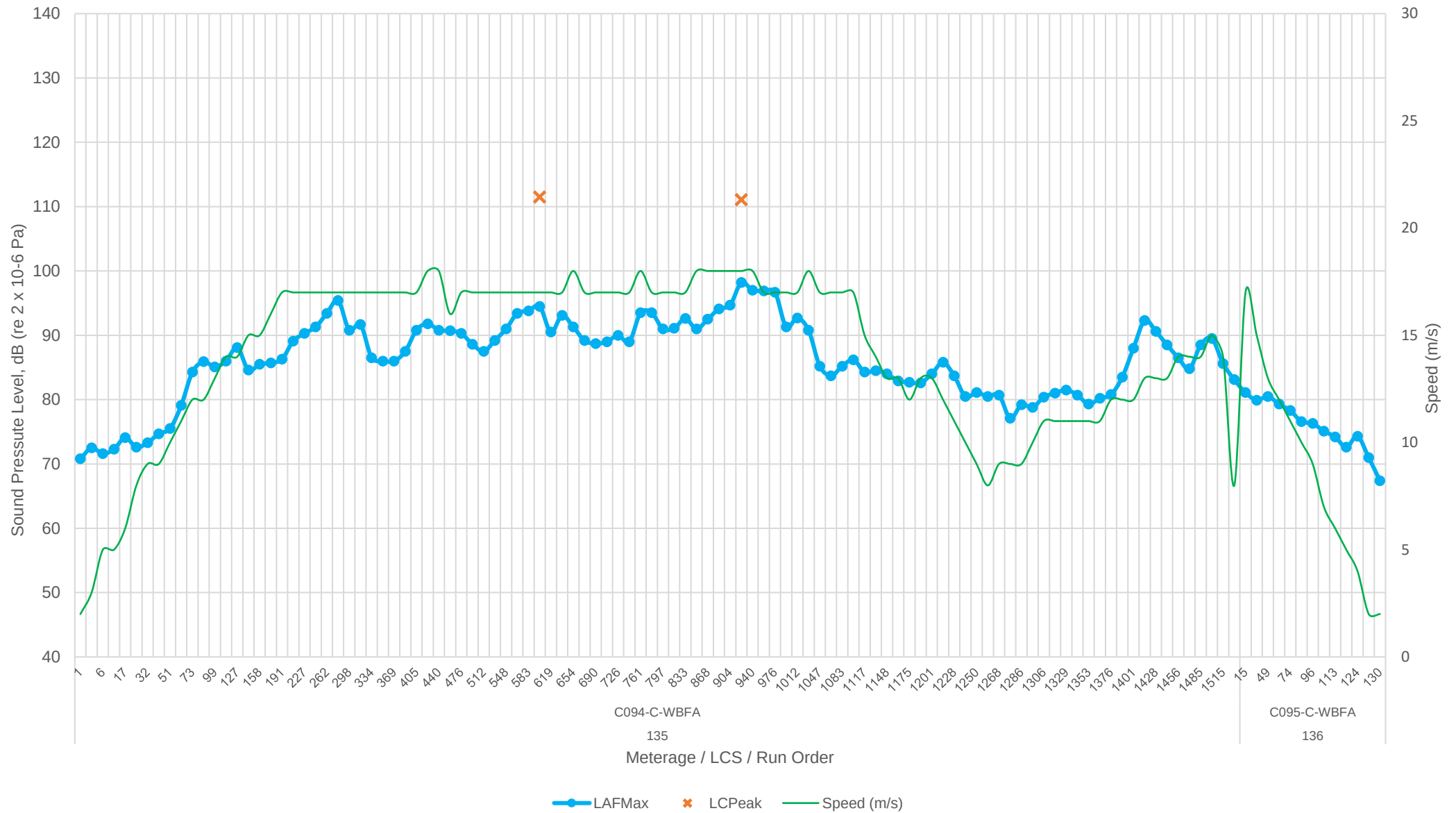


Figure C69 - Mile End (WB) to Bethnal Green (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

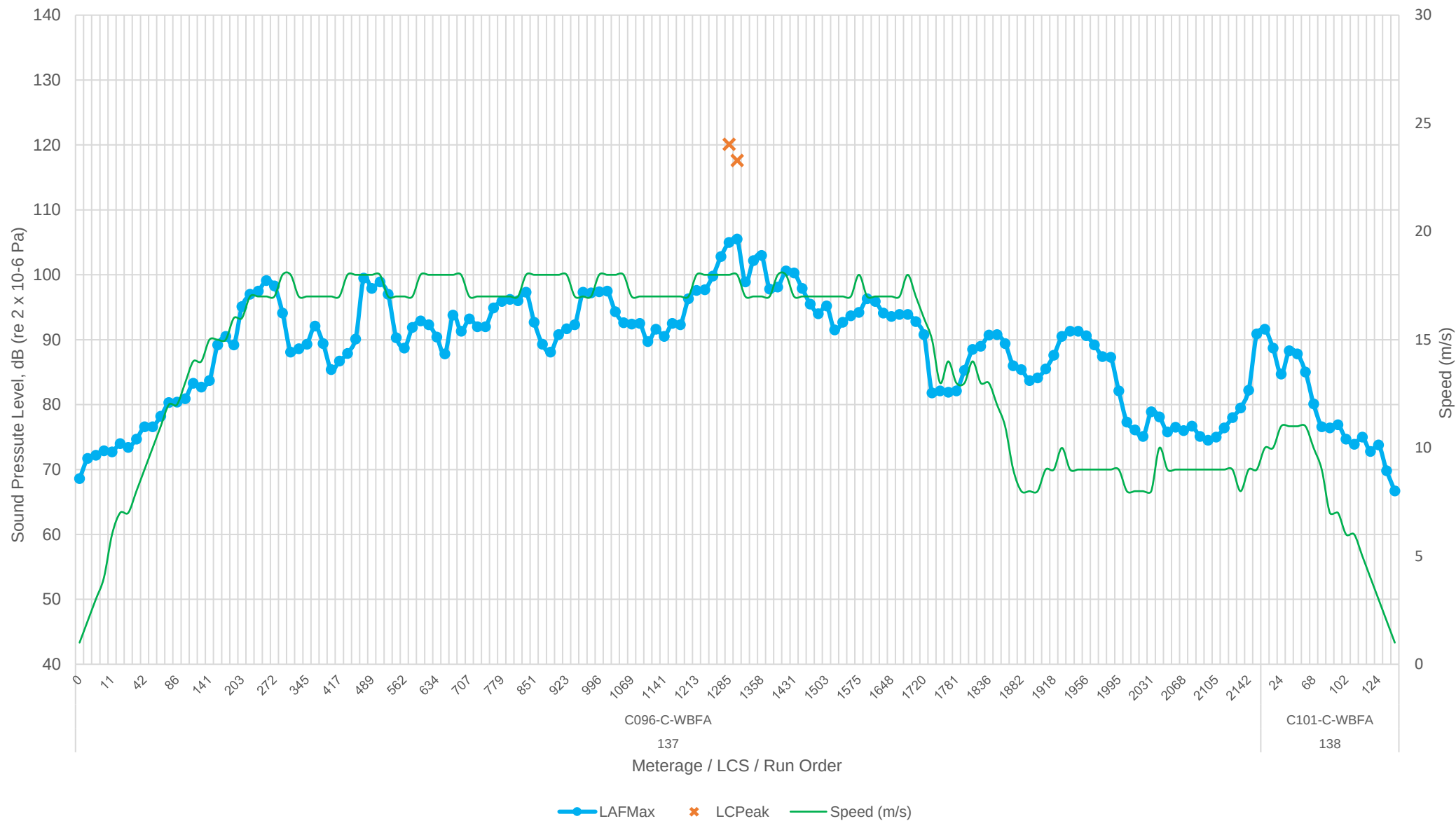


Figure C70 - Bethnal Green (WB) to Liverpool Street (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

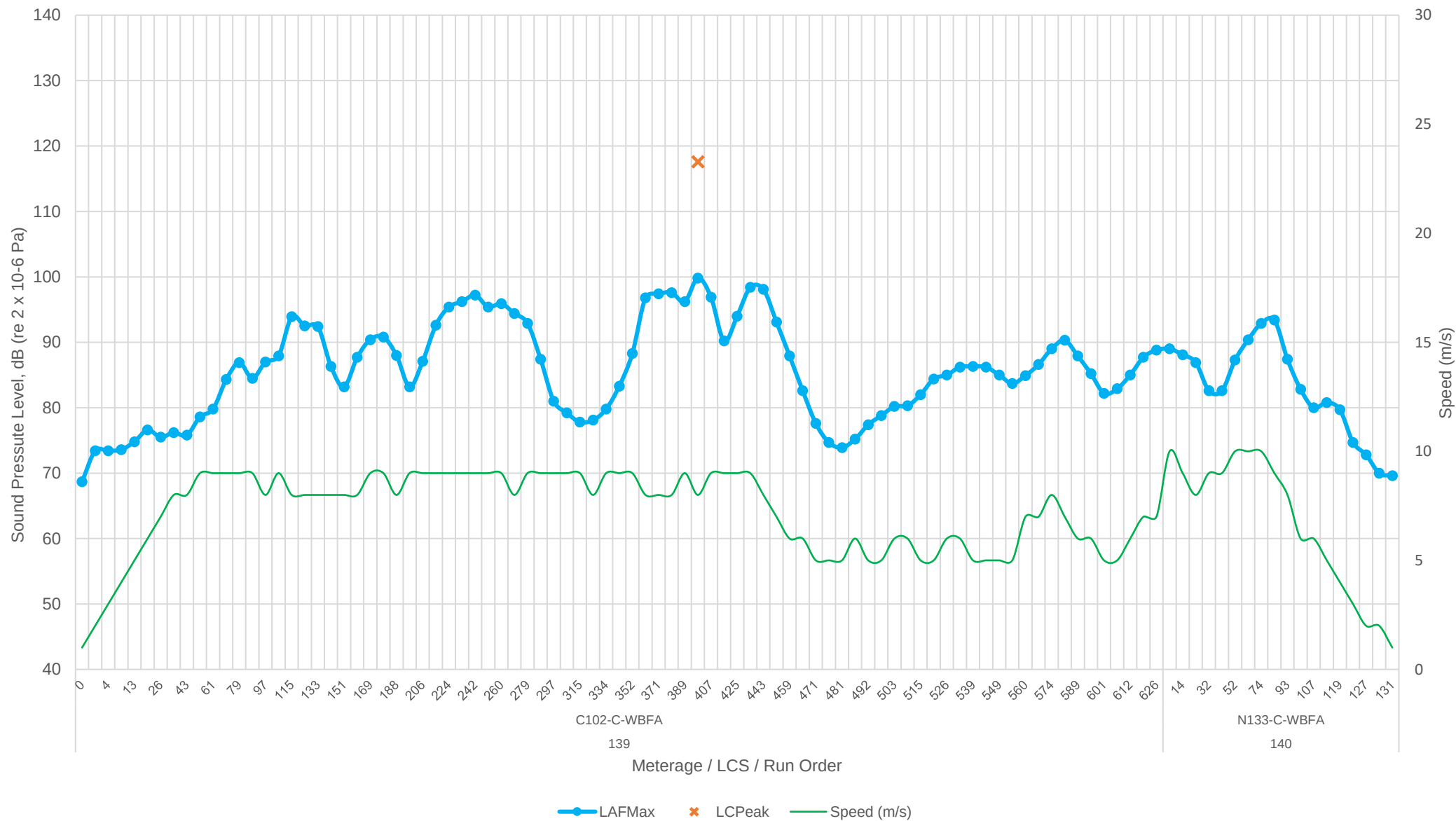


Figure C71 - Liverpool Street (WB) to Bank (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

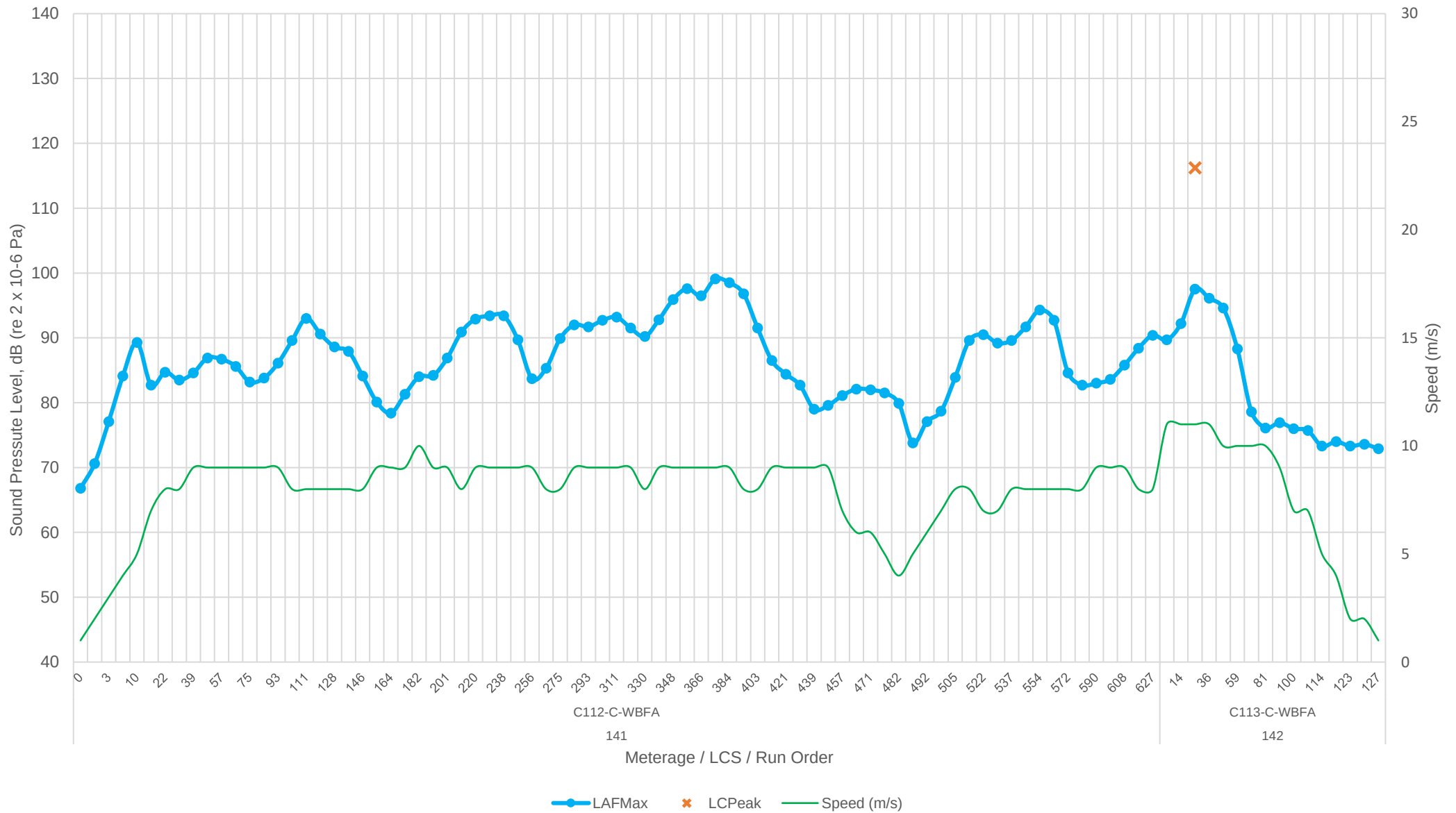


Figure C72 - Bank (WB) to St. Paul's (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

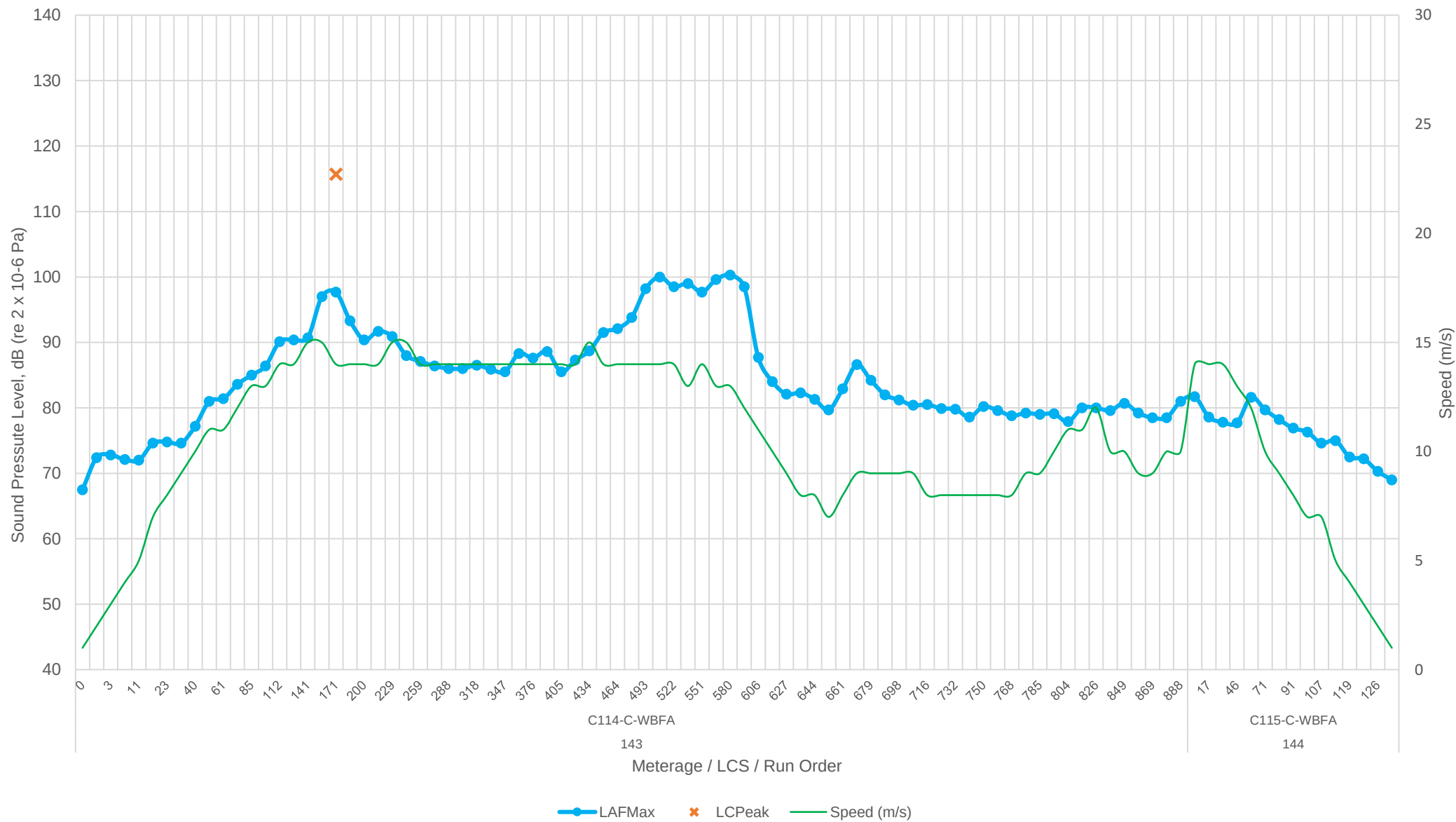


Figure C73 - St. Paul's (WB) to Chancery Lane (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

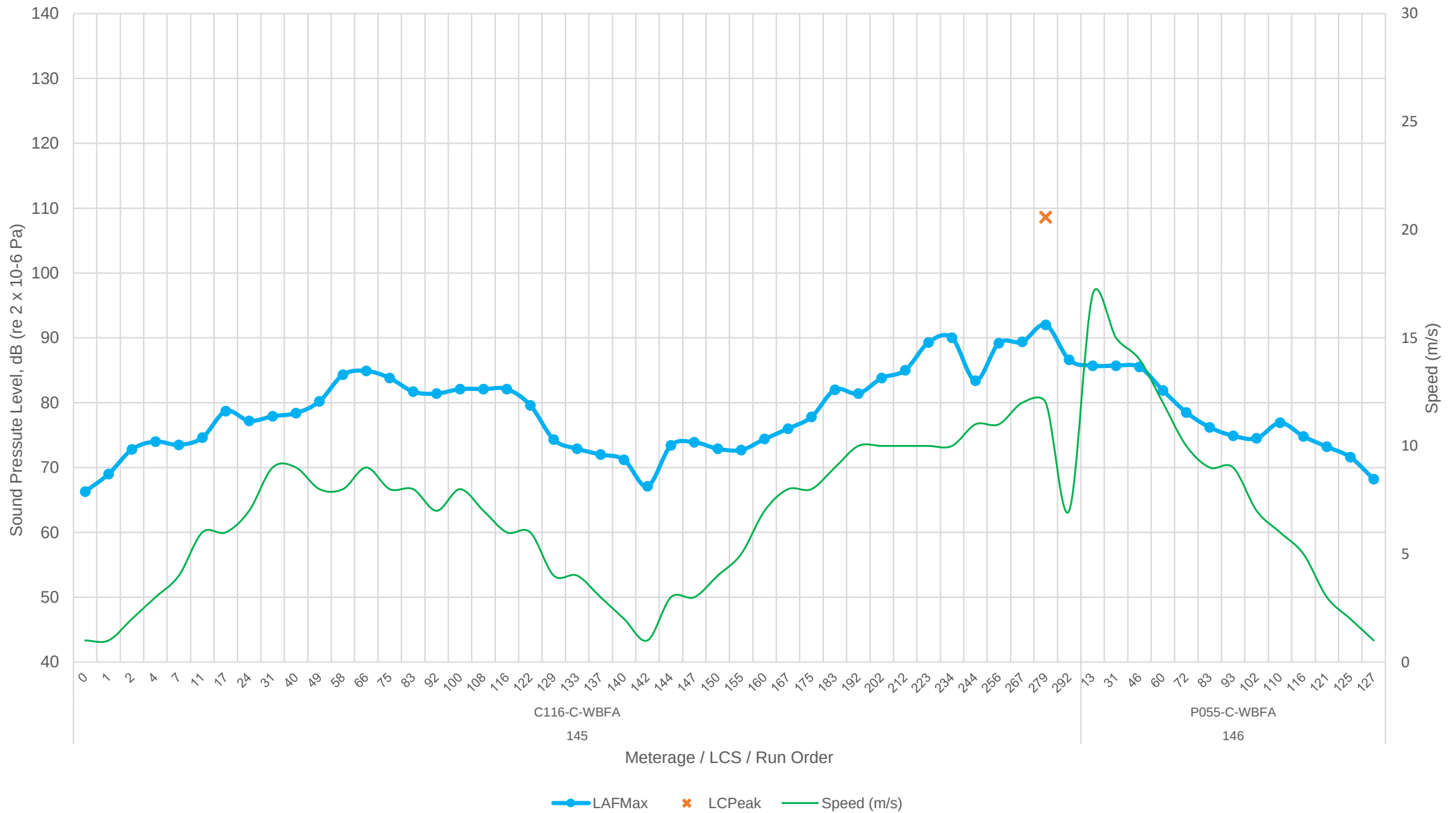


Figure C74 - Chancery Lane (WB) to Holborn (WB)



# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

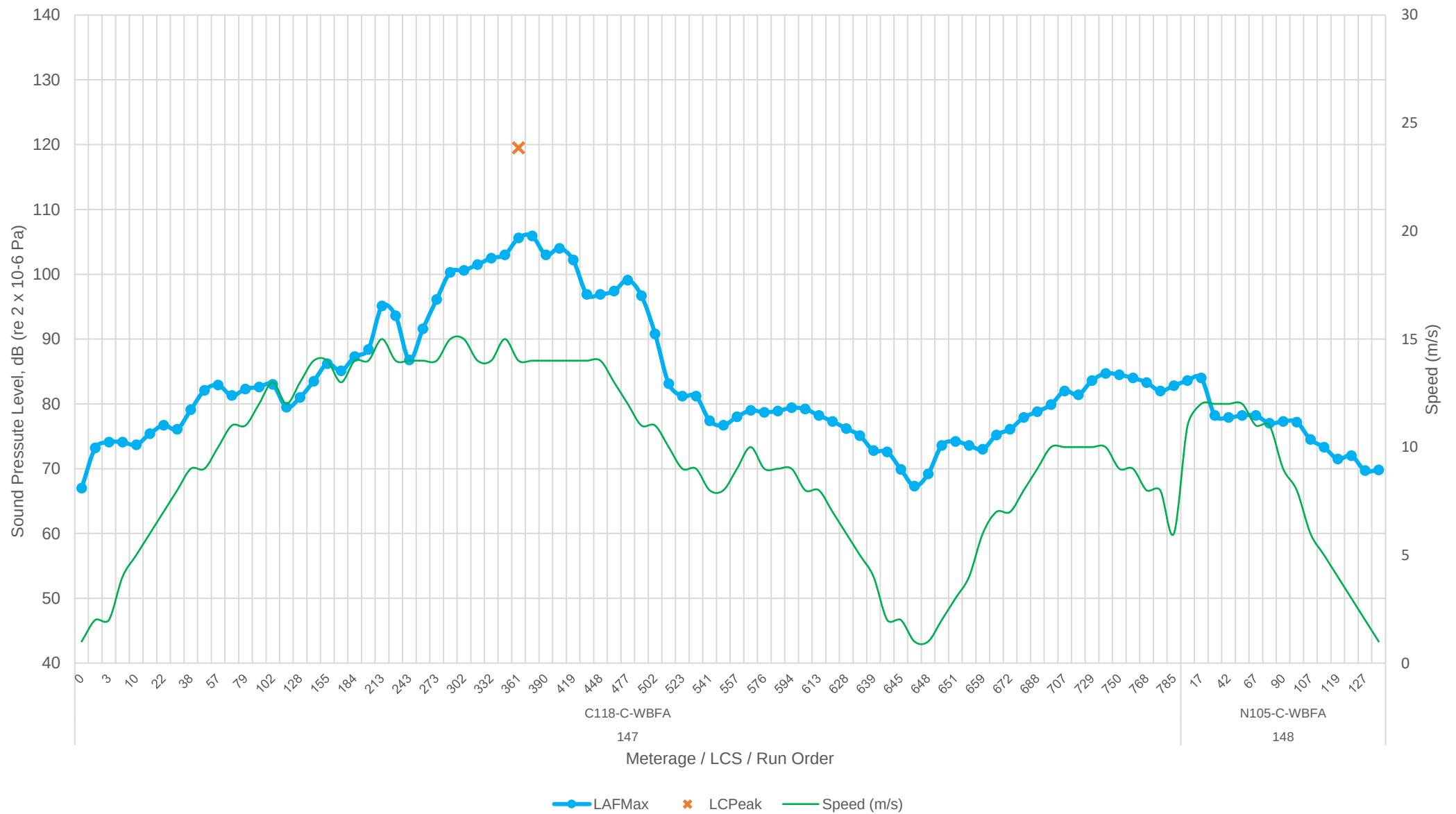


Figure C75 - Holborn (WB) to Tottenham Court Road (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

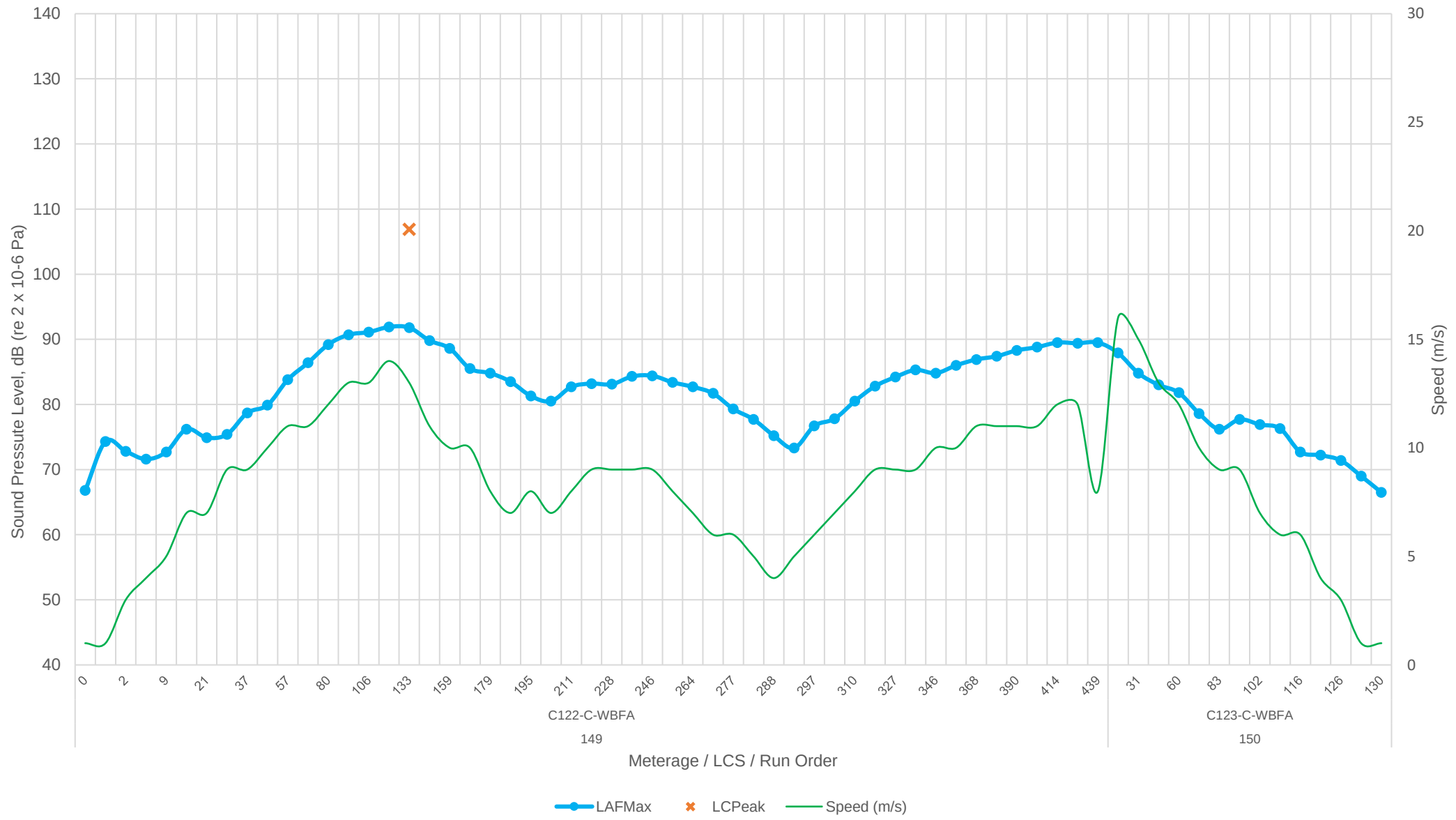


Figure C76 - Tottenham Court Road (WB) to Oxford Circus (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

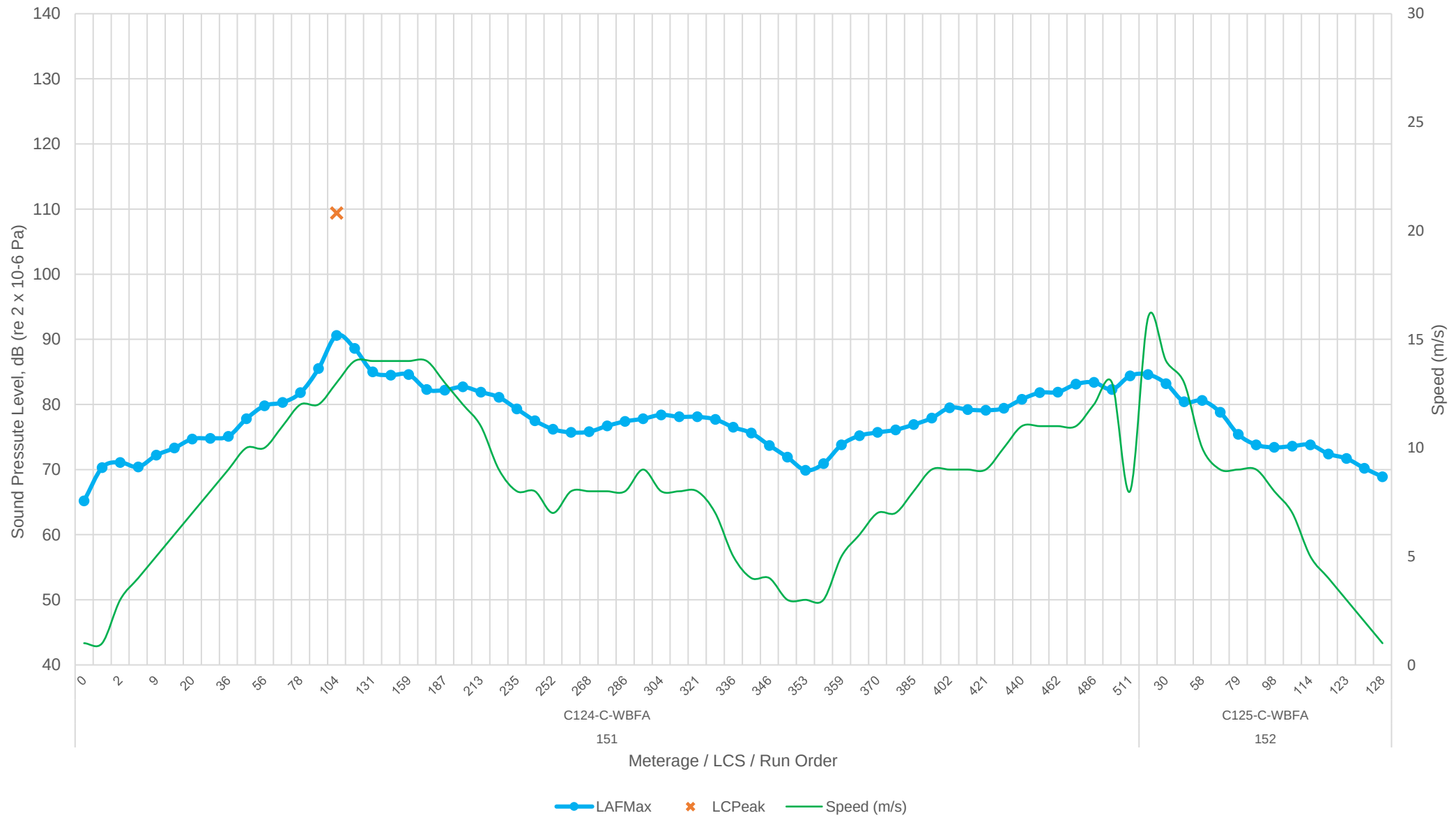


Figure C77 - Oxford Circus (WB) to Bond Street (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

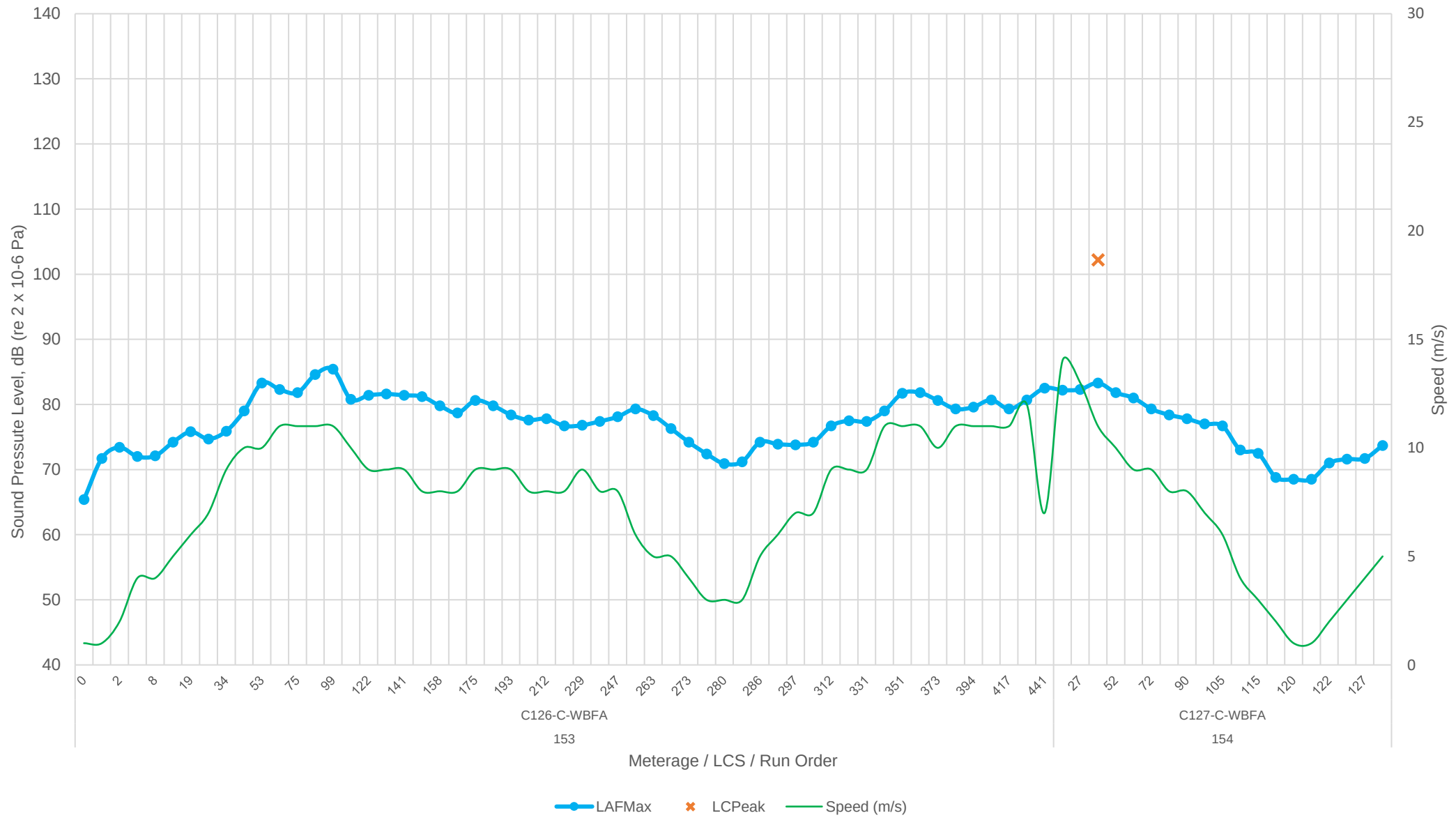


Figure C78 - Bond Street (WB) to Marble Arch (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

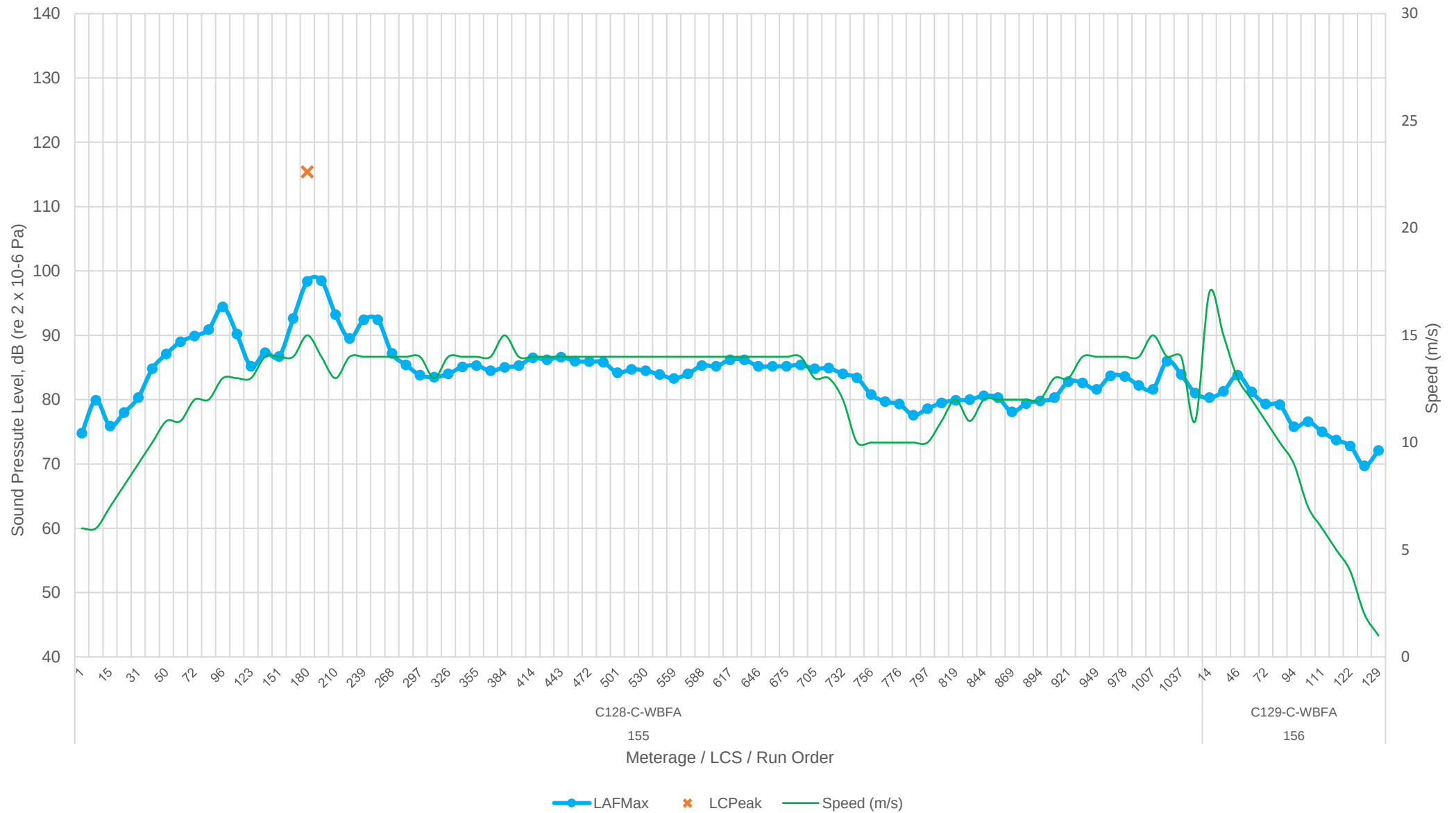


Figure C79 - Marble Arch (WB) to Lancaster Gate (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

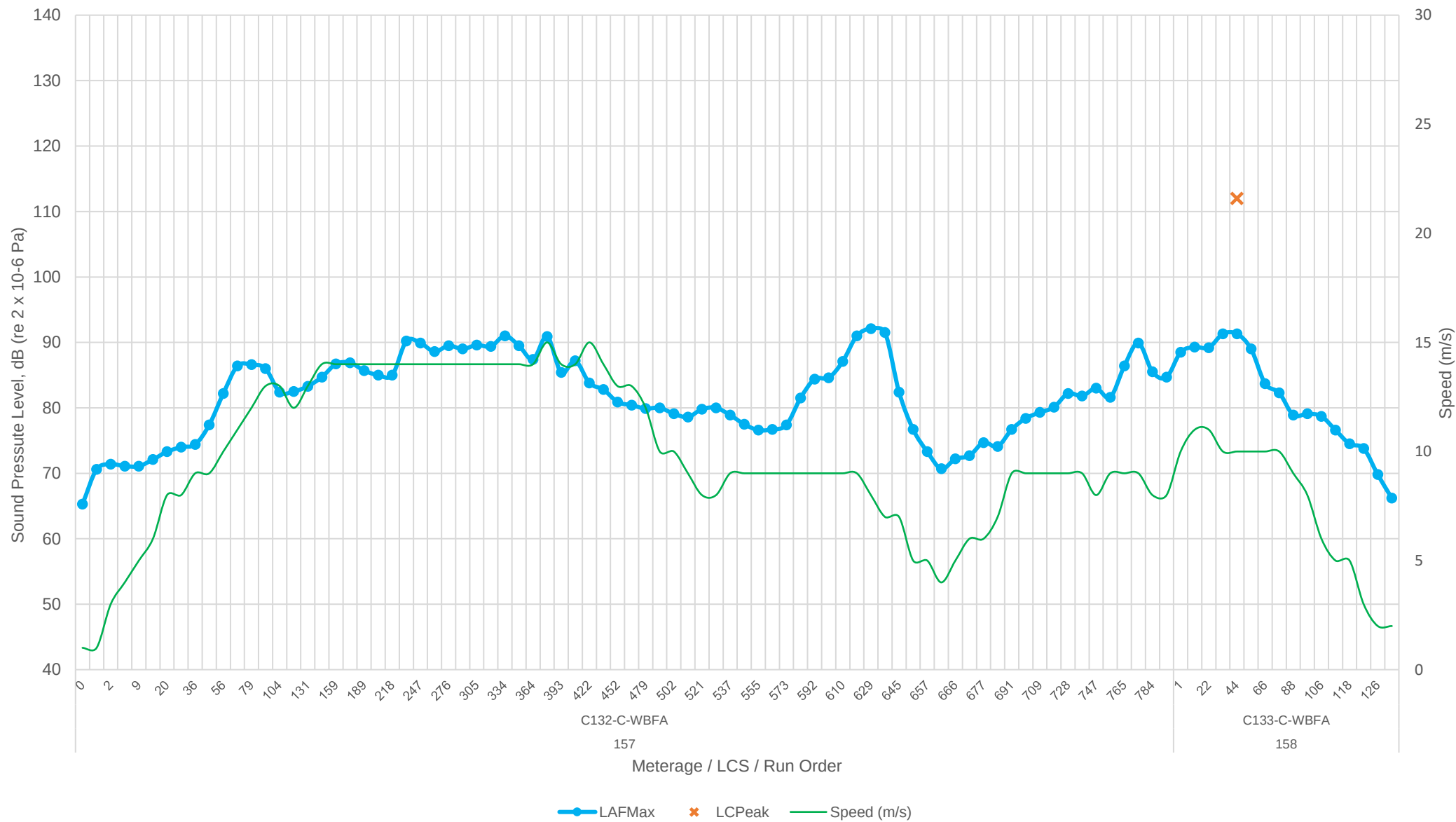


Figure C80 - Lancaster Gate (WB) to Queensway (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

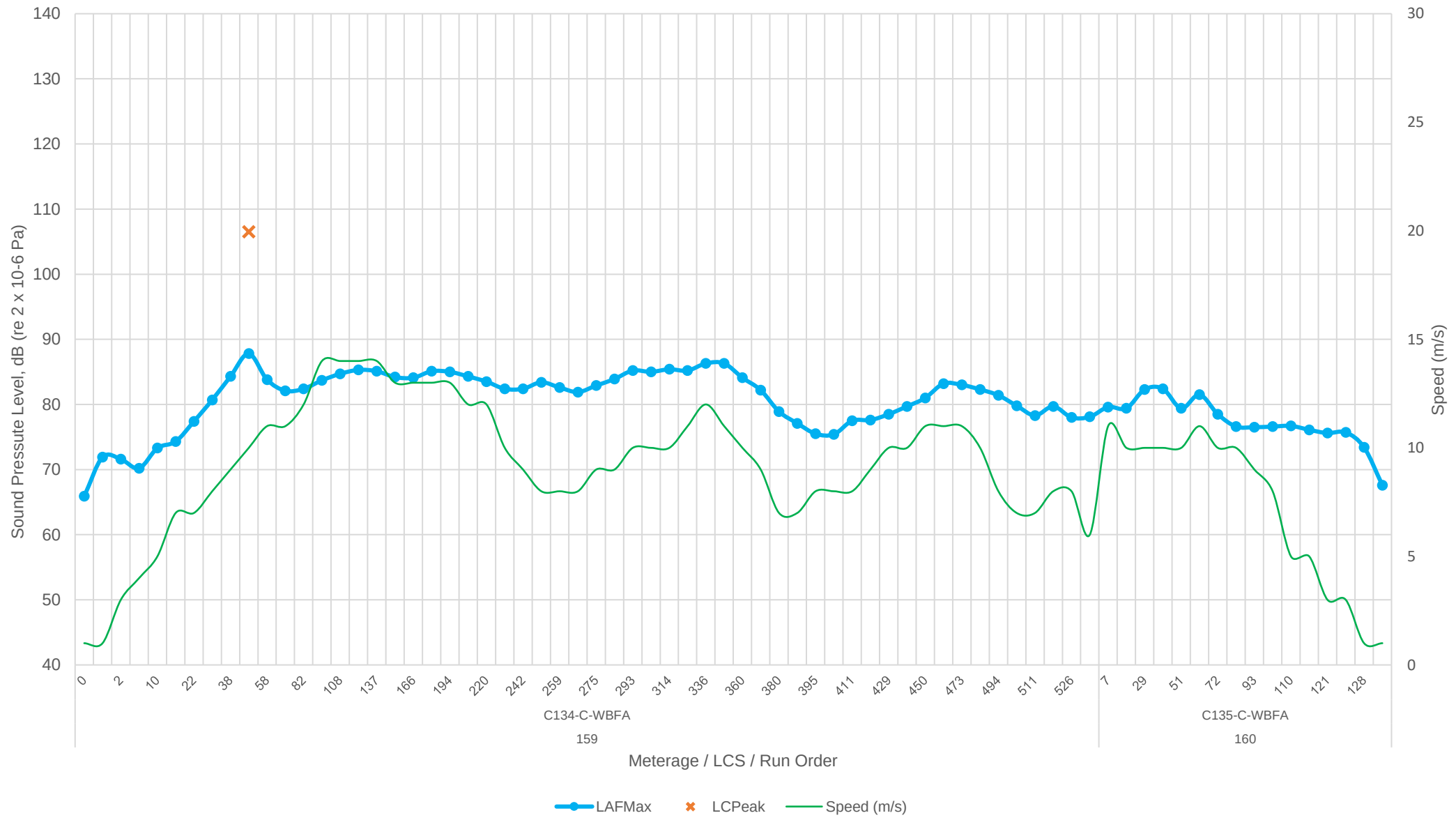


Figure C81 - Queensway (WB) to Notting Hill Gate (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

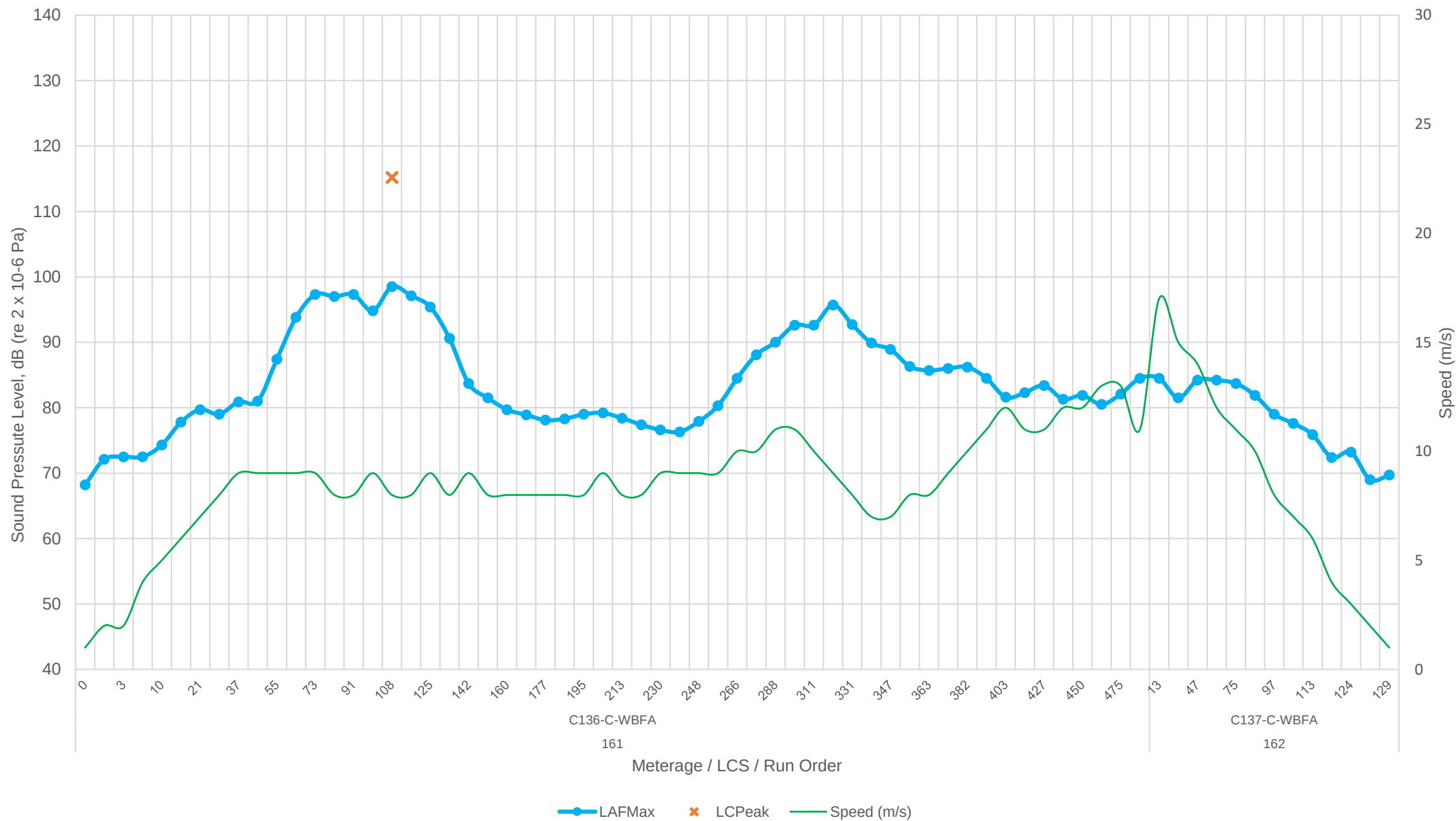


Figure C82 - Notting Hill Gate (WB) to Holland Park (WB)



Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

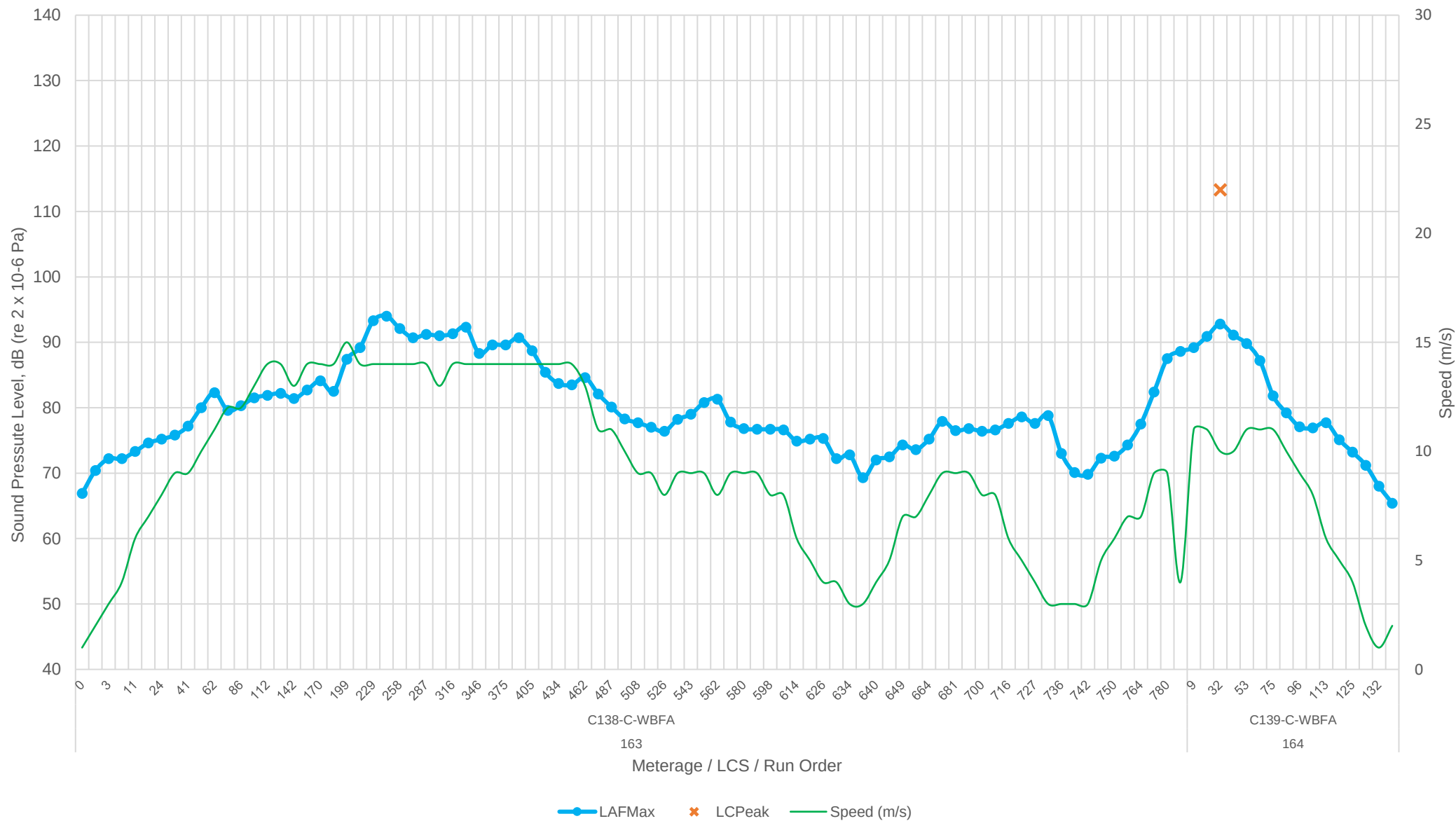


Figure C83 - Holland Park (WB) to Shepherd's Bush (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

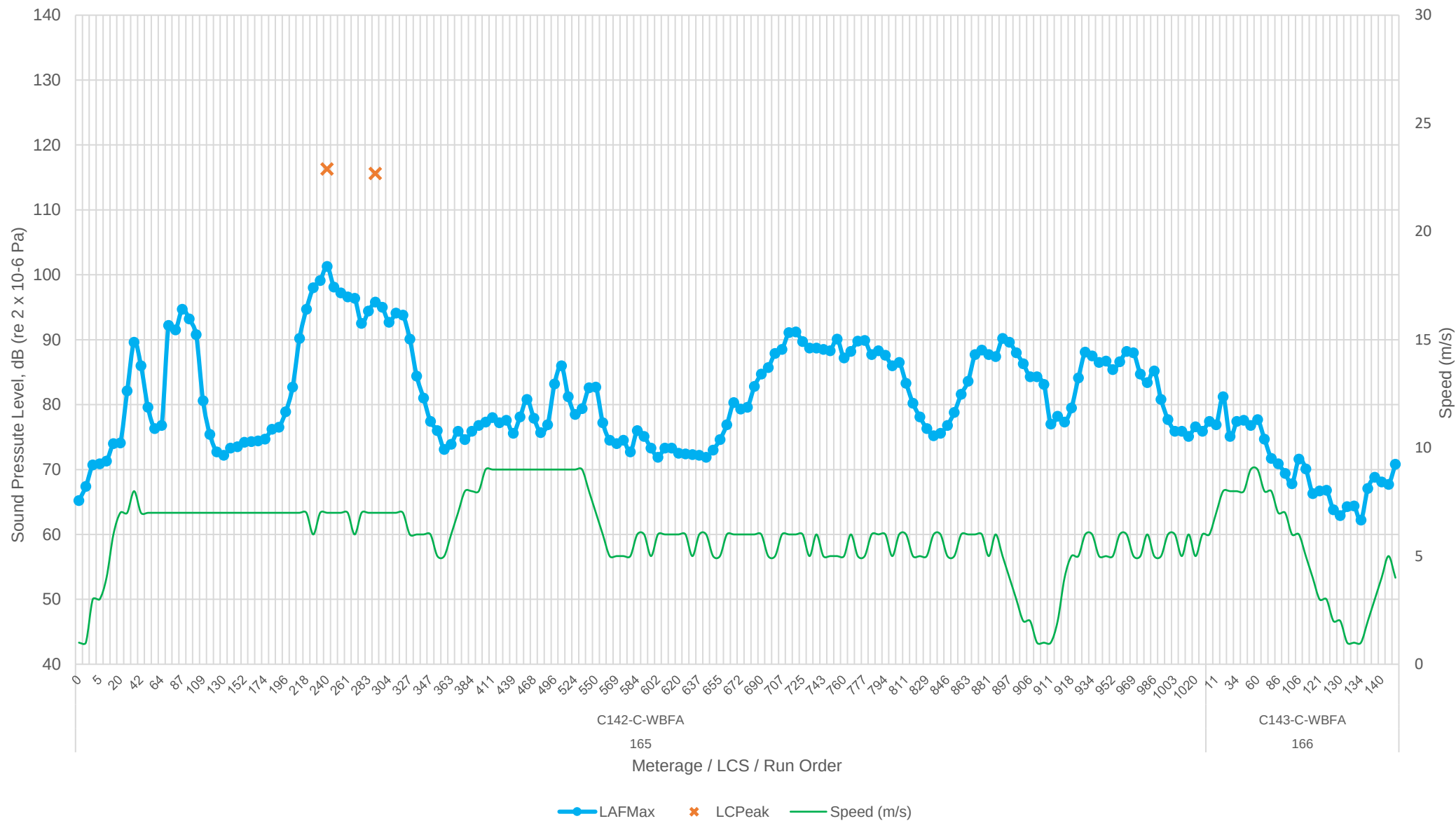


Figure C84 - Shepherd's Bush (WB) to White City (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

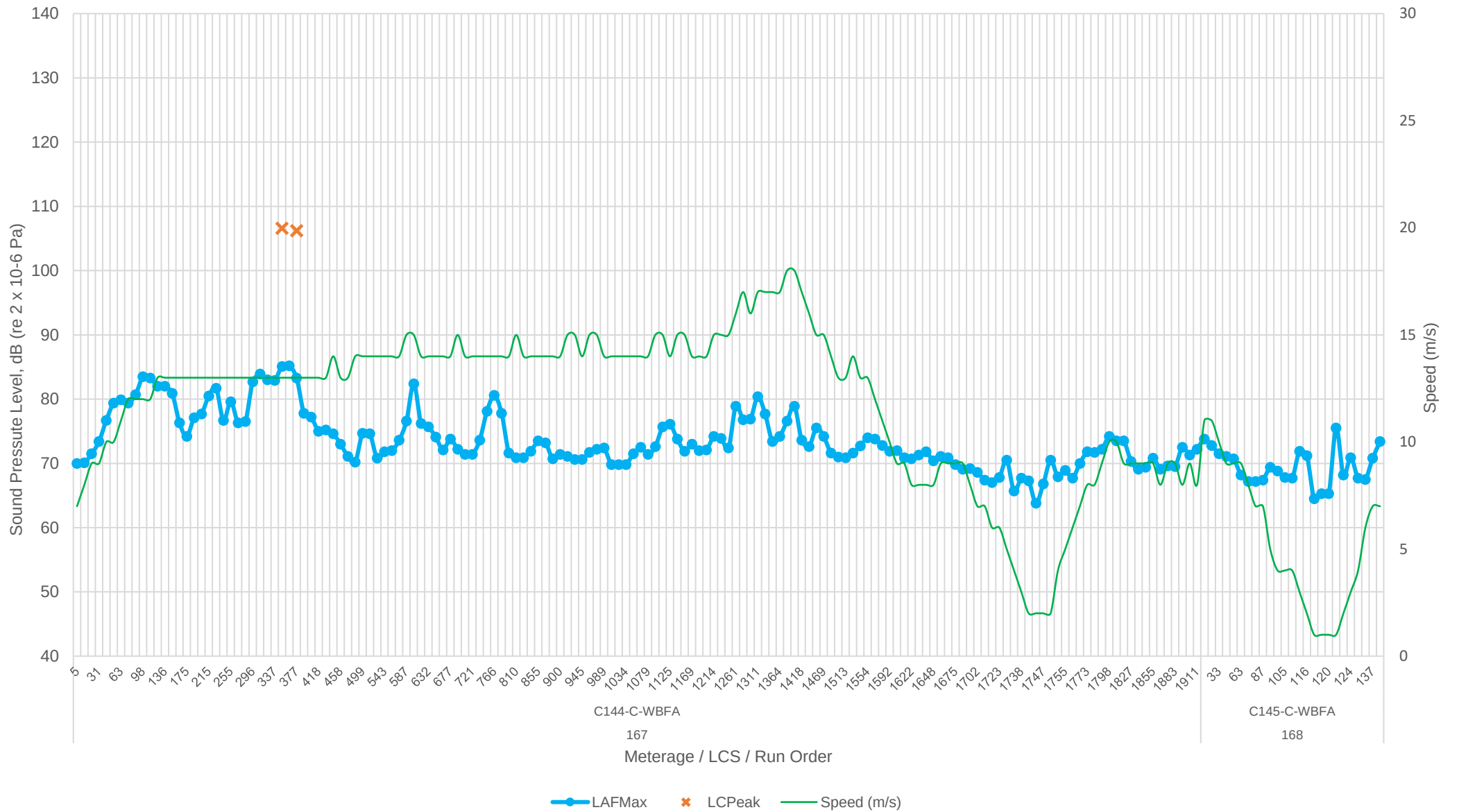


Figure C85 - White City (WB) to East Acton (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

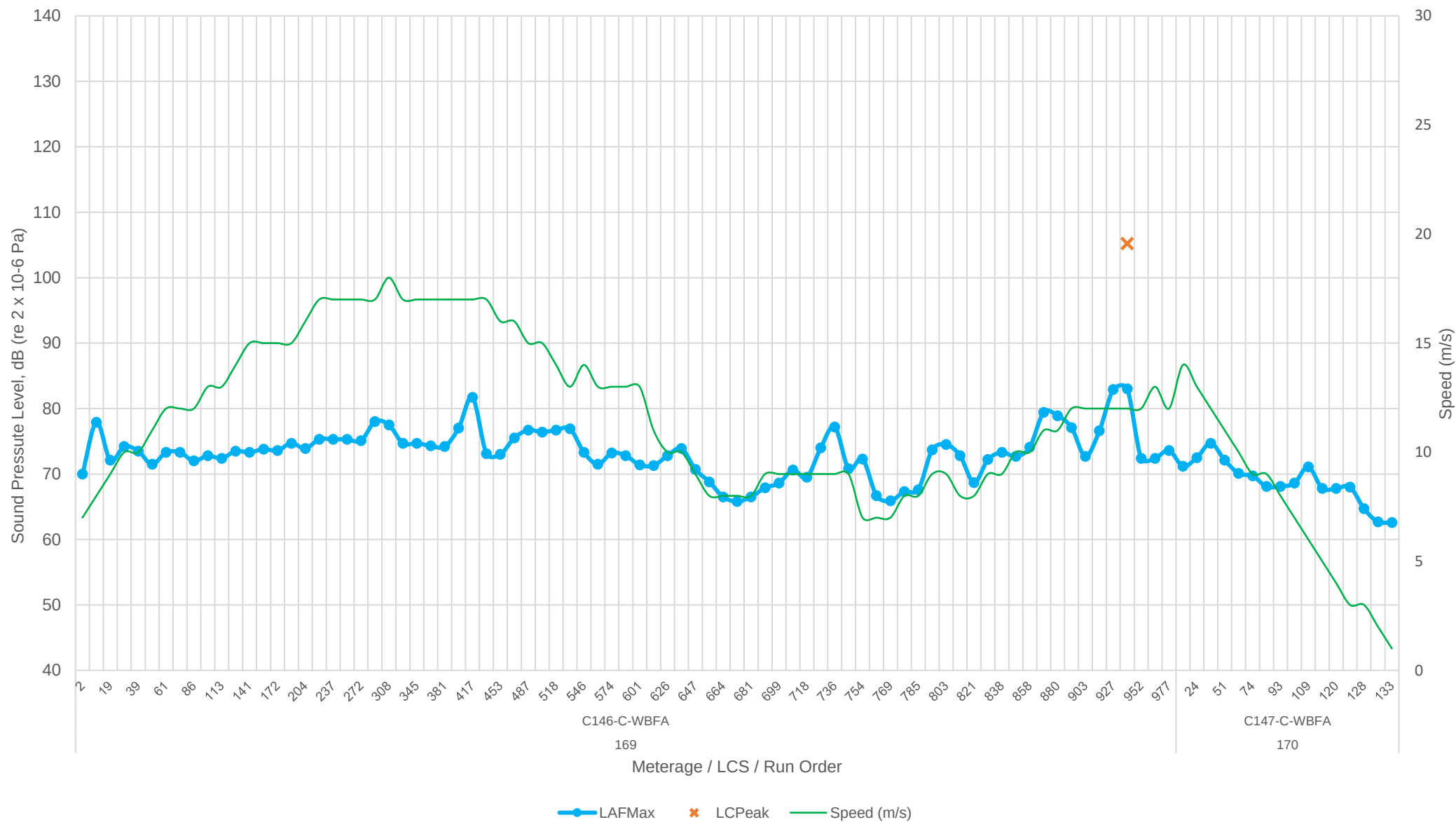


Figure C86 - East Acton (WB) to North Acton (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

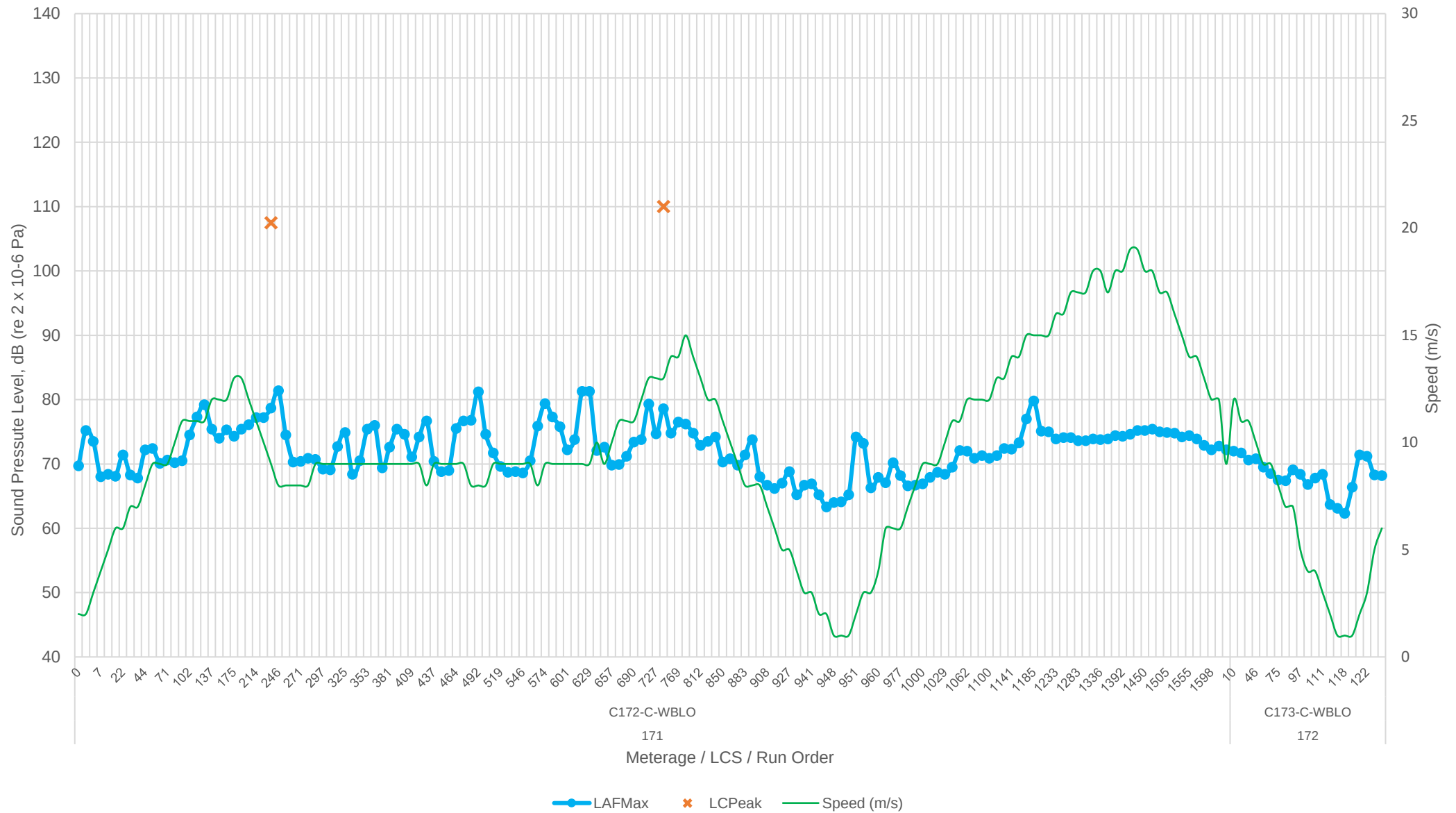


Figure C87 - North Acton (WB) to West Acton (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

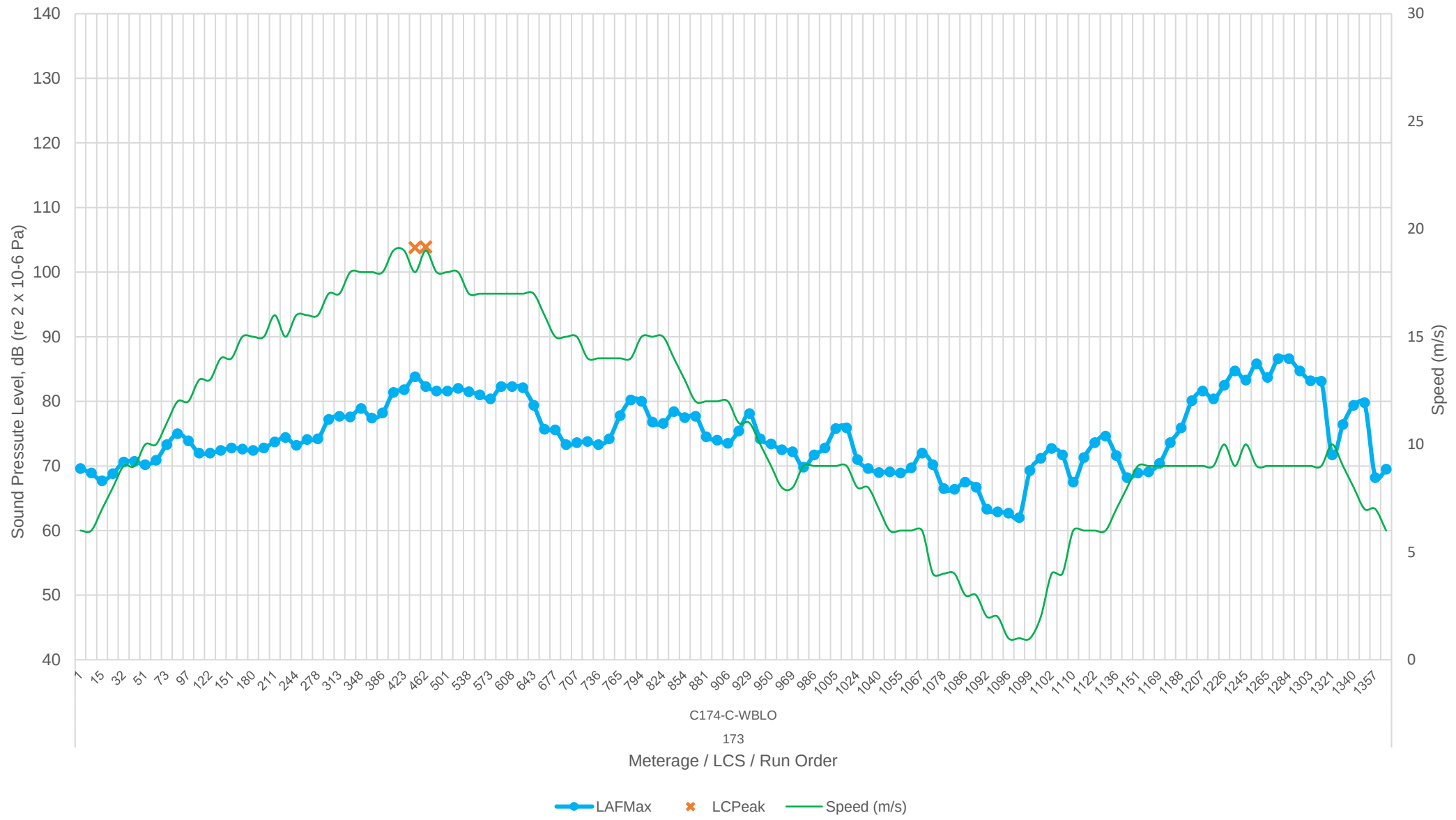
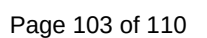


Figure C88 - West Acton (WB) to Ealing Broadway (Terminating)



# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

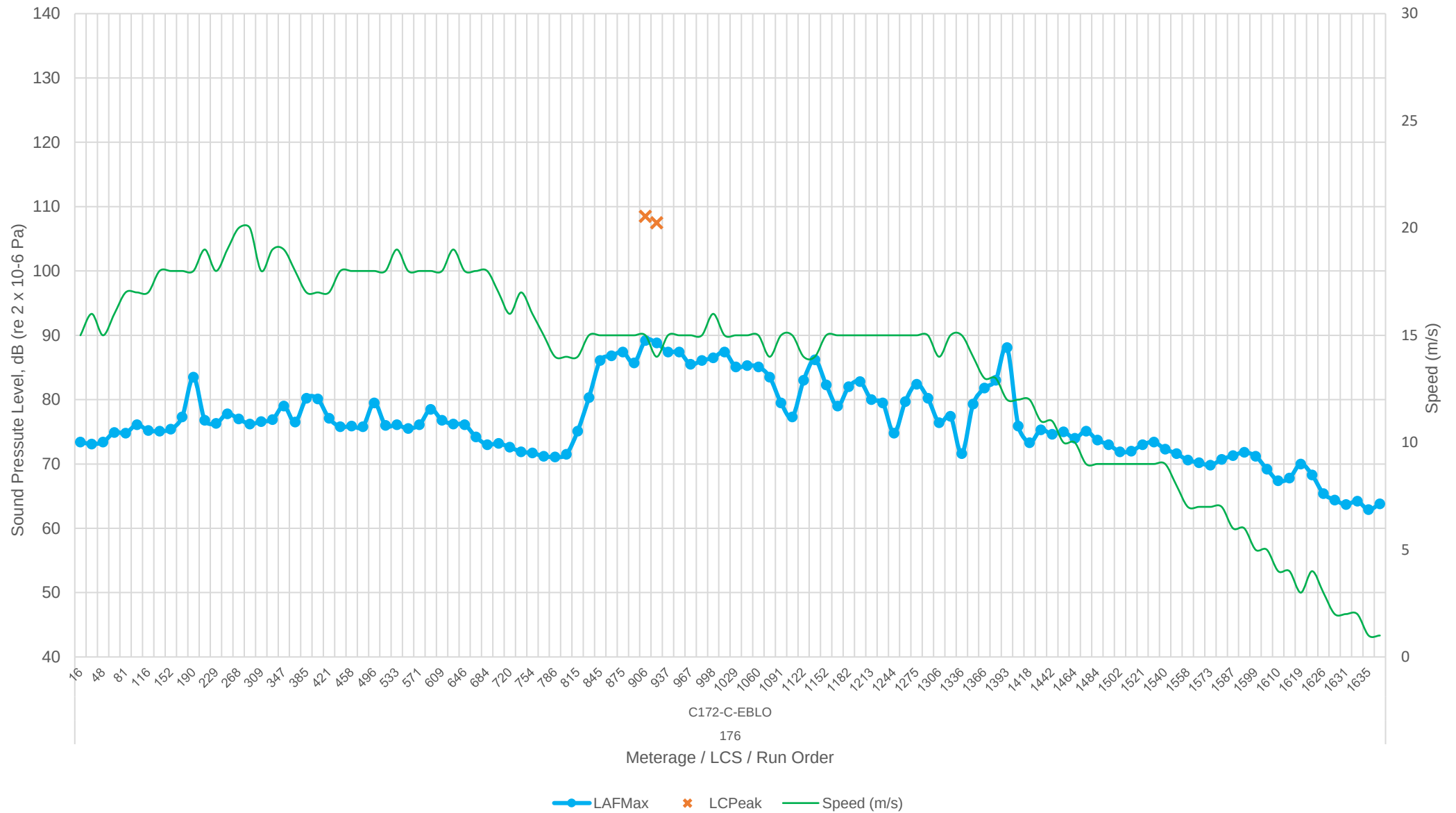


Figure C90 - West Acton (EB) to North Acton (EB)



Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

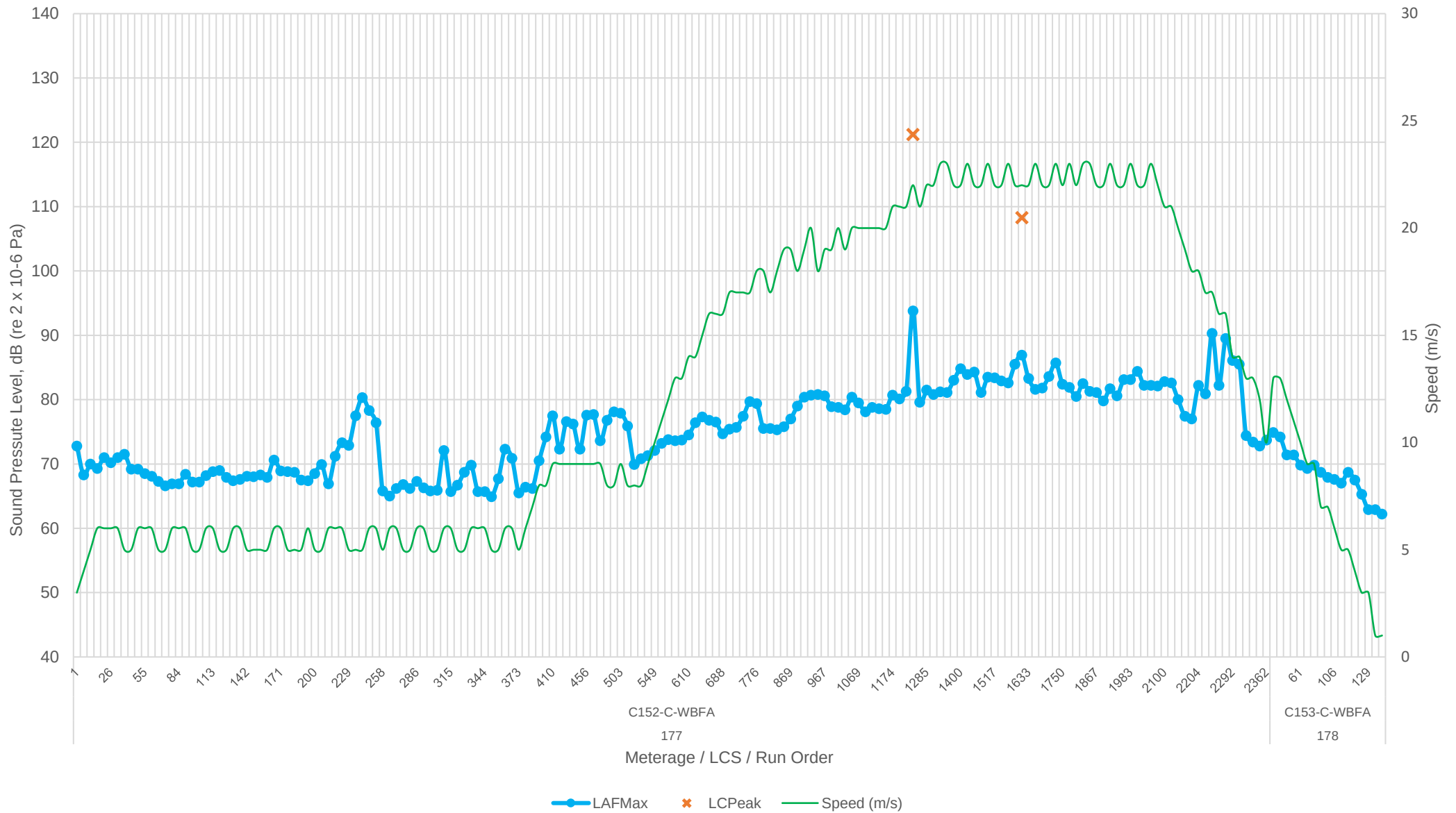


Figure C91 - North Acton (EB) to Hanger Lane (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

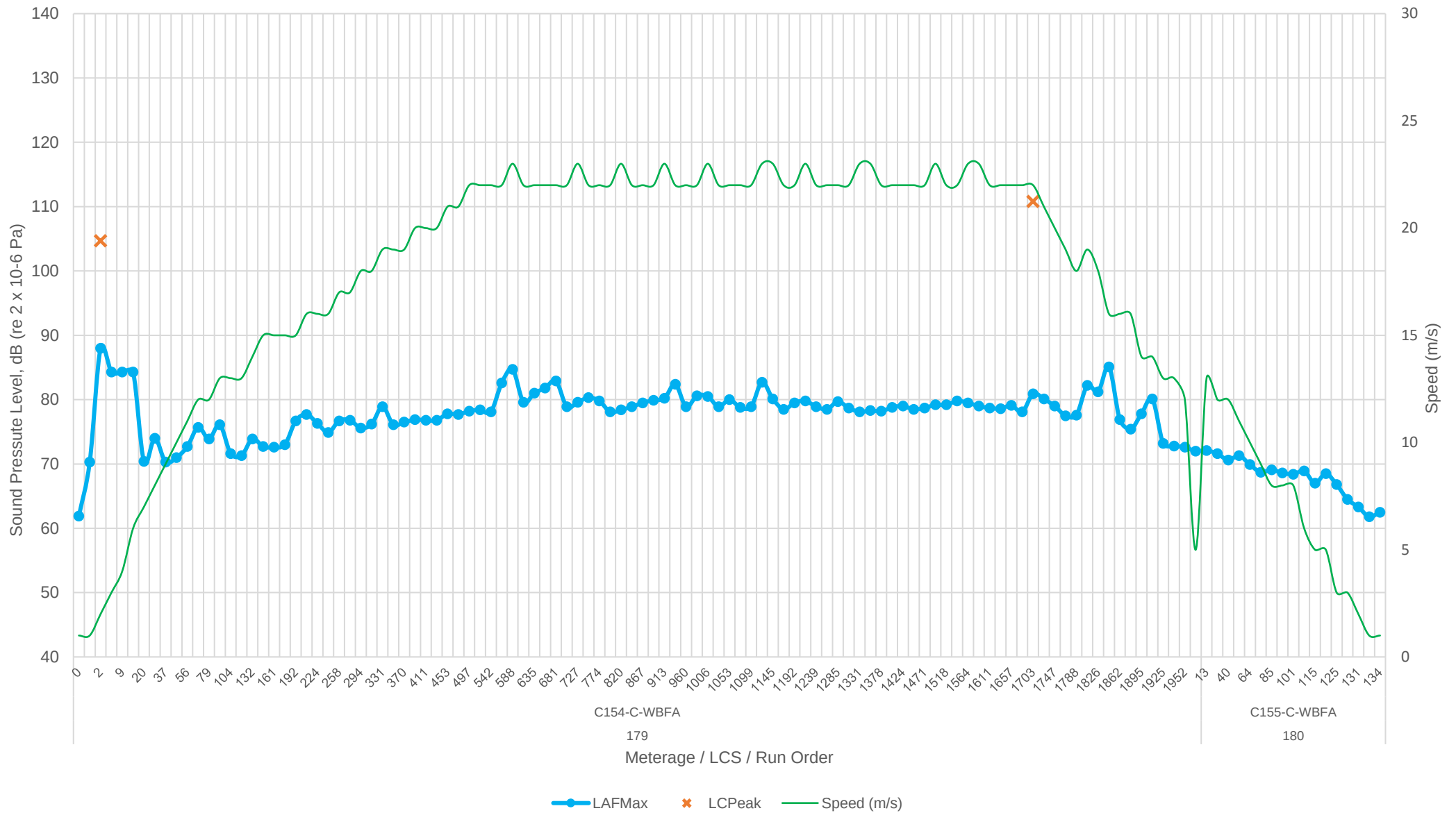


Figure C92 - Hanger Lane (WB) to Perivale (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

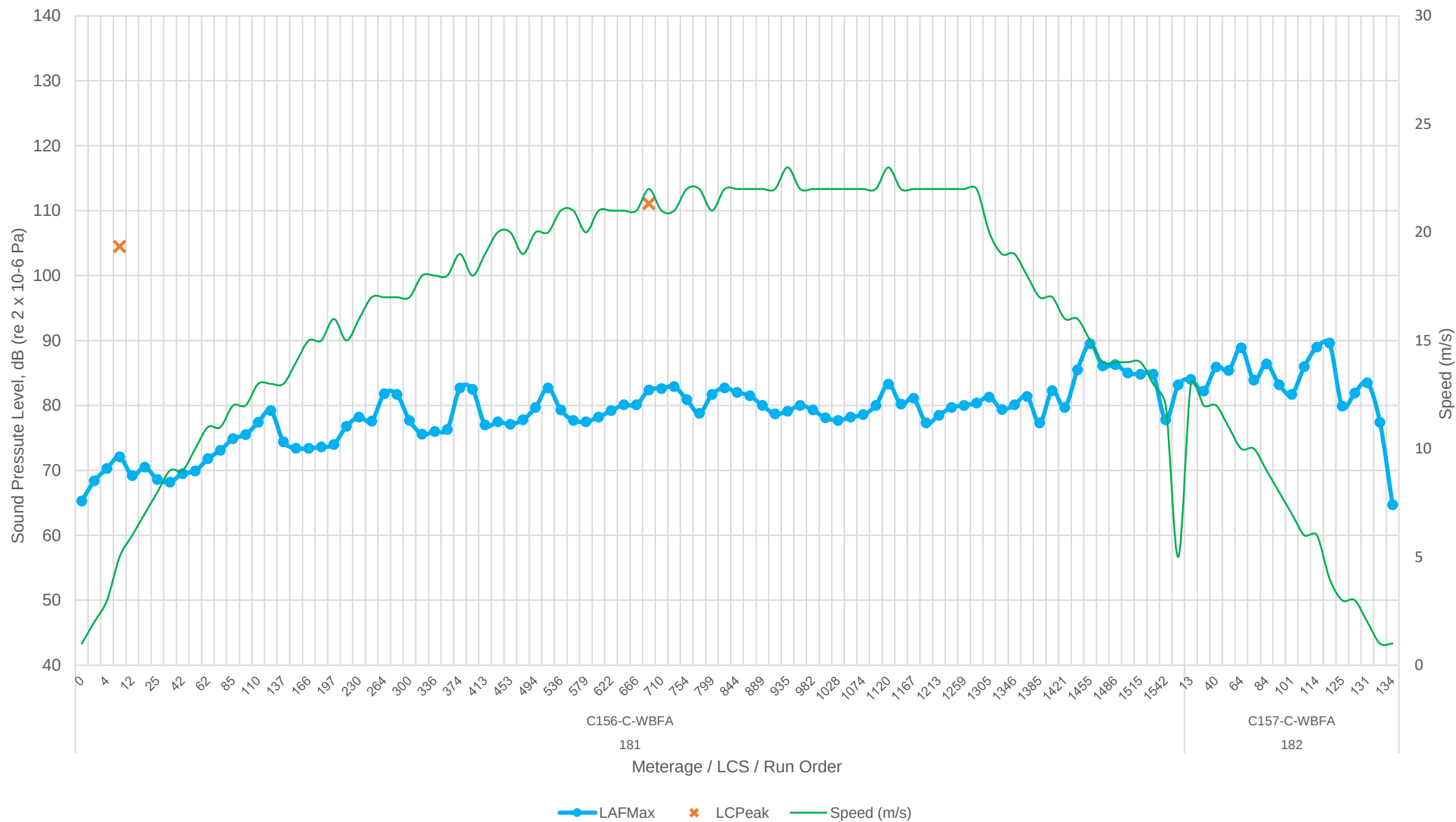


Figure C93 - Perivale (WB) to Greenford (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

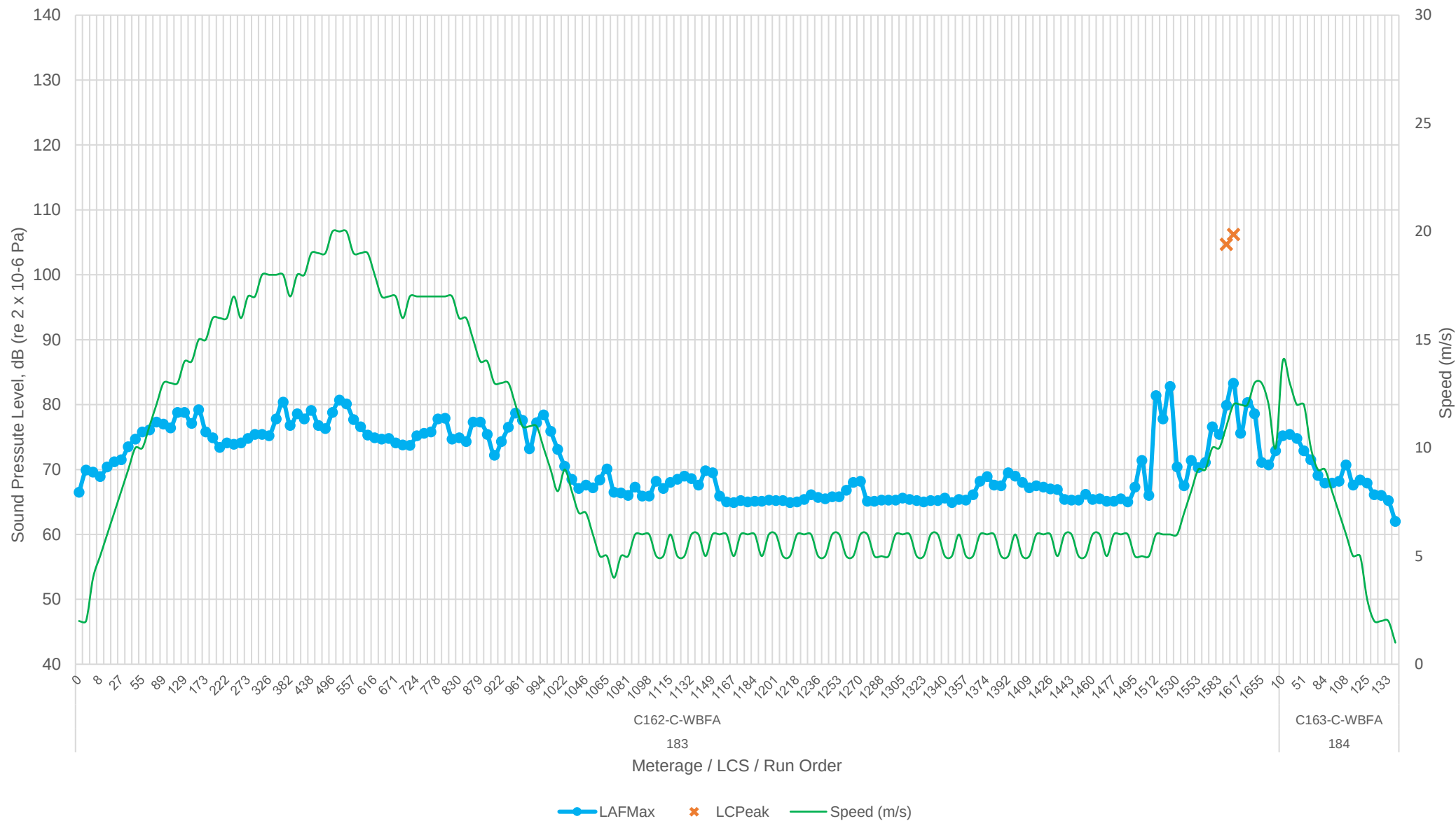


Figure C94 - Greenford (WB) to Northolt (WB)

# Acoustic Test Report: Interior Noise Measurement of a Test Vehicle

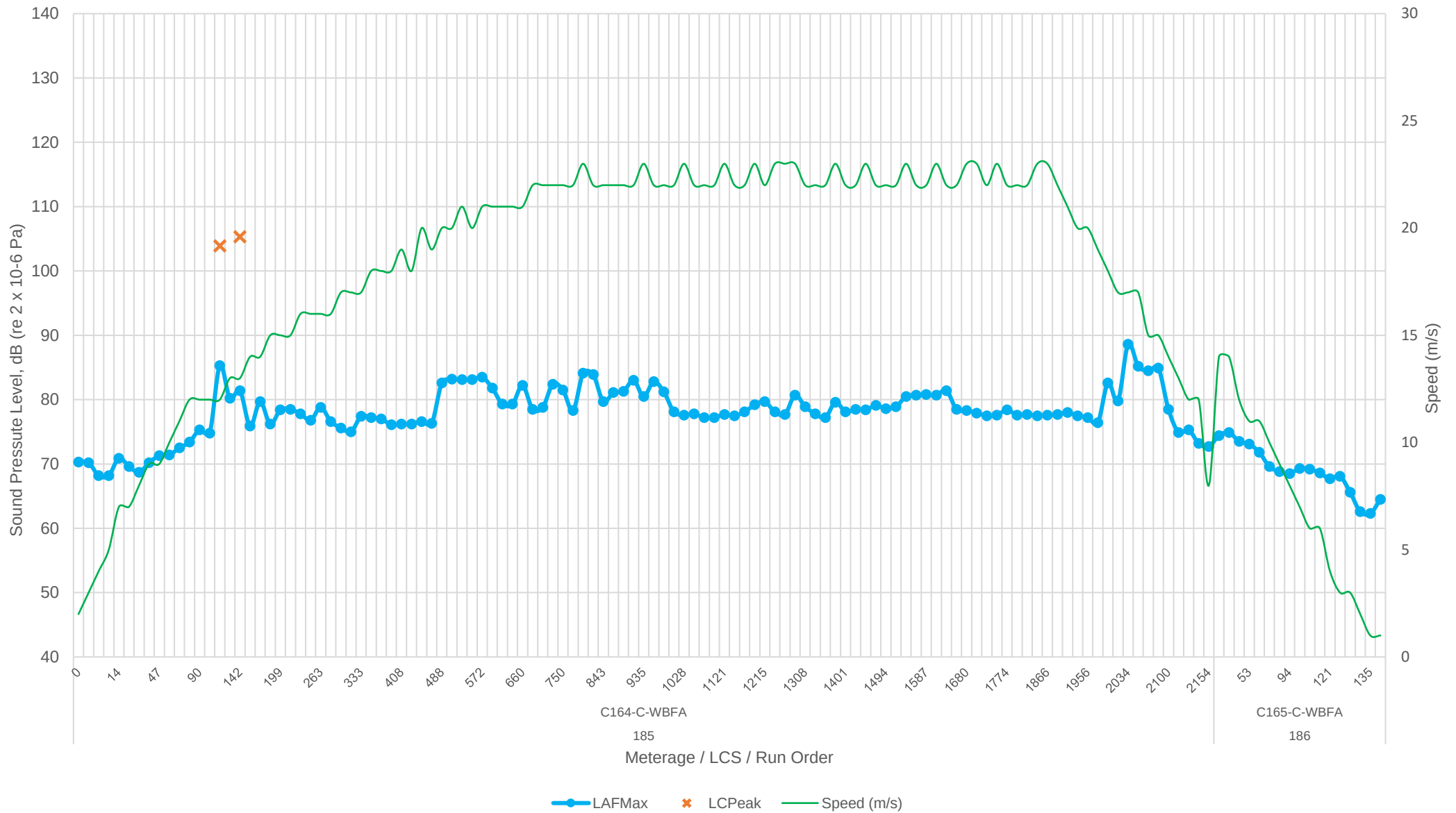


Figure C95 - Northolt (WB) to South Ruislip (WB)

Acoustic Test Report:  
Interior Noise Measurement of a Test Vehicle

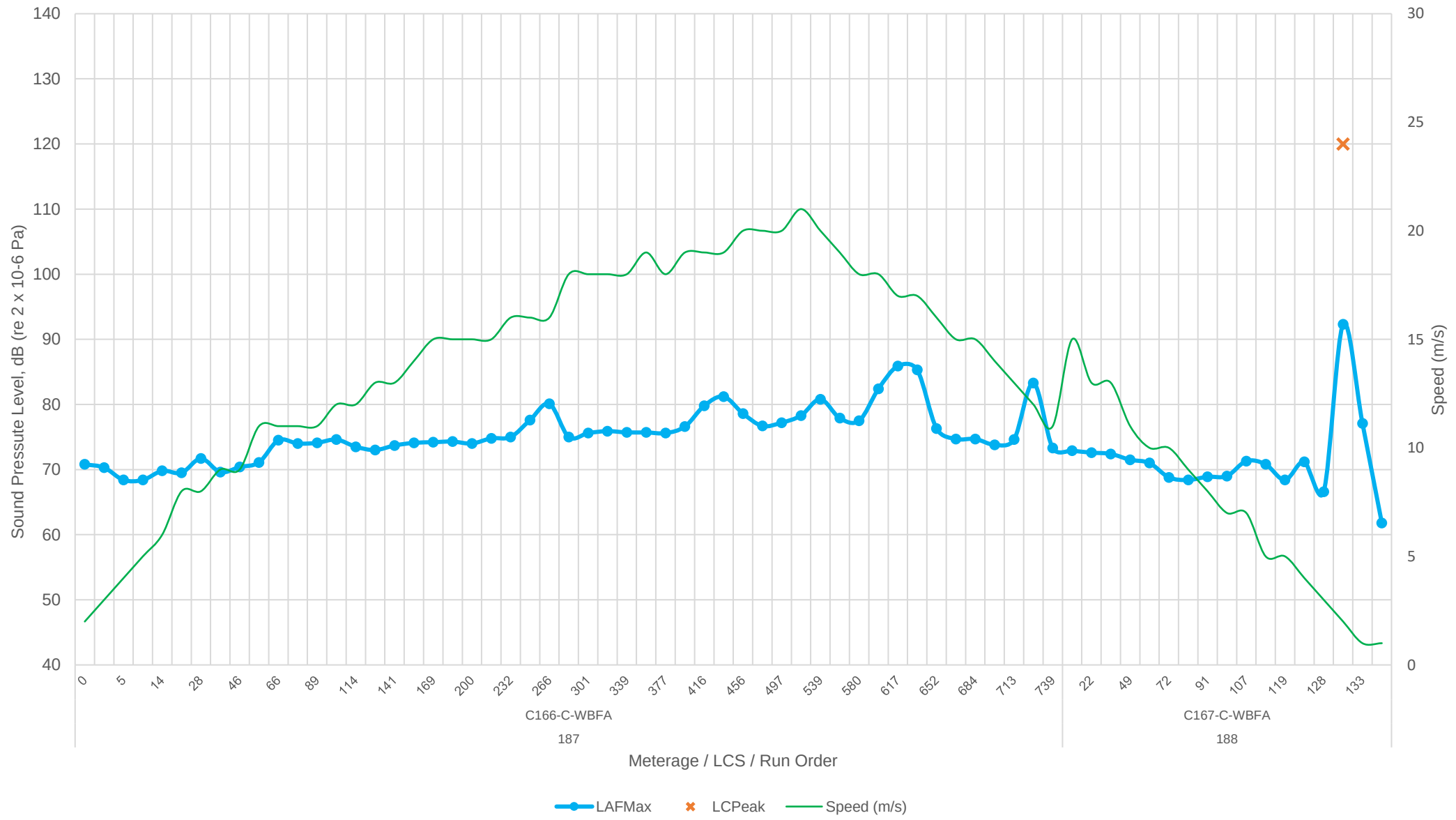


Figure C96 - South Ruislip (WB) to Ruislip Gardens (WB)